

The Connes–Kreimer Hopf Algebra and Hypersequent Proof Trees

Abstract

We ask what the Connes–Kreimer decomposition of hypersequent proof trees contributes to proof-theoretic semantics. Cutting an ancestry antichain detaches subproofs and leaves the typed context which uses them; filling the punctures reconstructs or varies the proof. A typed Faà di Bruno law relates retrospective cutting to prospective context composition. Two positive interpretations result. Backward-refinement histories modulo exchange of independent steps are precisely proof contexts, CK cuts are the frontiers of their presentation-causality ideals, and convolution powers of the all-ones and degree-one functionals count staged decompositions and complete construction schedules, respectively. Moreover, proof operations uniform under typed proof substitution are represented by open contexts. Filling repeated formal inputs by fresh probes and evaluating the CK coproduct by two characters gives the resource polynomial $\prod_i (1 + z_i)^{\nu_i}$; it detects linear, affine, relevant, erasing, and independently copying uses of supplied proofs. These results use cut support and multiplicity rather than rule names. The same construction then tests proof identity. Local rule-interface symmetries give CK-stable quotients, but neither a harmonious tensor reduction nor a standard permutation between sequentialisations of one MLL proof net can belong to any CK-stable proof congruence on this tree carrier. Component traces finally distinguish Communication from ordinary assembly and separate consequence-equivalent calculi with different direct Gentzen-Cut-free proof operations.

Introduction

Many central ideas in proof theory and proof-theoretic semantics—including harmony, polarity, proof-theoretic validity, and proof identity—have been articulated through the analysis of introduction and use rules for logical vocabulary [39, 18, 41, 37]. This paper studies a complementary aspect of proof structure. Once a sequent or hypersequent calculus and a material base are fixed, which families of subproof occurrences can be detached simultaneously, what typed residual context records their use, and which such decompositions survive when the proof is transformed or identified with another? Here a *material ground* is a tagged terminal proof object supplied by the base for an accepted primitive-boundary inference, rather than by a logical or structural rule. The base need not be monotone or closed under Weakening, Contraction, or composition.

The Connes–Kreimer admissible-cut construction gives an algebraic treatment of this question. We read a derivation as a rooted tree whose vertices are concrete rule occurrences and whose indexed child edges identify their premise positions. A CK-admissible cut selects an antichain of subproofs in the premise-ancestry order, so none contains another. Removing them leaves the root conclusion together with the surrounding derivation, punctured at the premise positions formerly occupied by those subproofs.

$$t = \underbrace{\frac{\pi : G_1 \quad \sigma : G_2}{G} a}_{a(\pi, \sigma)} \xrightarrow{\text{cut both premise edges}} \underbrace{\frac{\pi \sqcup \sigma}{P^c(t)}}_{P^c(t)} \otimes \underbrace{\frac{\square_1 : G_1 \quad \square_2 : G_2}{G} a}_{R^c(t)},$$

$R^c(t)[\square_1 := \pi, \square_2 := \sigma] = t, \quad R^c(t)[\square_1 := \pi', \square_2 := \sigma'] \text{ is defined when } \text{bd}(\pi') = G_1, \text{bd}(\sigma') = G_2.$

Figure 1: A CK cut and its proof-theoretic reading. The boundaries G_1, G_2, G may be hypersequents. Reinstating the selected proofs gives the distinguished reconstruction of t ; inserting other complete proofs of G_1, G_2 is filling, while inserting punctured contexts is context composition.

The construction has retrospective and prospective readings. Retrospectively, the cut records which subproofs a completed proof used and the rule context in which they occurred. Prospectively, the punctured context may be completed by any proofs of the required premises available in the calculus; the

proofs originally removed are one possible filling, not privileged contents of the punctures. Linearising all admissible cuts gives the CK coproduct, while multiplication forms the commutative forest of separate proof trees. The roots of such a forest can be flattened to one hypersequent boundary, but multiplication is neither an external rule nor a connected proof of that hypersequent. Successive cuts record nested stages of proof construction.

This factorisation has a causal reading, but a deliberately proof-specific one. Starting from an open conclusion, a proof context may be built by refining currently open premise positions. Refinements at independent positions commute, while a premise rule cannot be introduced before the rule whose premise it fills. The CK coproduct sees the same dependency in reverse: its terms are causal frontiers between a detached proof region and the context which uses it, and its convolution powers count staged decompositions. A second reading concerns hypothetical proof use. Once distinct open occurrences are allowed to name the same formal input, the coproduct records every independently detachable subfamily of the resulting proof copies. The history quotient uses strict interchange of independent context refinements; its CK incidence refinement and the resource statements use coproduct support and multiplicity, not classifications extracted from logical rule names.

Two small calculations preview the positive and negative payoffs. If a represented operation uses one formal proof twice, $R(x) = a(x, x)$, then the two singleton probe cuts and their simultaneous cut give

$$Q_R(z) = 1 + 2z + z^2;$$

the coefficient 2 counts cut addresses rather than reading a Contraction tag. Conversely, the two familiar sequentialisations of one MLL proof net have the same inputs and conclusion but expose different intermediate proof boundaries. The resource profile therefore agrees while raw CK factorisation separates them.

Hypersequents make the distinction between these operations particularly important. A connected proof of a displayed hypersequent is not the same proof object as a forest of independent proofs whose conclusions flatten to that hypersequent. Moreover, external structural rules may move, divide, or recombine component occurrences in ways which scalar tree data do not detect. We therefore retain at each rule vertex the component-incidence data recording which indexed premise-component occurrences contribute to which indexed conclusion components, and compose these relations along the punctured contexts exposed by CK cuts.

The logical and structural regime is supplied by the underlying calculus. Its logical rules, structural rules, normalisation conversions, and proposed proof equations are supplied by that calculus and justified on proof-theoretic grounds independently of the CK structure. The Hopf-algebraic analysis asks what those transformations preserve: their input proofs, their component action, and their decomposition into detached evidence and a surrounding proof context. This separates the proof-theoretic justification of an equation from the further question whether CK factorisation is well-defined on the resulting proof classes.

Section 1 constructs the proof-tree carrier, punctured derivations, and filling, and relates the one-hole case to the differential/zipper account of datatype contexts developed by McBride and Abbott et al. [34, 1]. Section 2 linearises CK cuts, proves a typed Faà di Bruno identity for context composition, recovers premise ancestry from iterated cuts, identifies refinement traces and their Hopf-convolution counts, and studies changes of material base. Section 3 adds component traces and the coideal criterion for proof equations, proves CK stability for local interface symmetries, and contrasts them with an MLL proof-net permutation. Section 4 separates factorisations internal to fixed input proofs from those contributed by a surrounding rule context, represents substitution-uniform hypothetical operations, extracts their CK resource-use polynomials, and applies the relative construction to local Gentzen-Cut reduction and harmony. Section 5 gives the specifically hypersequent application to Communication and analytic proof structure.

The paper can also be read payoff first. Readers primarily interested in the positive semantics may begin with the causal-schedule theorem in Theorem 2.12 and the resource-use theorem in Theorem 4.11. Readers primarily interested in harmony and proof identity may begin with the positive local-symmetry quotient in Proposition 3.12, the proof-net obstruction in Example 3.14, the tensor and tonk comparison in Proposition 4.16 and Example 4.17, and the Communication application in Section 5. They can then return to Sections 1–3 for the constructions supporting those tests. The dictionary below gives the intended proof-theoretic reading of the imported algebraic vocabulary; the formal definitions remain in the body of the paper, and the final notation index records symbols by section.

Reader's dictionary

The entries in this table are orientation rather than substitute definitions. In particular, algebraic inversion is not proof reduction, and a quotient becomes a semantics of proof identity only after the chosen proof calculus has supplied and proof-theoretically justified the equations being imposed.

algebraic or combinatorial term	proof-theoretic reading in this paper
forest, basis symbol $[F]$, product	A finite multiset of complete or punctured derivation trees. $[F]$ is its basis symbol; formal sums combine outcomes with coefficients, and multiplication juxtaposes independent proof blocks. Flattening their root boundaries displays a hypersequent profile, not a connected proof of that hypersequent.
CK cut and coproduct Δ_{CK}	A cut selects an ancestry antichain of subderivations. The coproduct is the formal sum of all pairs consisting of the detached forest and the punctured remainder.
ancestry incidence map, order character	Forgetting proof decorations sends the CK coproduct to the incidence coproduct of causal ideals. Convolution powers count causal layerings; restricting every layer to one vertex counts complete construction schedules.
counit ε_{CK}	The bookkeeping map which returns 1 on the empty forest and 0 on every nonempty forest; it is not a truth valuation.
puncture, filling fibre, corolla, telescope, zipper	Respectively: an open premise position; all compatible tuples of complete proofs which can fill the open positions; one rule occurrence with every premise open; the ordered open positions together with their boundary requirements; and the factorisation of a marked subproof into that subproof and its one-hole context. A nullary corolla has no open premises and is already a complete proof.
polynomial grammar and derivative	A sum over possible rule constructors, with a product over their premise slots. Its formal derivative marks one selected proof-input occurrence and retains the context surrounding it.
operad and context insertion	The operad is the typed associative composition of proof contexts with identities; it is a compact name for the insertion operations already present in the proof grammar.
Green series and Faà di Bruno identity	A Green series is the formal inventory of all contexts of one interface. The identity collects every way to read a composite context as inner proof contexts inserted into an outer context.
CK resource polynomial Q_R	Fill every occurrence of a formal proof input by a fresh probe, take the CK coproduct, and evaluate its two factors by probe and all-ones characters. Integrally, coefficients count simultaneous probe selections (with the whole-tree endpoint handling a nodeless identity) and hence independently copied, erased, or single uses of supplied proofs.
coaction, comodule, comodule map	A coaction is a coproduct-like observation which keeps the root-containing remainder—or the retained tuple of pointed inputs—in a distinguished right-hand factor. A comodule is a space carrying such an observation; a comodule map preserves it.
coideal, Hopf ideal, quotient	Together with counit compatibility, the coideal condition is precisely what lets a coproduct be computed on equivalence classes rather than representatives. A Hopf ideal is an algebra ideal which also satisfies this coideal condition, has zero counit, and is stable under the antipode; its quotient retains the Hopf structure.
local rule-interface symmetry	A presentation change confined to one rule occurrence: its boundary is fixed, while component action and formula incidences are transported by chosen occurrence reindexings; genuinely symmetric premise slots may be permuted or a local evidence tag forgotten. Such a change preserves ancestry and transports CK cuts one-for-one.
context-relative residue Ω_s ; protected replacement residue Res^{Prot}	The first subtracts cuts wholly internal to every designated input and retains cuts entering the fixed outer context. The second compares two maps after subtracting only the protected inputs' decompositions, so changes involving unprotected inputs remain. Its projection is complete for the displayed comparison ideal; full contextual closure is tested separately.

continued on next page

algebraic or combinatorial term	proof-theoretic reading in this paper
convolution and antipode	Convolution combines two maps by first decomposing an input, applying one map to each factor, and multiplying the results. The antipode supplies the recursive formal cancellation required for a convolution inverse; it does not invert proofs or reduce Gentzen Cuts.

1 Material inference and rooted proof trees

1.1 Boundaries, rule vertices, and component focus

Fix a finitary formula language $\mathcal{L}$ with a chosen primitive formula sublanguage $\mathcal{L}_{\text{pr}}$, often the atomic formulae. The primitive material objects below are not bare formulae: they are sequents or hypersequents all of whose formula occurrences lie in $\mathcal{L}_{\text{pr}}$. Write $M_f(X)$ for the finite multisets over X . Throughout, we treat formula contexts as finite multisets of formulae and hypersequents as finite nonempty multisets of sequents. A sequent is a pair $\Gamma \vdash \Delta$. A *proof-boundary hypersequent* is a finite nonempty multiset

$$G = S_1 \mid \cdots \mid S_n.$$

Write HSeq for their set, identifying a sequent with its singleton hypersequent. If $G = (S_j)_{j \in I_G}$, then $\text{Comp}(G) = I_G$ is its finite set of component *occurrences*. This indexing matters when two displayed components are equal: they remain two occurrences. A coherent reindexing is a bijection between equal multiset occurrences which preserves the displayed formula and component incidences.

Thus internal Exchange is already absorbed by equality of formula contexts, and external Exchange by coherent reindexing of component occurrences. Neither convention identifies the first and second premise slots of a binary rule. Nor does it add Weakening or Contraction; any such rule used by a proof is an explicit rule occurrence. Empty boundary hypersequents are excluded, while $\mathbf{1}$ will denote the empty proof forest.

A familiar component-local instance illustrates the data retained at a vertex:

$$\frac{\mathcal{G} \mid \Gamma, A \vdash B}{\mathcal{G} \mid \Gamma \vdash A \rightarrow B} (\rightarrow R). \quad (1)$$

The premise and conclusion are whole hypersequents. The displayed $\rightarrow R$ -occurrence acts on one component and carries every companion component in $\mathcal{G}$ unchanged. By an *external rule* here we mean one whose principal action changes or relates the hypersequent component structure, rather than a sequent rule acting within one component while carrying every companion component unchanged. Communication will provide the main example in Section 5.

Write HSeq_{pr} for hypersequents all of whose formulae lie in the primitive fragment, and fix a stock $\mathcal{R}$ of rule symbols, each with finite proof-rule arity.

The definitions permit arbitrary finite rule arity, although the displayed sequent and hypersequent rules are nullary, unary, or binary. Arity here counts immediate proof premises, not hypersequent breadth or the number of punctures accumulated through a proof.

Definition 1.1 (Concrete rule occurrence). A *concrete rule occurrence* is a finitary constructor with indexed premise slots

$$a = (\rho, \theta) : (G_1^a, \dots, G_{n(a)}^a) \longrightarrow G^a, \quad (2)$$

where ρ is the rule tag and θ is its fully instantiated formula and context data. The premises are ordered: for a binary rule they are its literal left and right proof premises. Write $\text{in}_i(a) = G_i^a$ and $\text{out}(a) = G^a$.

The instance data θ determine a component relation

$$\kappa_a \subseteq \left(\bigsqcup_{i=1}^{n(a)} \{i\} \times \text{Comp}(G_i^a) \right) \times \text{Comp}(G^a). \quad (3)$$

Read $((i, c), d) \in \kappa_a$ as saying that component occurrence c of premise i contributes tracked occurrence data to conclusion component d . For a logical or internal structural rule, θ selects one active component occurrence in each relevant premise and conclusion, together with its active and principal formula

occurrences. The relation κ_a includes the active action and the concrete bijections which carry passive component occurrences unchanged; equal passive components remain distinct occurrences. For such an external rule, θ may select several active components and κ_a may be a general finite relation. Any formula-occurrence incidence, discharge, or pairing used below is likewise read from the concrete instance. If a material rule tag does not determine such incidence, the required relation is included in θ ; it is not duplicated as extra decoration on every vertex. The occurrence is *primitive-boundary* when all G_i^a, G^a lie in HSeq_{pr} .

For example, if p, q, r are primitive formulae, a small material base might contain two tagged grounds

$$b_{pq} : () \longrightarrow (p \vdash q), \quad b_{qr} : () \longrightarrow (q \vdash r). \quad (4)$$

These are proof objects for the displayed primitive inferences. Their presence alone supplies neither a proof of $p \vdash r$ nor any weakened or contracted variant.

Definition 1.2 (Material base and the equipped calculus). A *material base* $\mathcal{B}$ is a tagged set of primitive-boundary material rule occurrences. Constructor tags are part of their identity, so several material inferences may have the same input–output profile. Its nullary subfamilies are

$$\mathcal{B}_G^{(0)} = \{b \in \mathcal{B} \mid n(b) = 0, \text{ out}(b) = G\}, \quad \mathcal{B}_0 = \coprod_G \mathcal{B}_G^{(0)}.$$

Here $\coprod$ is a disjoint union: it remembers the boundary G from which an occurrence came even when two displayed sets happen to coincide. No Weakening, Contraction, or Gentzen Cut occurrence is supplied merely by this definition. We distinguish material occurrences in $\mathcal{B}$, logical initials, and the allowed instances of logical or structural symbols in $\mathcal{R}$ by disjoint tags. A concrete occurrence *belongs to the equipped calculus* $(\mathcal{B}, \mathcal{R})$ when it is one of these occurrences. Thus $\mathcal{B}_0$ supplies terminal material grounds, while positive-arity members of $\mathcal{B}$ are primitive material inference operations in the same recursive rule grammar.

A *persistent material inclusion* $j : \mathcal{B} \hookrightarrow \mathcal{B}'$ identifies each old tagged material occurrence with one having the same ordered proof-input arity, concrete premise and conclusion boundaries, and component/formula-incidence data. Thus $\mathcal{B}'$ may add material grounds or primitive material operations, but it does not revise an old one. This extends the material base within the fixed language; it is not a formula-language extension. After identifying $\mathcal{B}$ with its image we write $\mathcal{B} \subseteq \mathcal{B}'$.

An arbitrary primitive material support relation

$$B \subseteq \mathbf{M}_f(\mathcal{L}_{\text{pr}}) \times \mathbf{M}_f(\mathcal{L}_{\text{pr}})$$

can be represented discretely by assigning each accepted pair $(\Gamma, \Delta) \in B$ one tagged nullary ground

$$\ulcorner \Gamma \vdash \Delta \urcorner : () \longrightarrow (\Gamma \vdash \Delta). \quad (5)$$

Here “nullary” refers only to proof-input arity: Γ remains part of the sequent boundary. This is a faithful discrete encoding of B : every accepted pair becomes a terminal proof generator, and no weakened, contracted, or composed pair is added automatically. Thus B determines the terminal part of the proof-tree carrier and the available filling fibres. It may, for example, contain $\Gamma \vdash \Delta$ while omitting $\Gamma, A \vdash \Delta$; the former ground cannot fill the latter boundary. A nonmonotone support relation therefore remains nonmonotone at the material level, while any monotonic closure of the equipped calculus must come from explicitly admitted material, structural, or logical rule occurrences. An inclusion of support relations gives a persistent inclusion of their grounds. Positive-arity material occurrences add proof-relevant internal structure which the discrete presentation omits.

1.2 Punctured derivations and filling fibres

Definition 1.3 (Proof tree and puncture). A (possibly punctured) proof tree is written recursively as the premise-slot-indexed term

$$t = a(u_1, \dots, u_{n(a)}),$$

and we use *term* for this constructor notation and *tree* for the rooted-tree presentation of the same object. The root is decorated by a concrete rule occurrence a , and each u_i is either a compatible rule-rooted proof term of boundary $\text{in}_i(a)$ or a vacant premise symbol in that position. Every vertex is an admitted concrete rule occurrence of the equipped calculus; punctures are vacant positions, not vertices. Thus the

carrier is calculus-relative: arbitrary decorated trees whose occurrences or boundaries are incompatible with $(\mathcal{B}, \mathcal{R})$ are excluded. Its root boundary is $\text{bd}(t) = \text{out}(a)$. Vertices are addressed by finite words of premise-slot indices; the root has address ϵ , and the i -th child appends i to its parent's address. Such a *premise-slot address* records successive choices of proof-premise slots from the root. It is a position in the proof tree, not a formula occurrence or a position inside a sequent. The same words address vacant child positions.

A *puncture* is a vacant child position $\square_p$ at an address p . If p is the i -th slot of the retained parent occurrence a , its required premise is inferred from that occurrence:

$$\text{req}_R(p) = \text{in}_i(a) \quad (6)$$

For the rule in (1), a puncture at its sole premise slot therefore requires the whole hypersequent $\mathcal{G} \mid \Gamma, A \vdash B$, while the retained occurrence identifies $\Gamma, A \vdash B$ as the active component and the members of $\mathcal{G}$ as unchanged companion components.

Write

$$\text{Punc}(R) = \{p \mid \square_p \text{ is a vacant position of } R\}$$

for the puncture addresses of the particular derivation R . Punctures belong to arbitrary open derivations; when a CK cut c is applied to a complete proof t , the newly punctured remainder satisfies $\text{Punc}(R^c(t)) = c$ under the retained addresses. A $(\mathcal{B}, \mathcal{R})$ -derivation is *complete* when $\text{Punc}(R) = \emptyset$ and otherwise is an *n -punctured derivation*, $n = |\text{Punc}(R)|$. Throughout, *proof* means a complete derivation; a punctured derivation viewed as an operation on its possible fillings is a *proof context*. Derivations are identified up to rooted-tree isomorphism preserving every concrete occurrence, premise-slot index, and coherent component reindexing.

Write $V(t)$ for the ordinary finite vertex set of a rule-rooted proof tree, and put $V(\square^G) = \emptyset$.

The lexicographic ordering of slot addresses gives the *premise telescope*: the ordered list of open positions and the hypersequent each position requires,

$$\text{Tel}(R) = (p_1 : \text{req}_R(p_1), \dots, p_n : \text{req}_R(p_n)). \quad (7)$$

Thus $\text{Tel}(R)$ is the ordered interface of the particular punctured proof tree R : each entry records an open tree address and the whole hypersequent boundary required there. These requirements are fixed by the retained, fully instantiated rule occurrences, so filling one puncture cannot change the requirement at another; in this precise sense the telescope is nondependent. An inserted context may itself introduce punctures, accounting for nested CK cuts. The same requirements serve as the typing colours in the algebraic presentation below. Optional variables $x_1, \dots, x_n$ label positions only when differently shaped contexts are compared over common formal inputs. They form a temporary many-sorted extension of the term grammar used to state substitution; they are not intrinsic edge labels, and erasing them returns the underlying context.

Write $\text{Prf}_{\mathcal{B}, \mathcal{R}}(G)$ for the set of complete proof trees with boundary G .

Definition 1.4 (Filling). For a punctured derivation R , its complete filling fibre—the set of all compatible tuples which can close its punctures—is the indexed Cartesian product of proof sets

$$\text{Fill}^{\mathcal{B}, \mathcal{R}}(R) = \prod_{p \in \text{Punc}(R)} \text{Prf}_{\mathcal{B}, \mathcal{R}}(\text{req}_R(p)). \quad (8)$$

This Cartesian product is unrelated to multiplication in the CK algebra. Evaluation grafts the selected proofs into their premise slots:

$$\text{ev}_R : \text{Fill}^{\mathcal{B}, \mathcal{R}}(R) \longrightarrow \text{Prf}_{\mathcal{B}, \mathcal{R}}(\text{bd}(R)), \quad (t_p)_p \longmapsto R[p := t_p]_p. \quad (9)$$

When the equipped calculus $(\mathcal{B}, \mathcal{R})$ is fixed, we abbreviate $\text{Fill}^{\mathcal{B}, \mathcal{R}}(R)$ to $\text{Fill}(R)$. For a complete proof t , a CK cut c supplies one point of $\text{Fill}(R^c(t))$: the indexed family of subproofs actually cut from t . Other points of the same Cartesian product give proofs with the same surrounding rule structure.

More generally, inserting punctured derivations of matching root boundary defines *context composition*. Addresses internal to an inserted context are prefixed by the slot address at which it was placed. The nodeless context $\square^G$ is its identity: it has boundary and sole requirement G , no rule vertex, and

$$\square^G[t] = t, \quad \text{Fill}^{\mathcal{B}, \mathcal{R}}(\square^G) \cong \text{Prf}_{\mathcal{B}, \mathcal{R}}(G). \quad (10)$$

It is a typed identity for context composition, not a generator of the CK algebra.

Proposition 1.5 (Associativity of context composition). *For each $p \in \text{Punc}(R)$, let S_p be a compatible context, and for each $q \in \text{Punc}(S_p)$, let $t_{p,q}$ be compatible with that puncture. Writing pq for concatenation of premise-slot addresses,*

$$(R[p := S_p]_p)[pq := t_{p,q}]_{p,q} \cong R[p := S_p[q := t_{p,q}]_q]_p. \quad (11)$$

Here $[p := S]$ denotes insertion, or substitution, at puncture address p . Insertions at distinct puncture addresses commute. When the inserted objects are complete proofs, this restricts to the evaluation maps of Definition 1.4.

Proof. Both sides are the same concrete-occurrence-decorated rooted tree: every $t_{p,q}$ occupies the prefixed address pq , and all other vertices retain their addresses. Concatenation of premise-slot words is associative, while insertions at incomparable addresses modify disjoint subtrees. $\square$

Backward proof search uses the same objects in the other direction. If $p \in \text{Punc}(R)$ and a is an occurrence of the equipped calculus with $\text{out}(a) = \text{req}_R(p)$, let $\text{Cor}(a) = a(\square_1, \dots, \square_{n(a)})$ be its one-vertex *corolla*: the rule occurrence with each of its premise positions left open (and hence a complete proof when a is nullary). The refinement

$$R[p := \text{Cor}(a)] \quad (12)$$

replaces one puncture by the premises of a . Proposition 1.5 says that solving those goals and then filling p agrees with filling p by the completed rule application. Retrospective decomposition and prospective refinement therefore meet in the same context calculus without identifying a puncture with any privileged filler.

The bridge to the standard typed-tree terminology can now be stated without interrupting that proof-theoretic construction. For a complete proof t , let $\mathcal{D}$ be the set of concrete rule occurrences and let $\mathcal{T} = \mathbb{N}_{>0} \times \text{HSeq}$. If $e_{v,i}$ joins the child in slot i to its parent v , put

$$\text{dec}(v) = a_v, \quad \text{type}(e_{v,i}) = (i, \text{in}_i(a_v)), \quad \text{out}(a_{vi}) = \text{in}_i(a_v). \quad (13)$$

Thus a complete proof is a $\mathcal{D}$ -decorated, $\mathcal{T}$ -typed tree in Foissy's sense [26], restricted by rule-boundary compatibility. The underlying CK tree is nonplanar; the first coordinate of the edge type retains the rule's premise slot, even when two premises have the same boundary. We draw children in slot order, but use the ordinary commutative CK product on typed and decorated forests, not a Loday–Ronco product. Punctured derivations are the corresponding proof contexts with vacant typed slots.

The full Foissy-style carrier may contain arbitrary combinations of its available decorations and edge types, independently of whether those combinations are derivations licensed by the chosen calculus. We use that construction as the ambient CK theorem, but not as the basis of proof objects: the set $\mathbf{T}$ below is the subcollection recursively generated by the admitted concrete occurrences of $(\mathcal{B}, \mathcal{R})$. Hypersequent boundaries provide its typing colours, while membership in the equipped calculus and the matching equations in (13) exclude ill-typed or unlicensed decorated trees. An open member can have an empty complete filling fibre; it is nevertheless a well-typed proof context rather than an ambient ill-typed tree.

1.3 One-hole proof contexts and context composition

The proof-theoretic content of this subsection is the zipper principle: a proof with one marked subproof factors uniquely as that subproof together with the one-hole context which uses it. The polynomial notation packages the recursive proof grammar and extends the same factorisation to several distinctly labelled punctures; it introduces neither new proof objects nor a semantics for the rules.

Let $\text{Ctx}(H; G)$ be the family of G -rooted one-punctured derivations with requirement H and a complete proof in every off-path premise. It includes the nodeless identity $\square^G$ when $H = G$.

An immediate *rule frame* from K to G consists of a concrete root occurrence in the equipped calculus, a chosen premise of boundary K , and complete proofs of all its other premises. The symbol $\coprod$ below denotes a disjoint union, so the chosen rule occurrence and premise slot remain identifiable:

$$\text{Frame}(K; G) = \coprod_{\substack{a \text{ belongs to } (\mathcal{B}, \mathcal{R}), \\ 1 \leq i \leq n(a), \text{ out}(a)=G, \text{ in}_i(a)=K}} \coprod \text{Prf}_{\mathcal{B}, \mathcal{R}}(\text{in}_j(a)). \quad (14)$$

Every one-hole context other than the nodeless identity has a unique final rule frame and a unique child context on the path to its puncture. Hence

$$\text{Ctx}(H; G) \cong \left(\begin{cases} \{\square^G\}, & H = G, \\ \emptyset, & H \neq G \end{cases} \right) \sqcup \coprod_{K \in \text{HSeq}} \text{Frame}(K; G) \times \text{Ctx}(H; K). \quad (15)$$

Thus a one-hole proof context is a finite typed path of immediate rule frames, retaining the completed side proofs at every step. Marking a subproof makes the elementary factorisation visible before any polynomial notation is introduced. In Figure 1, for example, marking π factors $a(\pi, \sigma)$ into the proof π and the one-hole context $a(\square_1, \sigma)$; filling that hole by π reconstructs the original proof. The resulting zipper decomposition—named for the marked subproof together with the context focused on its position—is the typed bijection

$$\{(t, v) \mid t \text{ is a complete } G\text{-proof and } v \text{ an } H\text{-subproof occurrence}\} \cong \text{Ctx}(H; G) \times \text{Prf}_{\mathcal{B}, \mathcal{R}}(H). \quad (16)$$

The forward map removes the marked subproof. Filling and marking the root of the inserted copy gives the inverse. In particular, removal at the root corresponds to the nodeless identity rather than a connected context.

The polynomial notation now packages the same concrete grammar. Here “polynomial” means a disjoint sum over possible final rule occurrences, with a Cartesian product over the ordered premises of each chosen rule. For a family $X = (X_G)_{G \in \text{HSeq}}$, put

$$(P_{\mathcal{B}, \mathcal{R}} X)_G = \coprod_{\substack{a \text{ belongs to } (\mathcal{B}, \mathcal{R}) \\ \text{out}(a)=G}} \prod_{i=1}^{n(a)} X_{\text{in}_i(a)}. \quad (17)$$

The family of slot-indexed proof terms with formal punctures is the least solution

$$\text{OpenPrf}_{\mathcal{B}, \mathcal{R}}(X) \cong \mu Y (X + P_{\mathcal{B}, \mathcal{R}}(Y)). \quad (18)$$

Thus μ denotes the family of finite well-founded terms generated by the displayed choices, not a further proof operation. The X -summand consists of explicitly sorted nodeless punctures. At every nonroot variable leaf below a rule occurrence, the same sort is instead inferred from the parent occurrence and its premise slot. Variables are used for substitution but contribute no vertex to the CK algebra. At $X = 0$, the grammar gives $\text{Prf}_{\mathcal{B}, \mathcal{R}}(G)$, the complete G -proofs. Under a constructor-faithful Curry–Howard interpretation, punctured derivations are sent to typed term contexts and filling to simultaneous term substitution. A general proof context need not be an *evaluation* context: the latter is a strategy-dependent one-hole subgrammar selected by an operational semantics.

Following McBride’s differential account of regular datatypes, taking a formal derivative marks one input occurrence in this sum–product grammar and retains the constructor data surrounding it [34, 1]. Accordingly the zipper bijection can be written

$$\text{Ctx}(H; G) \cong (\partial_H \text{OpenPrf}_{\mathcal{B}, \mathcal{R}, G})(0).$$

Here the formal derivative ∂_H selects one occurrence of an H -sorted variable in the sum-of-products grammar and retains its surrounding constructor data.

For a distinctly labelled family $\mathbf{H} = (x_1 : H_1, \dots, x_n : H_n)$, replace each coordinate X_G by the labelled sum

$$X_G^{\mathbf{H}} = \coprod_{\{i: H_i=G\}} X_{G, x_i},$$

and write ∂_{H_i, x_i} for differentiation in the indicated summand. Let $\text{Ctx}(\mathbf{H}; G)$ be the G -rooted contexts having exactly the labelled punctures $x_i : H_i$, with every other branch complete. The corresponding mixed derivative classifies these n -punctured derivations:

$$\text{Ctx}(\mathbf{H}; G) \cong \partial_{H_1, x_1} \cdots \partial_{H_n, x_n} \text{OpenPrf}_{\mathcal{B}, \mathcal{R}, G}(X^{\mathbf{H}}) \Big|_{X^{\mathbf{H}}=0}. \quad (19)$$

Distinct labels avoid symmetry factors when requirements coincide. Differentiation marks subderivation occurrences, not formula occurrences. The puncture addresses of one context form an antichain in the prefix order; nested detachments are represented by composition of punctured derivations.

The following family forgets the optional external labels x_i , but retains every intrinsic slot address and its lexicographic order.

For an ordered word $\mathbf{K} = (K_1, \dots, K_n)$ of premise requirements, write $\tau(C)$ for the word underlying $\text{Tel}(C)$ after its addresses are suppressed, and let $\mathcal{O}(\mathbf{K}; G)$ be the proof contexts of the equipped calculus with root boundary G and inferred requirement word $\tau(C) = \mathbf{K}$ in slot-address order. Put

$\mathcal{O}(); G = \text{Prf}_{\mathcal{B}, \mathcal{R}}(G)$, and let $\mathcal{O}^+(\mathbf{K}; G)$ contain the contexts with at least one rule vertex. The nodeless context $\square^G \in \mathcal{O}((G); G)$ is the identity. Simultaneous context composition has the typed form

$$D\langle C_1, \dots, C_m \rangle \in \mathcal{O}(\mathbf{K}_1 \cdots \mathbf{K}_m; G) \quad (20)$$

whenever

$$D \in \mathcal{O}((H_1, \dots, H_m); G), \quad C_j \in \mathcal{O}(\mathbf{K}_j; H_j).$$

Here juxtaposition concatenates ordered words. Proposition 1.5 supplies associativity and the nodeless contexts supply identities. Algebraically, this says that $\mathcal{O}$ is an **HSeq**-coloured nonsymmetric operad: “coloured” means that every insertion must match a premise requirement, “nonsymmetric” means that premise-slot order is retained, and “operad” means here only that compatible contexts may be inserted associatively with identities. We use the term as shorthand for the context-composition operation already defined; it introduces no new proof objects.

1.4 Proof forests and two root profiles

Let $\mathbf{T}$ be the set of isomorphism classes of complete and punctured rule-rooted derivation trees, so every member has positive vertex number, and $\mathbf{F} = \mathbf{M}_f(\mathbf{T})$ the commutative monoid of finite forests under $\sqcup$, with unit $\mathbf{1}$. Nodeless typed punctures remain context identities with an externally retained sort; they are not additional degree-zero forest generators. All its connected factors are derivations over the fixed equipped calculus. Its puncture set is the disjoint union of the puncture sets of its connected factors, with the factor occurrence retained; a labelling is a bijection from a finite label set to that disjoint union, and req_F is inherited factorwise. If $F = t_1 \sqcup \dots \sqcup t_m$, its labelled premise family is the disjoint union of the premise telescopes of its connected factors. The labels distinguish equal requirements without imposing an order on the forest factors. Define its *block-root profile* and flattened root by

$$\begin{aligned} \text{BRoot}(F) &= \{\text{bd}(t_1), \dots, \text{bd}(t_m)\} \in \mathbf{M}_f(\mathbf{HSeq}), \\ \text{Root}(F) &= \text{flat BRoot}(F) = \text{bd}(t_1) \mid \dots \mid \text{bd}(t_m). \end{aligned} \quad (21)$$

Here *flat* takes the multiset union of all displayed component occurrences in the block-root profile. A connected proof $z : S \mid T$ has block profile $\{S \mid T\}$, whereas a forest $p : S \sqcup q : T$ has profile $\{S, T\}$. Their flattened roots coincide. The comma inside a sequent, the bar inside a connected hypersequent boundary, a formula connective, and forest multiplication are four different operations. In particular, forest multiplication is not an implicit assembly rule: an assembly occurrence admitted by the calculus is a vertex producing one connected proof.

2 Punctured factorisation and substitution

2.1 CK cuts and punctured derivations

Let t be a complete or punctured derivation tree, with vertex positions $\text{Pos}(t)$ given by premise-slot addresses. A *CK-admissible cut* is a finite prefix-free set

$$c \subseteq \text{Pos}(t) \setminus \{\epsilon\}.$$

Prefix-free means that no selected position extends another; equivalently, no selected subderivation contains another selected one. For $p \in c$, write $t|_p$ for the subtree rooted at p . Detach these subtrees and leave a puncture at each same address:

$$\mathbf{P}^c(t) = (t|_p)_{p \in c}, \quad P^c(t) = \bigsqcup_{p \in c} t|_p, \quad R^c(t) = t[p \mapsto \square_p]_{p \in c}. \quad (22)$$

Order the finite set c lexicographically by its slot addresses. The same address therefore indexes a detached subtree and its puncture; no separate matching map is required. The indexed tuple $\mathbf{P}^c(t)$ is retained before its commutative forest image, even when selected subtrees are isomorphic.

Definition 2.1 (Address-indexed cut witness). The occurrence-level witness determined by c is

$$\mathbb{W}_c(t) = (c, \mathbf{P}^c(t), R^c(t)). \quad (23)$$

Its distinguished reconstruction composes $t|_p$ into $\square_p$ and recovers the original proof tree t . If t is complete, the tuple of detached subtrees is one point of $\text{Fill}(R^c(t))$. If t is punctured, the detached terms may themselves be punctured contexts; they are composable with the newly created punctures but need not belong to the complete filling fibre.

One selected premise edge creates one puncture, irrespective of how many components occur in its premise boundary. That boundary is recovered as $\text{req}_{R^c(t)}(p)$ from the retained parent rule. For a component-local rule, the same parent records the component at which the puncture is logically active.

Figure 1 shows the binary case. The prefix-free condition is the part needed beyond that picture: it makes the selected subtrees disjoint and preserves a literal address-by-address correspondence between the cut forest and the new punctures.

Coassociativity compares cuts performed at successive depths. To state the corresponding witness-level coherence, write $u < v$ when the occurrence u lies strictly inside the premise subtree rooted at v . Two disjoint CK cuts c, d are *noncrossing* when comparable cross-pairs have one consistent orientation:

$$\neg(\exists c_1, c_2 \in c, d_1, d_2 \in d : c_1 < d_1 \text{ and } d_2 < c_2). \quad (24)$$

Thus disjoint branches cause no conflict, while opposite nesting orders on two branches are excluded. Equivalently, after branches met by only one cut are ignored, one cut is everywhere weakly inside the other.

In this two-stage comparison, the second CK cut is transported after the first to the relevant detached or retained factors of its witness. Thus “either order” below compares two decompositions of one fixed tree, not two proof conversions.

Theorem 2.2 (Puncture factorisation and nested coherence). *For every derivation tree t and CK cut c :*

(i) *the distinguished reconstruction selected by $\mathbb{W}_c(t)$ reconstructs the original tree:*

$$R^c(t)[p := t|_p]_{p \in c} \cong t; \quad (25)$$

- (ii) *replacing the new punctures by proofs or punctured contexts admitted by the fixed calculus and having the inferred requirements gives a derivation of the same calculus with root $\text{bd}(t)$; if t and all replacements are complete, the result is complete;*
- (iii) *if two noncrossing cuts are performed in either order, the three vertex regions determined by the two cut levels, the resulting punctured root derivation, and their relative slot addresses are canonically the same;*
- (iv) *consequently, a fixed noncrossing hierarchy of address-indexed cuts is independent of parenthesisation, and staged context composition reconstructs t .*

Proof. For (i), the subtree at p is put back at the same slot address. Part (ii) follows because $\text{req}_{R^c(t)}(p) = \text{bd}(t|_p)$, as inferred from the retained parent rule; context composition preserves any old punctures.

For (iii), describe a two-stage witness by its two noncrossing cuts in the original tree. The three vertex regions are then independent of the order in which they are exposed, and address prefixing composes associatively. This proves (iii); induction using Proposition 1.5 proves (iv). $\square$

Corollary 2.3 (Strict interchange of independent refinements). *Let R be a proof context, let $p \neq q$ be two of its punctures, and let S, T be compatible proof contexts for those positions. Under the canonical transport of the old addresses,*

$$(R[p := S])[q := T] \cong (R[q := T])[p := S]. \quad (26)$$

The isomorphism fixes every old rule occurrence and prefixes the addresses internal to S, T by p, q . It therefore induces a bijection of all address-indexed CK cut witnesses and equality of the collected coproducts. More generally, finite refinement schedules which differ by adjacent exchanges at currently incomparable punctures produce the same proof context.

Proof. Substitution at distinct punctures acts on disjoint subterms, by Proposition 1.5. The resulting slot-indexed trees are therefore isomorphic before any CK cut is chosen, and the cut witnesses transport along that isomorphism. $\square$

This is a strict interchange law for independent proof-search obligations: they may be refined in parallel or in either schedule without changing the proof presentation or its CK decomposition. The noncrossing law in Theorem 2.2 likewise concerns repeated decomposition of one fixed proof. Neither statement by itself proves a commuting conversion between different proof trees; such a conversion may reverse rule ancestry and must instead pass the proof-identity and relative-residue tests below. The mixed derivative (19) classifies simultaneous antichain punctures; nested observations are represented by staged context composition.

Corollary 2.4 (Marked-subproof factorisation). *Let $\mathbf{H} = (x_1 : H_1, \dots, x_n : H_n)$, and let $\text{Mark}_{\mathbf{H}}(G)$ consist of a complete G -proof with labelled subproof positions $p_1, \dots, p_n$ forming an ancestry antichain, such that $p_i \neq \epsilon$ and $\text{bd}(t|_{p_i}) = H_i$. Detachment and filling give a bijection*

$$\text{Mark}_{\mathbf{H}}(G) \cong \text{Ctx}^+(\mathbf{H}; G) \times \prod_{i=1}^n \text{Prf}_{\mathcal{B}, \mathcal{R}}(H_i). \quad (27)$$

Here Ctx^+ denotes the contexts having at least one rule vertex. For $n = 1$ this is the nonroot sector of the zipper bijection (16).

Proof. Remove the marked subproofs and leave x_i at position p_i . Conversely, fill the displayed positions and mark the roots of the inserted copies. These operations are inverse before commutative forest collection. $\square$

2.2 Linearisation and the CK recursion

Linearisation changes the bookkeeping, not the proof objects: each forest becomes a basis symbol, and finite formal sums record alternative cuts and their multiplicities. Fix a commutative coefficient ring k with $1 \neq 0$, and put

$$H_{\text{CK}} = k[\mathbf{F}].$$

Because $\mathbf{T}$ was defined using the concrete occurrences admitted by $(\mathcal{B}, \mathcal{R})$, this is the calculus-relative algebra $H_{\mathcal{B}, \mathcal{R}}$; we write H_{CK} when emphasising its CK structure and $H_{\mathcal{B}, \mathcal{R}}$ when the dependence on $(\mathcal{B}, \mathcal{R})$ matters. This is the free k -module with basis symbols $[F]$, one for each forest, with multiplication induced by forest union. The coproduct below lists every admissible cut; each tensor records the detached and retained entries of one pair. Its counit is the bookkeeping map which selects the empty forest coefficient. Let $\mathcal{A}(t)$ be the prefix-free sets of nonroot slot addresses, including the empty set. For the empty cut set $P^\emptyset(t) = \mathbf{1}$ and $R^\emptyset(t) = t$. Adjoin one endpoint choice with factors $t, \mathbf{1}$.

A running factorisation. The general formula is easiest to read first on a three-rule proof. Let $b_S : S$ and $b_T : T$ be distinct nullary material grounds, let $m : (S, T) \rightarrow S \mid T$ be a binary assembly occurrence, and let $i : (S \mid T) \rightarrow U$ be unary. For $t = i(m(b_S, b_T))$, the nonempty cut choices select $m(b_S, b_T)$, either ground, or both grounds. Including the two algebraic endpoints gives

$$\begin{aligned} \Delta_{\text{CK}}[t] = & [t] \otimes \mathbf{1} + \mathbf{1} \otimes [t] \\ & + [m(b_S, b_T)] \otimes [i(\square_1)] \\ & + [b_S] \otimes [i(m(\square_{(1,1)}, b_T))] + [b_T] \otimes [i(m(b_S, \square_{(1,2)}))] \\ & + [b_S \sqcup b_T] \otimes [i(m(\square_{(1,1)}, \square_{(1,2)}))]. \end{aligned} \quad (28)$$

Each tensor says what was detached on the left and what remains, with typed holes, on the right. The middle right factors are one-puncture contexts; the last is a simultaneous two-puncture context. The $m(b_S, b_T)$ -term displays nested composition, while cut incidence recovers the premise ancestry $b_S, b_T < m < i$. Adding a new ground $b'_S : S$ makes it a possible filling of $i(m(\square_{(1,1)}, b_T))$ without changing this old cut. Finally, $m(b_S, b_T)$ has an assembly vertex, whereas $b_S \sqcup b_T$ merely juxtaposes two independent proof blocks.

Definition 2.5 (CK coproduct of proof trees). On a connected derivation tree,

$$\Delta_{\text{CK}}[t] = [t] \otimes \mathbf{1} + \sum_{c \in \mathcal{A}(t)} [P^c(t)] \otimes [R^c(t)], \quad \varepsilon_{\text{CK}}[F] = \begin{cases} 1, & F = \mathbf{1}, \\ 0, & F \neq \mathbf{1}. \end{cases} \quad (29)$$

Extend Δ_{CK} multiplicatively to forests and linearly to H_{CK} .

Let $\mathcal{A}^+(t) = \mathcal{A}(t) \setminus \{\emptyset\}$ be the proper CK cuts. Let D_t be the premise-ancestry poset of rule occurrences in t , ordered by the reflexive closure of strict subproof occurrence. The occurrence-level observation of t is the family $\{\mathbb{W}_c(t)\}_{c \in \mathcal{A}^+(t)}$, together with its iterated noncrossing cuts and filling maps. It retains the actual cut addresses and the aligned tuple of detached subtrees. The CK coproduct and its iterates are the collected linear images obtained after adjoining the two algebraic endpoints.

For a connected tree put

$$\overline{\Delta}_{\text{CK}}[t] = \Delta_{\text{CK}}[t] - [t] \otimes \mathbf{1} - \mathbf{1} \otimes [t] = \sum_{c \in \mathcal{A}^+(t)} [P^c(t)] \otimes [R^c(t)]. \quad (30)$$

This reduced coproduct is the collected proper-cut observation. The two removed tensors are necessary for the Hopf laws but are not nontrivial decompositions. A collected tensor does not by itself recover the address alignment, and equal tensors may collect with multiplicity even when their cut addresses differ. For later comparison, define the CK *cut-size polynomial*

$$\text{Cut}_t(q) = \sum_{c \in \mathcal{A}^+(t)} q^{|c|} \quad (31)$$

of a connected derivation tree t . It retains only the number of connected detached blocks in each proper cut, with multiplicity.

Cut witnesses are intrinsic to the proof presentation, whereas filling fibres depend on the fixed equipped calculus. A persistent material inclusion can therefore fix every old cut while adding alternative fillings. The finite witnesses are computed locally; the fibres are specified by their premise requirements and material base, not by enumerating every larger proof context.

Vertex number gives a connected grading: the empty forest is the sole degree-zero basis element. The standard CK theorem now says that the compatible forest product and cut coproduct then determine, recursively in vertex degree, a formal inverse called the antipode.

Corollary 2.6 (Hopf algebra of proof trees). *The operations above make $H_{\text{CK}} = H_{\mathcal{B}, \mathcal{R}}$ a connected graded Hopf algebra, graded by vertex number. Complete proofs alone are not closed under the coproduct, because a CK cut can create punctures in the retained factor.*

This is the standard typed and decorated CK construction [17, 25, 26]. Its restriction to the calculus-relative sector is closed because a CK cut partitions admitted concrete rule occurrences without changing any local instance. The closure is hereditary: cutting replaces occupied premise slots by vacancies without changing an occurrence or boundary, detached components are recursively admitted derivations, and pre-existing punctures remain punctures. This is why complete proofs alone are not the subcarrier, while complete and punctured admitted derivations are. The word “Hopf” adds a recursively determined antipode to the compatible product and coproduct. Connected grading makes that recursion terminate on each finite proof forest; the antipode is a formal convolution inverse, not an inverse proof or a normalisation step.

This establishes the calculus-relative instance of the ambient typed and decorated CK construction. The subsequent proof-theoretic arguments use its address-indexed cut coproduct, with premise requirements and slot positions retained at witness level.

Although the calculus-relative construction permits arbitrary finite proof-input arity, let $H_{\text{CK}}^{\leq 2}$ be spanned by forests whose concrete rule occurrences satisfy $n(a) \leq 2$. It is a Hopf subalgebra: forest union and CK cut preserve each occurrence and its fixed premise-slot arity, and connected grading restricts the antipode. This sector includes nullary grounds, unary rules, and binary rules. The bound concerns immediate proof-premise slots, not occupied graph valence, puncture number, hypersequent breadth, or active-component multiplicity [17, 26].

Lemma 2.7 (Final-corolla projection). *For a connected proof t , let $\text{Prem}(t)$ be the forest of its immediate proof children, let a_t be its final concrete rule occurrence, and let $\text{Cor}(a_t)$ replace those children by punctures in their ordered premise positions. If pr_1 fixes one-vertex connected trees and kills every other basis forest, then*

$$(\text{id} \otimes \text{pr}_1) \Delta_{\text{CK}}[t] = [\text{Prem}(t)] \otimes [\text{Cor}(a_t)]. \quad (32)$$

Schematically,

$$a_t(t_1, \dots, t_n) \longmapsto [t_1 \sqcup \dots \sqcup t_n] \otimes [a_t(\square_1, \dots, \square_n)].$$

The premise slots align each immediate proof with its puncture.

Proof. Exactly one cut leaves only the final vertex in the root factor: cut at the roots of all immediate proof children. For a nullary occurrence this is the empty cut. Every other retained factor has either zero or more than one vertex. $\square$

Extraction recovers the concrete final occurrence and its premise telescope $\text{Tel}(\text{Cor}(a_t))$. Its status as logical is determined independently by the rule family’s behaviour under inversion, use, and substitution.

2.3 Context composition and the typed CK Faà di Bruno identity

The proof-theoretic claim is that CK cutting reads typed context composition backwards: it exposes the inserted evidence and the outer context which uses it, while compatible composition rebuilds the original derivation. The family $\mathcal{O}$ organises these existing contexts; it does not replace the CK carrier. At witness level, every compatible decomposition

$$C = D\langle C_1, \dots, C_m \rangle \rightsquigarrow [C_1 \sqcup \dots \sqcup C_m] \otimes [D] \quad (33)$$

is obtained by cutting C at the roots of the positive-vertex inserted contexts, with identity contexts omitted from the forest factor. Conversely, every CK cut gives such an outer context and typed family of inner contexts. To sum this already established correspondence, use the coefficientwise forest-series completion

$$\widehat{H_{\mathcal{B}, \mathcal{R}}} = \prod_{F \in \mathcal{F}} k[F] \cong k^{\mathcal{F}},$$

Forest multiplication and the CK coproduct extend to the series below coefficientwise: a fixed finite forest has only finitely many subforest decompositions, and a fixed tensor has only finitely many assignments of its detached factors to the punctures of its retained context.

For a requirement word $\mathbf{K}$ and root boundary G , inventory all contexts of that interface in the formal Green series

$$\mathfrak{G}_{\mathbf{K}}^G = \delta_{\mathbf{K}, (G)} \mathbf{1} + \sum_{C \in \mathcal{O}^+(\mathbf{K}; G)} [C]. \quad (34)$$

The unit summand represents the externally typed nodeless identity $\square^G$; it is present only when $\mathbf{K} = (G)$ and is not a new CK generator.

“Faà di Bruno” names the substitution/decomposition law obtained by summing these presentations: every term chooses a retained outer context together with the compatible inner contexts inserted into its ordered punctures. The theorem below is the judgement-typed proof version of that law.

Theorem 2.8 (Typed CK-cut/context-composition bijection and Faà di Bruno identity). *Address-indexed CK cuts of a context $C \in \mathcal{O}^+(\mathbf{K}; G)$, with inherited punctures represented by typed identities, are in bijection with its compatible presentations*

$$C = D\langle C_1, \dots, C_m \rangle$$

as an outer typed context and an ordered family of inner contexts. Linearising this bijection gives, coefficientwise in the completed tensor product,

$$\Delta_{\text{CK}} \mathfrak{G}_{\mathbf{K}}^G = \sum_{m \geq 0} \sum_{\mathbf{H} = (H_1, \dots, H_m) \in \text{HSeq}^m} \sum_{\mathbf{K} = \mathbf{K}_1 \cdots \mathbf{K}_m} \left(\prod_{j=1}^m \mathfrak{G}_{\mathbf{K}_j}^{H_j} \right) \otimes \mathfrak{G}_{\mathbf{H}}^G. \quad (35)$$

The last sum is over decompositions into consecutive subwords in slot-address order; every $\mathbf{K}_j$ may be empty. For $m = 0$ it occurs only when $\mathbf{K} = ()$, and the empty product is $\mathbf{1}$. Here $\mathbf{H}$ is the premise-requirement word of the retained outer context, while $\mathbf{K}_j$ records the old punctures remaining inside its j -th inner context. The number m counts punctures of the outer context, not premises of one rule; it can be arbitrarily large even inside $H_{\text{CK}}^{\leq 2}$.

Proof. Let $C \in \mathcal{O}^+(\mathbf{K}; G)$ and let c be an ordinary CK cut, including the empty cut. Put $D = R^c(C)$, order its punctures $q_1 < \dots < q_m$, and set $H_j = \text{req}_D(q_j)$. If q_j was created by the cut, let $C_j = C|_{q_j}$; if it is an inherited puncture of C , let $C_j = \square^{H_j}$. The old punctures inside each C_j form one interval in slot-address order, so for $\mathbf{K}_j = \tau(C_j)$,

$$\mathbf{K} = \mathbf{K}_1 \cdots \mathbf{K}_m, \quad C = D\langle C_1, \dots, C_m \rangle, \quad P^c(C) = \bigsqcup_{|V(C_j)| > 0} C_j. \quad (36)$$

An empty $\mathbf{K}_j$ therefore records a detached complete proof with no old puncture; an inherited puncture contributes the singleton identity word (H_j) .

Conversely, from typed contexts

$$D \in \mathcal{O}(\mathbf{H}; G), \quad C_j \in \mathcal{O}(\mathbf{K}_j; H_j)$$

with positive-vertex D , form $C = D\langle C_1, \dots, C_m \rangle$ and cut at exactly the roots of the positive-vertex C_j . Those roots are nonroot and prefix-free. The two constructions are inverse at the ordered-witness level. The empty cut uses identity inner contexts, or $m = 0$ when C is complete; the whole-tree endpoint uses the identity outer context. The nodeless identity context has its single identity-by-identity decomposition.

Before forest collection the C_j remain ordered by the punctures of D . Passing to their commutative forest product may identify tensors, but the ordered decompositions then collect with exactly the CK cut multiplicities. Summing the bijection over all typed contexts proves (35). $\square$

Remark 2.9 (One-colour calibration). In a locally finite one-colour specialisation, write $\mathfrak{G}_n$ for the series of contexts with n punctures. Then (35) becomes

$$\Delta_{\text{CK}} \mathfrak{G}_n = \sum_{m \geq 0} \sum_{n_1 + \dots + n_m = n} \mathfrak{G}_{n_1} \cdots \mathfrak{G}_{n_m} \otimes \mathfrak{G}_m,$$

with the same nullary and identity conventions as in Theorem 2.8. After simultaneously forgetting judgement colours and rule decorations to the terminal one-colour operad, this is the usual Faà di Bruno Green-function coproduct. Thus the CK-cut/context-composition bijection used above is the standard Faà di Bruno mechanism for context composition [28, 33]; its application here consists in retaining premise-hypersequent colours, indexed proof inputs, concrete rule occurrences, and their hypersequent interpretation.

The local-finiteness qualification concerns numerical collection after colours and decorations have been forgotten. The typed identity itself is coefficientwise and needs no finite fibre for a global forgetful map: each concrete proof context retains its own rule decorations.

The terminal calibration is deliberately lossy. For fixed compatible inputs p, r , the proof trees $\text{Com}(p, r)$ and $\text{Asm}_1(p, r)$ have the same undecorated binary-root shape and hence the same cut-size polynomial. Proposition 5.1 nevertheless separates their immediate-input component traces as $\mathbf{K}_{2,2}$ and a bijection. The scalar one-colour formula calibrates decomposition multiplicities; judgement colours, concrete occurrences, and component transport carry the specifically hypersequent information.

The identity says that CK decomposition and typed context composition describe the same proof structure in opposite directions. A cut exposes the independently supplied evidence in the left tensor factor and the punctured rule context which uses it in the right; compatible composition reconstructs the original proof. Linearisation collects these address-indexed decompositions with their CK multiplicities. This is the proof-theoretic reading of the judgement-typed Faà di Bruno law inside the same CK Hopf algebra [27, 28, 33].

This also locates the later resource polynomial. The present Green series keeps its punctures as distinct ordered occurrences; repeated uses of one formal input do not yet exist. Section 4.3 adds a non-bijective alias map from those punctures to formal inputs. Its use monomial is the cardinal shadow of the mixed derivatives above, and the CK probe evaluation is its binomial translate. Thus context derivatives, Green-series substitution, and resource use are three levels of one occurrence calculus, but the resource polynomial is not literally a derivative of the unaliased Green series.

The colours are inferred premise hypersequents. Logical and structural rules, their instantiated formula data, and their component action remain in the vertex decorations; context composition creates no structural rule, and forest multiplication records only the independently detached proof blocks. Component relations compose towards the root as shown in Section 3.2.

Shared formula and context parameters are fixed by each concrete rule occurrence. Iterated composition records premise ancestry, but no later requirement depends on the particular proof chosen for an earlier puncture. Thus the telescope is nondependent for the present fully instantiated occurrence carrier. Harmony and proof identity remain independently justified; the later CK tests ask which of these typed decompositions they preserve.

The Faà di Bruno identity sums every compatible family of insertions at once. Its one-slot part is simpler: choose one required premise of an outer proof context and insert an inner context of the matching boundary. Summing that operation over all compatible punctures gives the following infinitesimal form of context composition.

Proposition 2.10 (One-slot insertion and singleton CK cuts). *Let*

$$\mathfrak{c}^+ = \bigoplus_{G \in \text{HSeq}} \bigoplus_{\mathbf{K} \in \text{HSeq}^*} k[\mathcal{O}^+(\mathbf{K}; G)]$$

be the linear span of positive-vertex proof contexts. For $S \in \mathcal{O}^+(\mathbf{K}; H)$ and $T \in \mathcal{O}^+((L_1, \dots, L_m); G)$, define, with ill-typed summands omitted,

$$[S] \triangleright [T] = \sum_{\substack{1 \leq i \leq m \\ L_i = H}} [T \circ_i S], \quad T \circ_i S = T(\square^{L_1}, \dots, \square^{L_{i-1}}, S, \square^{L_{i+1}}, \dots, \square^{L_m}). \quad (37)$$

The standard total-insertion theorem makes $\triangleright$ pre-Lie [13, 14]: its failure of associativity is symmetric in the two inserted contexts. We use that standard classification only as a name for the operation; the proof-theoretic content needed here is its exact pairing with one-address cutting. For a positive connected context U , put

$$\Delta_{\text{CK}}^{(1)}[U] = \sum_{p \in \text{Pos}(U) \setminus \{\epsilon\}} [U|_p] \otimes [R^{\{p\}}(U)], \quad (38)$$

the singleton-cut part of its reduced coproduct. For positive connected contexts S, T, U ,

$$\text{coeff}_{[U]}([S] \triangleright [T]) = \text{coeff}_{[S] \otimes [T]}(\Delta_{\text{CK}}^{(1)}[U]) = \text{coeff}_{[S] \otimes [T]}(\Delta_{\text{CK}}[U]). \quad (39)$$

Proof. An occurrence of U in $S \triangleright T$ selects a puncture of T at which S was inserted. Its address p satisfies $U|_p \cong S$ and $R^{\{p\}}(U) \cong T$. Conversely, filling the puncture of any singleton cut with its detached context reconstructs U by Theorem 2.2. Summing over p gives the first equality with the same address multiplicities. Since both tensor factors are positive and connected, no endpoint or multi-root cut can contribute $[S] \otimes [T]$, proving the second. $\square$

Equation (39) is the paper-specific content: prospective insertion into one typed proof obligation is the coefficient-transpose of retrospective cutting at one address. It is the $m = 1$ sector of Theorem 2.8, whereas Lemma 2.7 cuts all immediate children to retain only the final corolla. For a unary occurrence $e : G \rightarrow H$,

$$[C] \triangleright [\text{Cor}(e)] = [e(C)] = [B_e^+(C)], \quad (40)$$

so unary rule-rooting is right multiplication by its corolla in this inner-first convention; higher-arity rooting remains simultaneous typed context composition.

2.4 Iterated cuts and premise ancestry

For a proof forest F , let D_F be the finite premise-ancestry poset of its rule occurrences: $u \leq v$ when $u = v$ or u occurs in a premise subtree of v . Write $[n] = \{1 < \dots < n\}$. An n -stage cut is an address-resolved term of the $(n - 1)$ -fold iterated coproduct, hence an ordered list of n ancestry layers. Endpoint choices are allowed at every stage, so a layer may have degree zero. Assigning $f(v) = j$ below means that the occurrence v first appears in layer j . If empty layers were forbidden, the same construction would classify only surjective maps to $[n]$.

Proposition 2.11 (Iterated cuts record presentation ancestry). *For $n \geq 1$, before equal CK tensors are collected, address-indexed n -stage cuts of F are in bijection with weakly order-preserving maps*

$$D_F \longrightarrow [n].$$

They are equivalently flags of ancestry ideals: an ancestry ideal is a downward-closed subset, and a map f gives

$$\emptyset = I_0 \subseteq I_1 \subseteq \dots \subseteq I_n = D_F, \quad I_j = f^{-1}(\{1, \dots, j\}). \quad (41)$$

The flag says which rule occurrences have been exposed by each stage; a descendant cannot be exposed later than the rule which uses it. If one rule occurrence is exposed at each positive-degree stage, the one-vertex layers and their slot addresses reconstruct the raw proof presentation up to the isomorphisms of Definition 1.3. Exchanging adjacent incomparable stages does not change the reassembled proof.

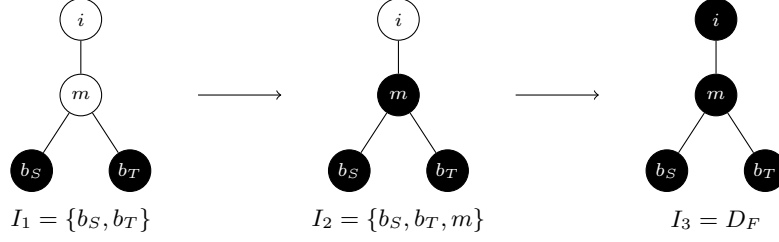


Figure 2: A flag of ancestry ideals for the running tree $i(m(b_S, b_T))$. Black vertices have been exposed by that stage. The leaves b_S, b_T may exchange order, but the rule m cannot precede its premises and i cannot precede m .

Proof. In the ideal form of the coproduct, an address-resolved iterated term has successive tensor layers $I_1, I_2 \setminus I_1, \dots, D_F \setminus I_{n-1}$. Conversely, successive frontier cuts of any flag produce a unique address-resolved term; coassociativity identifies its parenthesisations. Assigning each rule occurrence the first ideal in which it appears then gives the weakly order-preserving map, and initial preimages give the inverse flag. Refining the flag until one new vertex is added at each stage gives a linear extension of D_F , and Theorem 2.2(iv) reassembles its one-vertex layers independently of exchanges of incomparable stages. $\square$

This establishes presentation reconstruction and premise-ancestry recovery. The reconstruction claim concerns the uncollected, address-indexed witnesses: ordinary iterated coproduct endpoints contain the source forest itself, while collected tensors can identify distinct addresses. Theorem 2.8 records every one-stage typed context decomposition, including closed side proofs; iterated cuts additionally retain where those stages occur in a chosen proof tree. This allows the construction to be compared precisely with causal trace semantics.

2.5 Refinement traces and causal incidence

Fix an equipped calculus $C = (\mathcal{B}, \mathcal{R})$. A *backward-refinement history* of a finite context R is a sequence

$$\square^G = R_0 \longrightarrow R_1 \longrightarrow \dots \longrightarrow R_N = R \quad (42)$$

in which one current puncture is replaced at each step by the rule corolla occupying that address in R . The punctures of R itself are left unrefined. Thus every rule occurrence of R is introduced exactly once; the nodeless identity has the unique empty history. Let $\equiv_{\parallel}$ be generated by exchanging two adjacent steps which refine distinct punctures already present before either step. These are exactly the exchanges covered by Corollary 2.3. Write

$$\text{Hist}_C(\mathbf{H}; G) = \coprod_{R \in \text{Ctx}_C(\mathbf{H}; G)} \text{Hist}_C(R),$$

with histories taken up to the same coherent context isomorphisms as their stages.

To state the coalgebraic comparison, let $H_{\text{ForPos}}^{\mathbb{Z}}$ be the rooted-forest sub-Hopf algebra of the standard Hopf algebra of finite posets [2]. Its product is disjoint union and its coproduct splits a poset into a downward-closed ideal and the complementary upset:

$$\Delta_{\text{inc}}[P] = \sum_{I \in J(P)} [I] \otimes [P \setminus I], \quad (43)$$

where $J(P)$ is the set of downward-closed ideals and both factors carry the induced order. Empty factors give the two endpoints. Write $H_{\mathcal{B}, \mathcal{R}}^{\mathbb{Z}}$ for the integral form of the proof-tree Hopf algebra.

For the orientation used here, a downward ideal of a rooted-forest ancestry poset is a disjoint union of rooted descendant subtrees, while its complementary upset is the remaining root-containing pruned forest. Therefore these posets are closed under both factors of Δ_{inc} , which is why they form the stated sub-Hopf algebra rather than merely a subspace of the poset Hopf algebra.

Theorem 2.12 (Causal-trace representation and CK incidence semantics). *Let R be a connected finite proof context and F a finite proof forest.*

- (i) Histories of R are reverse linear extensions of D_R . The final-context map therefore induces a bijection

$$\text{Hist}_C(\mathbf{H}; G) / \equiv_{\parallel} \cong \text{Ctx}_C(\mathbf{H}; G), \quad (44)$$

where the left side ranges over histories ending in contexts of the displayed interface.

- (ii) If $|V(R)| > 0$, augment the CK cut choices of R by the whole-tree endpoint. For $c \in \mathcal{A}(R)$, send c to the vertex region detached by it,

$$I_c = \bigcup_{p \in c} V(R|_p), \quad (45)$$

sends the empty cut to $\emptyset$, sends the whole-tree endpoint to D_R , and gives a bijection with $J(D_R)$. For a nonempty proper ideal, the inverse cut is its antichain of maximal elements.

- (iii) Forgetting boundaries, rule tags, component data, and punctures but retaining premise ancestry defines a graded Hopf map

$$\text{Anc} : H_{\mathbf{B}, \mathbf{R}}^{\mathbb{Z}} \longrightarrow H_{\text{ForPos}}^{\mathbb{Z}}, \quad [F] \longmapsto [D_F]. \quad (46)$$

Thus the proof-tree CK coproduct is a typed, proof-relevant lift of the incidence coproduct of causal ideals.

- (iv) Let $\zeta([F]) = 1$ for every basis forest, and let

$$\omega_1([F]) = \begin{cases} 1, & |V(F)| = 1, \\ 0, & |V(F)| \neq 1. \end{cases} \quad (47)$$

Then, for $n \geq 1$,

$$\zeta^{*n}([F]) = \#\{f : D_F \rightarrow [n] \mid f \text{ is weakly order-preserving}\}. \quad (48)$$

If $N = |V(F)| > 0$, then

$$\omega_1^{*N}([F]) = |\text{LinExt}(D_F)|. \quad (49)$$

For connected positive-degree R , reversal gives a bijection between $\text{LinExt}(D_R)$ and its root-first refinement histories, and

$$1 + \text{Cut}_R(q) = \sum_{A \in \text{Ant}(D_R \setminus \{\rho_R\})} q^{|A|}, \quad \max_{c \in \mathcal{A}(R)} |c| = \text{width}(D_R \setminus \{\rho_R\}). \quad (50)$$

where ρ_R denotes the unique root occurrence of R . For a one-vertex context both sides of the width equation are 0.

Proof. Label a refinement step by the address of the vertex it creates in the final context. A parent rule must already be present before a rule can be inserted into one of its premise positions, so the resulting time order is a reverse linear extension of D_R . Conversely, every reverse linear extension gives a legal refinement sequence. Any two linear extensions of a finite poset are connected by adjacent exchanges of incomparable elements; strict refinement interchange identifies exactly those exchanges. This proves (i).

For $c \in \mathcal{A}(R)$ with R of positive degree, the union in (45) is downward closed. Every proper ideal is the downward closure of its maximal elements, which form a nonroot antichain; the empty and total ideals give the two exceptional choices. This proves (ii).

For a connected generator t , part (ii) identifies every coproduct term, including endpoints, with a unique ideal $I \in J(D_t)$. Writing c for the associated cut choice, including the endpoint conventions,

$$D_{P^c(t)} = I, \quad D_{R^c(t)} = D_t \setminus I.$$

Thus Anc preserves the coproduct on connected generators. Forest union becomes disjoint union of posets, so multiplicativity extends the equality to every forest. Connected gradedness then gives the antipode compatibility and proves (iii).

The functional ζ is a character and ω_1 is an infinitesimal character. Expanding an n -fold convolution expands the iterated CK coproduct. Proposition 2.11 identifies its address-resolved terms with weakly order-preserving maps to $[n]$, and ζ assigns 1 to every layer. Requiring all N layers to have degree one instead selects linear extensions and proves (49). Finally, proper cuts are precisely the nonempty antichains of nonroot vertices. Every antichain occurs, so the largest cut has the width stated in (50). $\square$

The integer in (48) is the order polynomial of the presentation-ancestry poset evaluated at n . After base change to a coefficient ring of positive characteristic it is observed modulo that characteristic; address-indexed witnesses retain the individual schedules before collection. No rule name contributes to (46): decorations supply the typed proof-theoretic lift, while the causal order is carried by CK cut support and convolution multiplicity.

The history/linear-extension correspondence, the cut/ideal bijection, and the resulting order-polynomial counts are standard trace and poset-Hopf mechanisms. Their role here is not a new concurrency theorem. The paper-specific content is their lift to typed proof contexts, where each frontier retains a detached proof forest, a punctured filling interface, material grounds, and hypersequent component data.

Corollary 2.13 (Antipode reciprocity for causal stages). *Work over a characteristic-zero coefficient ring. Let $\text{ord}_P(x)$ be the order polynomial occurring in (48), and let $\text{ord}_P^\circ(n)$ count strictly order-preserving maps $P \rightarrow [n]$. For $N = |V(F)|$ and $n \geq 1$,*

$$((\zeta \circ S)^{*n})([F]) = \text{ord}_{D_F}(-n) = (-1)^N \text{ord}_{D_F}^\circ(n). \quad (51)$$

Thus the convolution inverse supplied by the antipode performs the standard incidence-Hopf inclusion-exclusion which replaces weak causal layerings by signed strict layerings. It is not a proof reduction or an inverse proof.

Proof. Put $\eta = \zeta - \varepsilon_{\text{CK}}$. This functional vanishes in degree zero and is locally nilpotent for convolution on each fixed finite-degree forest. Consequently the binomial identity

$$\zeta^{*m}([F]) = \sum_{k \geq 0} \binom{m}{k} \eta^{*k}([F])$$

is degreewise finite. Work first in the integral form. For positive integers m the left side is $\text{ord}_{D_F}(m)$ by Theorem 2.12; the finite binomial expression and the order polynomial are therefore the same polynomial over $\mathbb{Q}$. Evaluating that integer-valued identity at negative integers and then extending scalars to the chosen coefficient ring gives the claimed identity. Since $\zeta \circ S = \zeta^{*-1}$, taking $m = -n$ gives the first displayed equality; compare the Hopf-algebraic treatment of poset chain enumerators in [20]. Stanley's order-polynomial reciprocity [42] gives the second equality. $\square$

The trace principle itself is established concurrency mathematics. Event structures represent computations by event occurrences, causal prerequisites, and conflict [36]; for a fixed completed context, D_R is only the conflict-free event-poset part of that picture, not a global event-structure semantics of proof search. Benson's shuffle/cut bialgebras algebraise interleavings and word-like traces [10]; their operations differ from the forest juxtaposition and ancestry cutting used here. More closely, Behr and Kock identify reorderings of sequentially independent rewrites by shift equivalence and construct a cocommutative Hopf algebra of causal tracelets [9]; their coproduct splits a tracelet normal form into subsets of primitive connected components. The generally noncocommutative coproduct here instead cuts within one proof-dependency tree and returns a detached typed proof forest together with the punctured context which uses it. Castellan and Yoshida construct causal invariants for unit-free MALL whose equality characterises commuting conversions [15]. By contrast, D_R records explicitly *presentation causality*: the MLL obstruction below shows that raw CK may retain ancestry forgotten by an accepted proof permutation. Hopf algebra has also entered denotational proof semantics through shuffle actions used for noncommutative full completeness [11]; that construction acts on denotations, whereas the present CK basis consists of proof presentations and its coproduct decomposes them.

This completes the central sequence from cut through the CK coproduct to typed context composition and causal incidence. We next apply it to changes of material base before turning to proof identity.

2.6 Persistent and revisory material change

Material change affects the prospective side of the construction. A persistent extension leaves every old proof and its cuts untouched but may supply new evidence for an old puncture. Withdrawing a previously admitted ground is different: affected proof presentations must be removed by a quotient, after which their surviving contexts can expose replacement obligations.

Proposition 2.14 (Persistent material inclusion and filling). *Let $\mathcal{B} \subseteq \mathcal{B}'$ be a persistent material inclusion, preserving every old concrete occurrence. Reinterpretation of an existing $(\mathcal{B}, \mathcal{R})$ -proof as a $(\mathcal{B}', \mathcal{R})$ -proof commutes with filling: for $\mathbf{t} \in \text{Fill}^{\mathcal{B}, \mathcal{R}}(R)$,*

$$j_*(\text{ev}_{\mathcal{R}}(\mathbf{t})) \cong \text{ev}_{j_*\mathcal{R}}(j_*\mathbf{t}). \quad (52)$$

Thus enlarging the material base may add positive-arity material rules as well as nullary grounds, and hence new proofs, contexts, and possible fillers, while preserving every puncture position and surrounding rule context of an old proof. Every filling operation anchored over the old base is natural over its persistent extensions.

Proof. Every old occurrence remains in the enlarged calculus. Both sides substitute the same proofs into the same premise slots. $\square$

Definition 2.15 (Persistent calculus inclusion and conservativity). With the boundary language and colours fixed, a *persistent calculus inclusion*

$$\iota : (\mathcal{B}, \mathcal{R}) \hookrightarrow (\mathcal{B}', \mathcal{R}')$$

consists of a persistent material inclusion together with instance-preserving injections of the old logical initials and concrete rule occurrences into the new ones. These injections preserve constructor tags, ordered premise positions, boundaries, component focus, and derived component relations, and hence induce a recursive reinterpretation map $\iota_* : \text{Prf}_{\mathcal{B}, \mathcal{R}}(G) \rightarrow \text{Prf}_{\mathcal{B}', \mathcal{R}'}(G)$.

Given a class $\mathcal{C}$ of old boundaries, the inclusion is *$\mathcal{C}$ -consequence-conservative* when

$$\text{Prf}_{\mathcal{B}', \mathcal{R}'}(G) \neq \emptyset \implies \text{Prf}_{\mathcal{B}, \mathcal{R}}(G) \neq \emptyset \quad (G \in \mathcal{C}). \quad (53)$$

For a selected family D of constructor tags common to the two calculi, write

$$\text{Prf}_{\mathcal{B}, \mathcal{R}}^{-D}(G) = \{t \in \text{Prf}_{\mathcal{B}, \mathcal{R}}(G) : t \text{ contains no occurrence tagged } D\}. \quad (54)$$

The inclusion is *D -free consequence-conservative* when the same inhabited-fibre condition holds for the D -free subsets. For the family of formula-mediated Gentzen-Cut instances we say *Gentzen-Cut-free*.

By (10), consequence conservativity preserves inhabitedness of old one-puncture filling fibres while allowing new proofs in them. Premise nonmonotonicity instead concerns $\Gamma \mapsto \Gamma, \Upsilon$ inside one support relation. A persistent inclusion extends the stock of admitted evidence; a revisory change which withdraws an old material occurrence is treated below by a quotient. Thus the support relation may be nonmonotone in its formula premises while the causal-incidence and resource-use results remain valid over every fixed equipped calculus. If adding information defeats an old ground, or if a negative rule condition ceases to hold after extension, the change is not persistent and no forward Hopf map is asserted. The fixed-calculus theorems remain intact; only naturality across that revisory change is unavailable.

The following theorem packages one repeated observation. Calling the assignment a *functor* says only that reinterpreting an old proof along successive persistent inclusions agrees with reinterpreting it once along their composite. The substantive claim is that this reinterpretation also preserves every CK cut and filling operation.

Theorem 2.16 (Persistent calculus inclusions and factorisation naturality). *Fix the boundary language $\mathcal{L}$, and let $\text{Calc}_{\mathcal{L}}$ be the category whose objects are equipped calculi $(\mathcal{B}, \mathcal{R})$ and whose arrows are persistent calculus inclusions from Definition 2.15. The assignment*

$$(\mathcal{B}, \mathcal{R}) \mapsto H_{\mathcal{B}, \mathcal{R}}, \quad (\iota : (\mathcal{B}, \mathcal{R}) \hookrightarrow (\mathcal{B}', \mathcal{R}')) \mapsto (\iota_* : [F] \mapsto [\iota_* F]) \quad (55)$$

is a functor to connected graded Hopf k -algebras and injective graded Hopf maps. At witness level, for every old proof t and CK cut c ,

$$P^c(\iota_* t) = \iota_* P^c(t), \quad R^c(\iota_* t) = \iota_* R^c(t). \quad (56)$$

Thus ι_ identifies every old address-indexed and iterated cut, while embedding its old evaluation family into the possibly larger filling family in the extended calculus. The maps also commute with Root, BRoot and typed filling. Recursive reinterpretation induces compatible injections $\text{OpenPrf}_{\mathcal{B}, \mathcal{R}}(X) \hookrightarrow \text{OpenPrf}_{\mathcal{B}', \mathcal{R}'}(X)$ and hence inclusions on labelled punctured-context coefficients.*

For fixed $\mathcal{R}$, the subcategory in which only $\mathcal{B}$ varies is the material-extension category $\text{Mat}_{\mathcal{L}, \mathcal{R}}$. The preceding functor restricts to the material-base functor used below.

Proof. Injectivity of ι on the full tagged constructor signature makes ι_* injective on rule-rooted trees and forests. An old proof has exactly the same vertex order after reinterpretation. Every prefix-free cut set therefore gives the same detached and retained factors at the same addresses, proving (56) and

$$\Delta_{\text{CK}\iota_*} = (\iota_* \otimes \iota_*)\Delta_{\text{CK}}, \quad \varepsilon_{\text{CK}\iota_*} = \varepsilon_{\text{CK}}.$$

Forest union and vertex degree are unchanged, so ι_* is an injective graded bialgebra map; connected gradedness makes it commute with the antipodes. Identity and composition of persistent inclusions act identically on every old rule occurrence, proving functoriality. The root assertions are immediate. The proof of Proposition 2.14 applies verbatim: both sides insert the same reinterpreted terms into the same preserved premise slots. Recursive reinterpretation injects every old punctured proof and every labelled context into the larger grammar; marking literal puncture occurrences commutes with that inclusion, which gives the coefficient inclusions. $\square$

Logically, a persistent extension leaves the decomposition of every old hypersequent proof unchanged. It may only supply further proofs of premise requirements already determined by the retained rule contexts.

Corollary 2.17 (Base-extension naturality of proofs and filling). *For each boundary G , the assignment*

$$P_G(\mathcal{B}) = \text{Prf}_{\mathcal{B}, \mathcal{R}}(G), \quad P_G(j) = j_*,$$

defines a covariant functor $P_G : \text{Mat}_{\mathcal{L}, \mathcal{R}} \rightarrow \text{Set}$. Fix an anchor base $\mathcal{B}_$, let $\text{Ext}(\mathcal{B}_*)$ be its category of persistent extensions, and let R_* be an n -punctured derivation admitted over $\mathcal{B}_*$, with root boundary G and labelled premise telescope $(x_i : G_i)_{1 \leq i \leq n}$. For each persistent extension $\mathcal{B}$ of $\mathcal{B}_*$, write $R_{\mathcal{B}}$ for its reinterpretation. Filling defines a natural operation on this extension category,*

$$\text{ev}_{R_*} = (\text{ev}_{R_{\mathcal{B}}})_{\mathcal{B} \in \text{Ext}(\mathcal{B}_*)} : \prod_i P_{G_i} |_{\text{Ext}(\mathcal{B}_*)} \Longrightarrow P_G |_{\text{Ext}(\mathcal{B}_*)}. \quad (57)$$

Explicitly, for every persistent inclusion $j : \mathcal{B} \hookrightarrow \mathcal{B}'$,

$$j_*(\text{ev}_{R_{\mathcal{B}}}^{\mathcal{B}}(\mathbf{t})) \cong \text{ev}_{R_{\mathcal{B}'}}^{\mathcal{B}'}(j_*\mathbf{t}). \quad (58)$$

Every address-indexed cut and the collected coproduct of every old evaluation are transported by the same reinterpretation.

Proof. The functor laws are those of recursive reinterpretation. Equation (58) is Proposition 2.14 applied simultaneously to the displayed punctures. Theorem 2.16 gives the factorisation assertions. $\square$

Every old filling embeds into the corresponding filling set over a persistent extension; additional fillers may become available while the anchored operation and all its old factorisations are preserved. This naturality maps proofs forward along material-base extensions in the same direction as the uniformity clauses of base-extension semantics, but its values are proof presentations rather than support judgements. The discrete presentation in (5) resembles a sequent-based atomic system only at boundary level: Γ is stored inside the nullary boundary $\Gamma \vdash \Delta$, not exposed as a list of proof premises. Positive arity can represent primitive atomic rules with proof premises, but neither presentation by itself defines semantic support. This distinction is especially relevant to recent sequent-calculus formulations of base-extension semantics [40, 8, 30, 31].

A genuine adequacy result would additionally have to specify

- (a) the class of semantic atomic bases and their admissible extensions;
- (b) the atomic derivability relation generated at each base;
- (c) support clauses for the chosen logical vocabulary, including any resource splitting and hypersequent bars;
- (d) soundness and completeness against inhabited proof fibres; and
- (e) when an operation natural over all base extensions is represented by one anchored proof context.

The last point is not formal: extension naturality alone may give a family of support implications without selecting a single uniform context. If an independently defined semantics proves $\mathcal{B} \Vdash G \iff \text{Prf}_{\mathcal{B}, \mathcal{R}}(G) \neq \emptyset$ with the required resource indices, then an anchored context supplies the forward extension-uniform implication from support of its premises to support of its conclusion. The missing converse would be a representation theorem: every suitably natural support operation is induced by one anchored context. Proving items (a)–(d) and that representation theorem in item (e) is additional proof-theoretic work, not

a consequence of CK naturality. The fixed-calculus results also permit a support relation nonmonotone in formula premises; a semantic change which defeats an old ground is revisory rather than an arrow of the persistent extension category.

At the one-hole level, context composition and filling already have the familiar categorical shape. Fix $C = (\mathcal{B}, \mathcal{R})$, and write $\text{Prf}_C, \text{Ctx}_C$ for its proof and context families. The one-hole sector forms the category $\text{Use}_C(H, G) = \text{Ctx}_C(H; G)$, with puncture insertion as composition and $\square^G$ as identity, and it acts covariantly on proof fibres by filling. Persistent calculus inclusions induce faithful identity-on-objects functors between these categories and commute with their actions. This is the unary part of context composition, not a category taking arbitrary derivations as arrows and formula-mediated Gentzen Cut as composition.

The next result is an extensionality criterion for raw proof-context presentation, not a proposed identity of proofs: distinctly tagged generic fillers separate contexts.

Theorem 2.18 (Generic-filler separation of proof contexts). *Let $R, R' \in \text{Ctx}_C(\mathbf{H}; G)$ be finite proof contexts with the same root and a chosen alignment $\mathbf{H} = (x_1 : H_1, \dots, x_n : H_n)$ of all their punctures. The following are equivalent:*

- (i) $R \cong R'$ by an isomorphism inducing the chosen puncture alignment;
- (ii) for every persistent calculus extension $j : C \hookrightarrow C'$ and every compatible tuple $\mathbf{t} = (t_i)$ of complete C' -proofs,

$$(j_* R)[t_i/x_i]_i \cong (j_* R')[t_i/x_i]_i; \quad (59)$$

- (iii) the equality in (59) holds in the one persistent formal rule-signature extension C^π obtained by adjoining pairwise distinct fresh nullary probe occurrences

$$\pi_i : () \longrightarrow H_i$$

and taking $t_i = \pi_i()$.

The probe tags are indexed by puncture labels, so equal requirements $H_i = H_j$ remain distinguishable.

Proof. Part (i) is preserved by reinterpretation and filling, giving (ii), and (ii) gives (iii). Conversely, every fresh tag π_i occurs exactly once in each completed proof and nowhere in either old context. An isomorphism in (iii) must preserve that tag, hence its label x_i and, when it is nonroot, its parent premise slot. Removing the matched probe leaves (or restoring the root identity context) recovers an isomorphism $R \cong R'$ inducing the chosen alignment. $\square$

The theorem makes filling jointly separating over persistent extensions: a raw punctured proof presentation is determined by its uniform action on possible reasons, and one finite tuple of generic reasons suffices. The probe extension is metasyntactic and need not be consequence-conservative. This is generic-element separation in the free proof-term grammar: under a faithful Curry–Howard assignment the same fresh constants separate typed term contexts. It is a syntactic genericity result, not relational parametricity. Proof identity coarser than presentation is treated below by quotienting the same context-composition operations.

Write $\Sigma_{\mathcal{B}, \mathcal{R}}$ for the full tagged signature of material occurrences, logical initials, and admitted logical or structural rule instances. For any tagged constructor signature Σ , write H_Σ for its proof-tree Hopf algebra. If D is a selected family of material, logical, or structural constructor tags, let

$$I_D = \text{span}_k \{[F] : F \text{ contains an occurrence tagged by } D\}. \quad (60)$$

An algebra ideal is called a *Hopf ideal* when coproduct, counit, and antipode pass to the quotient. Equivalently for the present connected graded setting, cutting an element of the ideal always leaves at least one tensor factor in the ideal, and the counit vanishes there. The next result is the standard decorated-tree instance of that condition; its proof-theoretic use follows the statement.

Theorem 2.19 (Constructor deletion as a Hopf quotient). *The subspace I_D is a homogeneous Hopf ideal, and deletion of the tagged instances induces*

$$H_{\Sigma_{\mathcal{B}, \mathcal{R}}}/I_D \cong H_{\Sigma_{\mathcal{B}, \mathcal{R}} \setminus D}. \quad (61)$$

This is the standard decorated-tree constructor-ideal result [25, 26]: a CK cut places every selected vertex in one of its two factors, and the surviving forest basis is precisely the basis over the signature with D deleted.

Its use below is proof-theoretic: constructor deletion lets full and selected proof fibres be compared without identifying either with mere derivability support.

Corollary 2.20 (Effect on filling fibres). *For every D -free punctured proof R , constructor deletion restricts its filling fibre to the tuples whose completed proof contains no occurrence tagged by D . In particular, when D is the family of Gentzen-Cut instances, the quotient is the Hopf algebra of Gentzen-Cut-free proof presentations.*

A completed filling is D -free exactly when its fixed context and all its fillers are D -free.

The quotient gives a retrospective answer: it removes a proof presentation once a forbidden constructor has been found in it. There is also a prospective question—which proof contexts can be built without that constructor in the first place? The ordinary recursive grammar of proof trees answers it by the following standard polynomial fixed point.

Corollary 2.21 (Prospective constructor-free grammar). *Let $\Sigma^- = \Sigma_{\mathcal{B}, \mathcal{R}} \setminus D$, and let P^- be the many-sorted polynomial (17) with the D -tagged constructor summands deleted. Let q_D be the quotient map of Theorem 2.19. Write $\mathcal{O}_{\Sigma^-}$ for the typed proof contexts over this truncated signature and put*

$$W_G^- = \coprod_{\mathbf{K} \in \text{HSeq}^*} \mathcal{O}_{\Sigma^-}(\mathbf{K}; G).$$

Let $\text{Id}_G = \{\square^G\}$ and let $\text{Id} = (\text{Id}_G)_{G \in \text{HSeq}}$ be the many-sorted family of identity contexts. Final-root decomposition gives the initial-algebra equation

$$W^- \cong \text{Id} + P^-(W^-). \quad (62)$$

This is the ordinary recursive constructor-free proof grammar (or, equivalently, its standard many-sorted combinatorial Dyson–Schwinger equation) [32]. For every positive-vertex basis proof or context T ,

$$T \text{ is } D\text{-free} \iff T \in W^- \iff q_D[T] \neq 0. \quad (63)$$

Define its profile-refined Green family by

$$\mathfrak{N}_{\mathbf{K}}^{G,-} = \delta_{\mathbf{K},(G)} \mathbf{1} + \sum_{C \in \mathcal{O}_{\Sigma^-}^+(\mathbf{K}; G)} [C].$$

Under the quotient identification of Theorem 2.19,

$$q_D(\mathfrak{G}_{\mathbf{K}}^G) = \mathfrak{N}_{\mathbf{K}}^{G,-};$$

applying $q_D \otimes q_D$ to (35) gives the same identity with every $\mathfrak{G}$ replaced by $\mathfrak{N}^-$, without a new combinatorial proof.

The quotient is retrospective: it rejects a presentation in which a deleted constructor occurs. The fixed point is prospective: it enumerates the contexts still available to proof search. For $D = \text{GCut}$, the support of $\mathfrak{N}_0^{G,-}$ is precisely the complete Gentzen-Cut-free G -proof fibre. Neither construction rewrites a rejected proof; typed reducts, termination, and confluence remain data of the underlying calculus.

Corollary 2.22 (Material withdrawal and revision). *If $D \subseteq \mathcal{B}$ is a subfamily of material occurrences, then*

$$H_{\mathcal{B}, \mathcal{R}} / I_D \cong H_{\mathcal{B} \setminus D, \mathcal{R}}. \quad (64)$$

More generally, if $\mathcal{B} = \mathcal{B}_{\text{keep}} \sqcup D \rightsquigarrow \mathcal{B}' = \mathcal{B}_{\text{keep}} \sqcup A$ with common occurrences carrying identical concrete instances, the change factors as

$$H_{\mathcal{B}, \mathcal{R}} \twoheadrightarrow H_{\mathcal{B}_{\text{keep}}, \mathcal{R}} \hookrightarrow H_{\mathcal{B}', \mathcal{R}}. \quad (65)$$

The composite sends a proof using D to 0; alternative evidence is described separately by the replacement context below.

This combines constructor deletion with the persistent inclusion $\mathcal{B}_{\text{keep}} \hookrightarrow \mathcal{B}'$ of Theorem 2.16.

Theorem 2.23 (Ground withdrawal exposes a replacement context). *Let $D \subseteq \mathcal{B}$ be a finite family of tagged nullary material occurrences, and let t be a connected complete proof which uses D but is not itself a one-vertex D -ground. The occurrences*

$$L_D(t) = \{v \in V(t) : a_v \in D\}$$

where a_v is the concrete rule occurrence decorating v , form a CK cut. We identify a nullary occurrence a_v with its one-vertex proof term $a_v()$. Cutting there detaches the D -grounds and leaves the D -free punctured derivation $\text{Rep}_D(t)$:

$$\left[\bigsqcup_{v \in L_D(t)} a_v \right] \otimes [\text{Rep}_D(t)]. \quad (66)$$

At each address v , the requirement is $\text{req}_{\text{Rep}_D(t)}(v) = \text{out}(a_v)$. If $\mathcal{B}' = (\mathcal{B} \setminus D) \sqcup A$ is a revised base, then

$$\text{Fill}^{\mathcal{B}', \mathcal{R}}(\text{Rep}_D(t)) \cong \prod_{v \in L_D(t)} \text{Prf}_{\mathcal{B}', \mathcal{R}}(\text{out}(a_v)). \quad (67)$$

The bijection is natural under further persistent base inclusions.

Proof. Nullary grounds are leaves, so their distinct nonroot occurrences are incomparable. The CK cut at those occurrences replaces them by punctures at the same addresses and detaches exactly the displayed forest. A filling is therefore precisely an independent tuple of proofs with the required boundaries; restriction to the inserted subtrees is the inverse. Filling by the original grounds reconstructs t , and Proposition 2.14 gives naturality. $\square$

The quotient of Theorem 2.19 says which old proofs survive withdrawal; this theorem retains more intensional information about an affected proof. Its detached-ground factor presents a proof-replacement obligation whose possible solutions are determined by the revised material base.

Restricting every connected boundary and rule instance to singleton hypersequents recovers the ordinary sequent-proof sector. It is closed under CK cuts, filling, and iterated cuts and therefore spans a Hopf subalgebra. The Communication occurrence studied in Section 5, whose conclusion has two active components, marks the proper hypersequent extension.

Proof identity is supplied independently in the next section.

3 Structural profiles and proof identity

Keep three levels distinct.

- (i) *Proof grammar.* Weakening, Contraction, Splitting, Communication, and similar admitted rules are concrete proof vertices. Internal and external Exchange are instead absorbed by the multiset and coherent-reindexing conventions.
- (ii) *Proof conversion.* The proof theory may independently accept conversions or equations between presentations.
- (iii) *Factorisation.* The CK coproduct descends to the resulting proof classes only when cutting equivalent representatives gives equivalent collected factors—the coideal condition formalised below.

Neither CK multiplication nor the CK coproduct is a structural rule. Multiplication juxtaposes independent proof blocks; the coproduct opens a proof presentation already built.

For any basis span M of proof forests, including a constructor-deletion quotient identified with its surviving forest basis, write M^{cl} for the span of complete forests and $M_{\text{cl}}^{\text{con}}[G]$ for the span of connected complete proofs with root boundary G .

3.1 Connected proof and independent forest

The block-root profile from (21) distinguishes a connected proof of $S \mid T$ from a forest containing independent proofs of S and T , even though their flattened boundaries agree.

Proposition 3.1 (Connected realisation is not forest factorisation). *Let z be a one-vertex connected proof of $S \mid T$, and let x, y be distinct one-vertex proofs of S, T . Then*

$$\text{Root}(z) = \text{Root}(x \sqcup y) = S \mid T, \quad \text{BRoot}(z) = \{S \mid T\} \neq \{S, T\} = \text{BRoot}(x \sqcup y),$$

while

$$\begin{aligned} \Delta_{\text{CK}}[z] &= [z] \otimes \mathbf{1} + \mathbf{1} \otimes [z], \\ \Delta_{\text{CK}}[x \sqcup y] &= [x \sqcup y] \otimes \mathbf{1} + \mathbf{1} \otimes [x \sqcup y] + [x] \otimes [y] + [y] \otimes [x]. \end{aligned} \quad (68)$$

Consequently the equation $[z] = [x \sqcup y]$ does not preserve CK factorisation over a nonzero coefficient ring.

Proof. One-vertex connected trees are primitive and the coproduct is multiplicative. Modulo $[z] - [x \sqcup y]$, the endpoint terms cancel while the two mixed tensors remain. They record the separate detachability of the forest factors. $\square$

Forest multiplication is therefore vertex-free juxtaposition, whereas an admitted assembly rule contributes a connected vertex and its own cut. Proof-theoretically, the former says only that two proofs are being considered side by side; the latter records that the calculus performed a specific operation on them. This distinction is finer than the hypersequent displayed at the root. We call an admitted binary occurrence

$$\text{Asm}_\downarrow : (\mathcal{G}, \mathcal{H}) \longrightarrow \mathcal{G} \mid \mathcal{H} \quad (69)$$

a *connected bar-assembly*: it joins two premise proofs below one rule vertex and transports their displayed components bijectively.

To observe all cuts while keeping the root-containing proof context in a distinguished position, remove the whole-tree endpoint from the coproduct. This is the role of a *coaction*: it is bookkeeping for a decomposition with one factor still living in the chosen space of rooted proofs, not a further rule of the calculus. For a hypersequent G , let C_G^{con} be the span of connected rule-rooted derivation trees, complete or punctured, with boundary G , and let $C_G^{\text{cl}} \subseteq C_G^{\text{con}}$ be the subspace spanned by the complete ones. Put

$$\delta_G : C_G^{\text{con}} \longrightarrow H_{\text{CK}} \otimes C_G^{\text{con}}, \quad \delta_G[t] = \Delta_{\text{CK}}[t] - [t] \otimes \mathbf{1}. \quad (70)$$

Every right-hand retained factor in $\delta_G[t]$ still has root G . Formally, a left H_{CK} -coaction is a map $\delta : C \rightarrow H_{\text{CK}} \otimes C$ satisfying

$$(\Delta_{\text{CK}} \otimes \text{id})\delta = (\text{id} \otimes \delta)\delta, \quad (\varepsilon_{\text{CK}} \otimes \text{id})\delta = \text{id}.$$

These equations say that staged observation is coherent and that the empty detached factor changes nothing. Coassociativity and the counit law for Δ_{CK} give them for δ_G . A space equipped with such a map is called a *comodule*; a comodule map preserves the observation.

3.2 Local structural action of rule vertices

Component-local logical rules, internal structural rules, and external hypersequent rules all occur as concrete connected vertices. Their behaviour is read from the instantiated rule and its component focus rather than from the spelling of the tag. For $G \in \text{HSeq}$, let $\text{br}(G)$ be its number of displayed components. For $a : (G_1, \dots, G_{n(a)}) \rightarrow G$ define its proof-factor and hypersequent-component defects by

$$\sigma_{\text{pf}}(a) = 1 - n(a), \quad \sigma_{\text{hs}}(a) = \text{br}(G) - \sum_{i=1}^{n(a)} \text{br}(G_i). \quad (71)$$

The first counts how many proof factors a rule joins beneath one root; the second counts its net change in displayed hypersequent components. The derived relation κ_a in (3) records the non-scalar movement, merger, splitting, deletion, or creation of component occurrences. Active and principal formula behaviour is already part of the concrete instance a and need not be duplicated in the CK carrier.

The following familiar cases show what the two numbers see and why the relation κ_a is also needed.

concrete rule occurrence	σ_{pf}	σ_{hs}	further incidence needed
binary component-local logical rule (singleton-hypersequent boundaries)	-1	-1	active/principal formula in the concrete instance
connected bar-assembly	-1	0	two premise blocks assembled below one root
Communication (Section 5)	-1	0	formula-occurrence-bijective but crossed component transport
unary Splitting	0	+1	one-to-many component transport
external Weakening adding one component	0	+1	unsupported output component
external Contraction	0	-1	many-to-one component or occurrence incidence

The scalar pair checks boundary arithmetic. Explicit connected bar-assembly and separated-tail Communication have the same pair, while their component relations distinguish independent assembly from crossed transport. External Exchange is the coherent reindexing already built into hypersequent component occurrences. If a binary component-local instance repeats g passive components in both premises, then $\sigma_{\text{hs}} = -(g+1)$; the first row records only the singleton-boundary calibration $g=0$.

Lemma 3.2 (Euler form of the scalar defects). *Let R be a nonempty finite punctured proof forest with $r \geq 1$ connected roots. Then*

$$\begin{aligned} \sum_{v \in V(R)} \sigma_{\text{pf}}(a_v) &= r - |\text{Punc}(R)|, \\ \sum_{v \in V(R)} \sigma_{\text{hs}}(a_v) &= \text{br}(\text{Root } R) - \sum_{p \in \text{Punc}(R)} \text{br}(\text{req}_R(p)). \end{aligned} \quad (72)$$

Thus the total scalar defects are determined by the root and premise telescope. Their useful intensional information is their distribution among rule vertices and ancestry layers; the non-scalar component action lies in κ .

Proof. The premise edges between rule vertices number $|V(R)| - r$, and the remaining premise positions are the $|\text{Punc}(R)|$ punctures. Hence $\sum_v n(a_v) = |V(R)| - r + |\text{Punc}(R)|$, proving the first equation. In the second, the breadth of every internal premise boundary occurs once with a minus sign at its parent and once with a plus sign at the child root; only the forest roots and puncture requirements remain. $\square$

For the empty forest, put $V(\mathbf{1}) = \text{Punc}(\mathbf{1}) = \emptyset$ and $\text{br}(\text{flat } \emptyset) = 0$; both identities then extend to $r = 0$. There is no nonempty root-free forest of nodeless context identities in the CK carrier, because $\square^G$ is an identity for context composition rather than a forest generator.

Accordingly, the carrier records component-local formula action through active and principal incidence, internal structural action through multiset-context incidence, and external structural action through component transport. Any nonnullary rule also joins its premise proof blocks beneath a new vertex, so these recorded behaviours may overlap. Whether a rule family is logical or harmonious is determined independently by its substitution, inversion, and use behaviour.

For a relation $L \subseteq X \times Y$, *functional* means that each x is related to at most one y , *inverse-functional* that each y has at most one antecedent x , *total* that every x has an image, and *surjective* that every y has an antecedent. Functionality forbids one-to-many splitting, inverse-functionality forbids many-to-one merger, totality forbids erasure, and surjectivity forbids unsupported creation.

For a punctured derivation R , its *composite component trace* κ_R is obtained by putting identity relations at its punctures and composing the local component relations κ_a towards the root. Thus, for a root occurrence a ,

$$\kappa_{a(R_1, \dots, R_{n(a)})} = \kappa_a \circ \bigsqcup_{i=1}^{n(a)} \kappa_{R_i}. \quad (73)$$

Also put

$$\sigma_{\bullet}(R) = \sum_{v \in V(R)} \sigma_{\bullet}(a_v), \quad \bullet \in \{\text{pf}, \text{hs}\}. \quad (74)$$

Lemma 3.3 (Composition of component traces). *Let R be a punctured derivation with labelled positions $x_j : K_j$, where $K_j = \text{req}_R(p_j)$ is inferred from the corresponding slot position, and let t_j be compatible proof contexts, including complete proofs as the filling case. Then*

$$\begin{aligned} \kappa_{R\langle t_j \rangle} &= \kappa_R \circ \bigsqcup_j \kappa_{t_j}, \\ \sigma_{\bullet}(R\langle t_j \rangle) &= \sigma_{\bullet}(R) + \sum_j \sigma_{\bullet}(t_j), \quad \bullet \in \{\text{pf}, \text{hs}\}. \end{aligned} \quad (75)$$

Consequently a composite of functional traces cannot split an input, a composite of inverse-functional traces cannot merge two inputs, a composite of total traces cannot erase one, and a composite of surjective traces cannot create an unsupported output.

Proof. Structural induction substitutes each filler trace at the identity trace of its puncture. Relational composition preserves the four stated properties, and the vertex sets of the outer context and fillers are disjoint. $\square$

Proposition 3.4 (Occurrence-resolved component semantics). *Let $\widetilde{\text{HSeq}}$ be the family of hypersequent boundaries with displayed component occurrences resolved, and let $\tilde{O}$ be the corresponding lift of the context family before coherent reindexing. For*

$$C \in \tilde{O}((K_1, \dots, K_n); G)$$

put

$$\tilde{\kappa}_C \subseteq \left(\bigsqcup_{i=1}^n \{i\} \times \text{Comp}(K_i) \right) \times \text{Comp}(G). \quad (76)$$

Then $C \mapsto \tilde{\kappa}_C$ preserves typed identities and context composition:

$$\tilde{\kappa}_{D\langle C_1, \dots, C_m \rangle} = \tilde{\kappa}_D \circ \bigsqcup_{j=1}^m \tilde{\kappa}_{C_j}. \quad (77)$$

Consequently, equality of $\tilde{\kappa}$ -relations is a typed context congruence. Persistent calculus inclusions preserve and reflect this congruence on old contexts.

This is the composition equation of Lemma 3.3 before coherent reindexing; persistence fixes every old local relation and hence its composite.

Corollary 3.5 (Local-to-global component-action criterion). *Fix a family $\mathcal{S}$ of concrete rule occurrences and generate all compatible derivations from their full-premise corollas and typed identities, including the complete corollas of nullary occurrences. For each property*

$$\mathcal{P} \in \{\text{functional, inverse-functional, total, surjective}\},$$

every generated derivation has a $\mathcal{P}$ -component trace if and only if every κ_a , $a \in \mathcal{S}$, has property $\mathcal{P}$. The equivalence is invariant under coherent component reindexing.

Proof. The forward direction evaluates the claim on the full-premise corolla $\text{Cor}(a)$, whose trace is κ_a . The reverse direction is Lemma 3.3, since disjoint union and relational composition preserve each displayed property. $\square$

For a complete proof the trace domain is empty. Functionality, inverse-functionality, and totality are therefore vacuous; surjectivity may still detect an output component with no premise-component source. Corollary 3.5 concerns every derivation generated by a rule family. It does not classify a single composite: a nonfunctional local trace may be followed by a rule whose net composite is functional.

A coherent reindexing only renames input and output component occurrences, so the carrier remembers the relation only up to those renamings. This is the invariance used by the CK-resolved profile below.

Identifying contexts exactly when their net component relations agree, after coherent renaming, gives the compositional abstraction which forgets everything except component transport. It is not by itself an identity of proofs: its nullary sector identifies all complete proofs of one root because their relation has empty domain. Any extension to forest equations must still pass the CK coideal criterion if recursive factorisation is to survive.

By Lemma 3.2, total scalar defects are boundary data. Their local distribution is a consistency check, while the composed relations carry the non-scalar component action.

behaviour visible in κ_R	some local rule trace κ_a must be	
one input reaches two outputs	nonfunctional	
two inputs reach one output	not inverse-functional	
an input disappears	nontotal	
an output has no source	nonsurjective	(78)

Thus a composite proof context cannot acquire splitting, merger, erasure, or unsupported creation merely by nesting otherwise incidence-preserving rules. The trace locates the kind of local structural action which must occur somewhere in its ancestry.

Proposition 3.6 (CK-resolved component profile). *Let t be a connected complete proof. For each proper CK cut c , let*

$$Q_c = \bigsqcup_{p \in c} \{p\} \times \text{Comp}(\text{bd}(t|_p)), \quad O_t = \text{Comp}(\text{bd}(t)).$$

The cut addresses identify Q_c with the component occurrences expected at the punctures of $R^c(t)$. Composing through the retained derivation gives

$$\kappa_t^c \subseteq Q_c \times O_t.$$

The addressed family, called the CK-resolved component profile,

$$\text{CT}(t) = \{(c, Q_c, \kappa_t^c, O_t) : c \in \mathcal{A}^+(t)\} \quad (79)$$

is unchanged by coherent reindexing of multiset-occurrence representatives. Forgetting the relations and replacing each cut by $q^{|c|}$ recovers $\text{Cut}_t(q)$. If the final occurrence a has at least one premise and c_{fin} consists of the roots of its immediate children, then

$$\kappa_t^{c_{\text{fin}}} \cong \kappa_a \quad (80)$$

under the ordered-premise identification. Under context composition, every retained relation is obtained by the composite law of Lemma 3.3.

Proof. The cut positions supply the domain, and recursive trace composition gives κ_t^c . At the immediate-child cut the retained context is the one-vertex final corolla, giving κ_a . Reindexing invariance and the context-composition claim follow from coherent reindexing and relational composition. $\square$

Corollary 3.7 (Persistent naturality of the CK-resolved profile). *Let $\iota : (\mathcal{B}, \mathcal{R}) \hookrightarrow (\mathcal{B}', \mathcal{R}')$ be a persistent calculus inclusion and let t be an old complete proof. Under the canonical identification of its preserved boundary component occurrences,*

$$\text{CT}(\iota_* t) = \text{CT}(t).$$

Thus extension preserves not only every old cut and retained context but also the component relation observed at that cut.

Proof. Theorem 2.16 preserves the cut addresses and both CK factors. Persistence preserves every old local relation κ_a ; Lemma 3.3 therefore gives the same composite κ_t^c for each cut. $\square$

This couples a component relation to the CK location at which it is observed. A persistent extension may add new fillers and new contexts, but it cannot reroute the components of an old proof at any old CK cut. Final-corolla extraction and fully refined iterated cuts expose the full concrete rule instance together with its causal position. The collected CK tensors are their coarser linear image.

3.3 Rule application, conversion, and structural equations

At the level of an individual rule, distinguish three operations.

- (i) *Rule application.* For an admitted occurrence e , $B_e^+(\mathbf{t}) = e(\mathbf{t})$ adds e as a new root below its ordered premise proofs. In the unary case this is the one-slot insertion of (40); higher arity uses simultaneous typed composition.
- (ii) *Conversion.* A proof conversion replaces one realised rule context F by another F' , transporting punctures, component indices, and the fillings available in the fixed calculus.
- (iii) *Equation.* Declaring $[F] = [F']$ symmetrically and contextually generates an algebra ideal. Whether the CK coproduct descends to the resulting proof classes is a further question.

The distinction is already exact for a unary occurrence $e : G \rightarrow H$. The root-preserving coactions satisfy the affine recursion

$$\delta_H[B_e^+(t)] = (\text{id} \otimes B_e^+) \delta_G[t] + [t] \otimes [\text{Cor}(e)]. \quad (81)$$

The first summand transports every cut internal to the premise; the second is the unique new cut immediately below e . A conversion which preserves this collected root-preserving decomposition instead intertwines the two coactions—that is, it is a comodule map—and has no extra corolla term. Formula (81) applies without changing its form to a unary component-local logical rule, internal structural rule, Splitting, external Weakening, or external Contraction. Binary Communication and explicit connected bar-assembly obey the corresponding multi-input grafting recursion, whereas definitional forest juxtaposition obeys multiplicativity.

Internal Exchange is absorbed by the multiset formula contexts. External Exchange acts by coherent reindexing of the component occurrences of a hypersequent multiset. Neither operation forgets the indexed premise order of a proof rule: left and right inputs remain distinct unless that particular rule schema carries an explicit input symmetry. The connected bar-assembly occurrence used below remains a connected rule vertex; identifying its application with independent juxtaposition would be a further equation subject to the coideal criterion.

3.4 Proof congruences and the coideal diagnostic

By puncture factorisation, the CK coproduct records decompositions into detached proof forests and retained proof contexts. If an equation is intended to preserve this collected factorisation, the coproduct must be independent of the chosen representative. That requirement is precisely coideal descent. Algebraically, “coideal” names the condition that the coproduct of every equation lies in $I \otimes H_{\mathcal{B}, \mathcal{R}} + H_{\mathcal{B}, \mathcal{R}} \otimes I$, where I is the generated equation ideal, so it becomes zero after both tensor factors are quotiented. In proof-theoretic terms, an equation passes the coideal test exactly when it remains valid after either side is inspected as detached evidence together with the punctured context which used it. An identity intended to abstract from a chosen scheduling presentation need not meet it; failure then diagnoses the dimension forgotten rather than refuting the identity.

Proof identity is supplied independently; the Hopf structure tests its factorisation stability.

Definition 3.8 (Proof congruence). A *compatible input labelling* of punctured forests F, F' is a finite set I with bijections $\ell_F : I \rightarrow \text{Punc}(F)$ and $\ell_{F'} : I \rightarrow \text{Punc}(F')$ such that

$$\text{req}_F(\ell_F(i)) = \text{req}_{F'}(\ell_{F'}(i)) \quad (i \in I).$$

An *aligned proof equation* consists of two forests admitted by the fixed calculus, with the same flattened root and, when they are punctured, a chosen compatible input labelling. The chosen labelling aligns common variables across differently shaped contexts; it is not an intrinsic name on either proof edge.

A *proof congruence* E is the equivalence relation $\sim_E$ on the underlying unlabelled forests generated by a family of aligned proof equations and their instances under forest multiplication, uniform substitution of formula and context metavariables whenever the resulting concrete occurrences remain in the calculus, and compatible context composition along the specified alignments, hence in particular complete filling. The alignment is retained while contextual instances are formed and then forgotten on the CK basis. Thus different alignments of equal-requirement punctures need not generate the same equation. Put

$$I_E = \text{span}_k\{[F] - [F'] : F \sim_E F'\} \subseteq H_{\mathcal{B}, \mathcal{R}}. \quad (82)$$

No preservation of vertex number or block-root profile is assumed.

Flattened root is nevertheless invariant under every proof congruence. Aligned generators have equal flattened root; forest multiplication adds the same companion boundary to both sides, context composition retains the outer root, and uniform substitution preserves equality of the instantiated roots. Thus

$$F \sim_E F' \implies \text{Root}(F) = \text{Root}(F'). \quad (83)$$

The same induction preserves the complete open interface

$$\text{Int}(F) = (\text{Root}(F), \{\{\text{req}_F(p) : p \in \text{Punc}(F)\}\}), \quad F \sim_E F' \implies \text{Int}(F) = \text{Int}(F'). \quad (84)$$

Indeed, the compatible input labelling of every generator is a requirement-preserving bijection of its punctures, and every closure operation adds, fills, or substitutes matching requirements on the two sides. These elementary sector conditions become decisive when a proposed proof identity exposes detached factors with different boundaries or open requirements.

Because boundary hypersequents are nonempty and $\mathbf{1}$ has no flattened root, aligned generators never relate $\mathbf{1}$ to a nonempty forest; the displayed closure operations do not create such a relation. Hence $\varepsilon_{\text{CK}}(I_E) = 0$ is automatic for proof congruences as defined here. It is retained in the next proposition to display the full algebraic quotient criterion.

Thus E -equivalent proofs may replace one another at a puncture by contextual closure. Agreement of their cut decompositions is the additional condition tested next.

Proposition 3.9 (Filling and the universal CK quotient of proof classes). *For each fixed labelled context $R : (x_i : H_i)_{i=1}^n \rightarrow G$, let E also denote its restrictions to the displayed complete-proof fibres. Contextual closure makes*

$$\overline{\text{ev}}_R : \prod_{i=1}^n \text{Prf}_C(H_i)/E \longrightarrow \text{Prf}_C(G)/E, \quad ([t_i]_E)_i \longmapsto [R[t_i/x_i]_E]_E \quad (85)$$

well-defined. If $R \sim_E R'$ through a specified compatible input alignment, their descended evaluation maps agree after the corresponding reindexing of factors. Put $A_E = H_{\mathcal{B}, \mathcal{R}}/I_E$ and let $q_E : H_{\mathcal{B}, \mathcal{R}} \rightarrow A_E$ be the algebra quotient. Here a bialgebra carries the compatible product, coproduct, and counit before an antipode is added. “Connected filtered” allows the equations to collapse exact vertex degrees while retaining the increasing degree filtration needed for the recursive antipode. With this terminology, the following are equivalent:

(i)

$$\Delta_{\text{CK}}(I_E) \subseteq I_E \otimes H_{\mathcal{B}, \mathcal{R}} + H_{\mathcal{B}, \mathcal{R}} \otimes I_E, \quad \varepsilon_{\text{CK}}(I_E) = 0. \quad (86)$$

- (ii) A_E has the necessarily unique induced bialgebra structure for which q_E is a bialgebra map;
 (iii) with the quotient vertex filtration

$$\mathcal{F}_n A_E = q_E \left(\bigoplus_{d \leq n} (H_{\mathcal{B}, \mathcal{R}})_d \right),$$

A_E is a connected filtered Hopf algebra, q_E is a Hopf map, and every Hopf map $f : H_{\mathcal{B}, \mathcal{R}} \rightarrow K$ satisfying

$$F \sim_E F' \implies f([F]) = f([F'])$$

factors uniquely through q_E as a Hopf map.

When these conditions hold, every finite iterated coproduct is well-defined on proof classes. The universal property concerns collected CK factorisation; occurrence-level invariance additionally requires the address-indexed evaluations to be representative-independent.

Proof. Filler congruence gives the well-defined map (85); aligned contextual closure identifies the maps arising from aligned E -equivalent contexts. The equivalence of (i) and (ii) is the standard quotient criterion. Under (i), the quotient of the vertex grading is a connected filtration; the standard connected-filtered bialgebra recursion supplies its unique antipode. Algebraic quotient universality and surjectivity of q_E give the Hopf factorisation in (iii). Conversely, the kernel of a Hopf map satisfies (i). $\square$

Contextual closure always makes prospective filling well-defined on E -classes. The aligned one-hole quotient Use_C/E and its action on proof classes therefore exist independently of CK stability. Retrospective recursive decomposition descends exactly under the equivalent conditions of Proposition 3.9.

If those conditions fail, no Hopf quotient has kernel exactly I_E . Any Hopf quotient enforcing the equations must make further identifications or annihilations, and E itself does not choose them. For an algebra equation outside the proof-congruence format, failure of the counit condition can rule out any Hopf quotient imposing it at all.

Definition 3.10 (Proper-factor multiplicity). Work over $\mathbb{Z}$. Let $\mathbb{F}_{\mathcal{B}, \mathcal{R}}$ be the monoid of forests admitted by the calculus. For E -classes U, V , let

$$N_F^E(U, V) = \text{coeff}_{U \otimes V}((q_E \otimes q_E) \overline{\Delta}_{\text{CK}}[F]), \quad (87)$$

where $q_E : \mathbb{Z}[\mathbb{F}_{\mathcal{B}, \mathcal{R}}] \rightarrow \mathbb{Z}[\mathbb{F}_{\mathcal{B}, \mathcal{R}}/E]$ is the quotient map and $\overline{\Delta}_{\text{CK}}[F] = \Delta_{\text{CK}}[F] - [F] \otimes \mathbf{1} - \mathbf{1} \otimes [F]$ for $F \neq \mathbf{1}$, as in (30), with value 0 on $\mathbf{1}$. Equivalently, $N_F^E(U, V)$ is the multiplicity of proper CK factorisations of F whose detached factor lies in U and whose punctured remainder lies in V . For connected trees these are exactly the proper CK cuts.

Proposition 3.11 (Proper-factorisation criterion for CK-stable proof identity). *Under the hypotheses of Definition 3.10, the equivalent quotient conditions of Proposition 3.9 hold if and only if*

$$F \sim_E F' \implies N_F^E(U, V) = N_{F'}^E(U, V) \quad \text{for every } U, V. \quad (88)$$

Proof. For $F \sim_E F'$, the two endpoint differences $([F] - [F']) \otimes \mathbf{1}$ and $\mathbf{1} \otimes ([F] - [F'])$ already lie in $I_E \otimes H_{\mathcal{B}, \mathcal{R}} + H_{\mathcal{B}, \mathcal{R}} \otimes I_E$. Thus coideal descent is equivalent to equality of the quotient reduced coproducts. The tensors $U \otimes V$ form a basis of the quotient module over $\mathbb{Z}$, so equality is coefficientwise. Counit compatibility is exactly the excluded empty/nonempty identification. $\square$

This is the class-level proof-identity criterion: synonymous proof presentations retain recursive decomposition exactly when they have the same multiplicities of synonymous detached evidence and synonymous punctured remainders. Over a coefficient ring of positive characteristic, the analogous linear criterion compares these multiplicities only after reduction in that ring; integral counts avoid such cancellation. In particular, every 1-versus-0 obstruction below persists over the fixed coefficient ring k , since $1_k \neq 0$.

The criterion has nontrivial positive instances. The first is the class of proof equations which alter the local presentation of one rule occurrence without changing its interface or its place in premise ancestry.

Call concrete occurrences

$$a : (G_1, \dots, G_n) \longrightarrow G, \quad a' : (G'_1, \dots, G'_n) \longrightarrow G'$$

locally interface-isomorphic when there are a permutation $\sigma \in \mathfrak{S}_n$, equalities $G_i = G'_{\sigma(i)}$ for every i and $G = G'$, and chosen bijections between the component occurrences of every matched input pair and of the output. The chosen reindexings must transport κ_a to $\kappa_{a'}$, including the active, principal, discharge, and formula-occurrence incidences carried by the concrete instances. Thus the occurrences express the same local proof operation up to presentation of its premise slots and component occurrences. The choices matter when an equal hypersequent has repeated components. Let $\mathcal{S}$ be a family of such local isomorphisms, closed under inverse, composition, and every permitted uniform substitution. It supplies the aligned corolla equations

$$a(x_1, \dots, x_n) \sim_{\mathcal{S}} a'(x_{\sigma^{-1}(1)}, \dots, x_{\sigma^{-1}(n)}), \quad (89)$$

where the displayed permutation is the puncture alignment. Write $E_{\mathcal{S}}$ for their proof-congruence closure. Component relations will be compared modulo the occurrence reindexings generated by $\mathcal{S}$; this is their $\mathcal{S}$ -orbit semantics.

Proposition 3.12 (Local interface symmetries are CK-stable). *The proof congruence $E_{\mathcal{S}}$ satisfies the proper-factorisation criterion (88). More strongly, every local symmetry move induces a prefix-order isomorphism of vertex addresses which transports its CK cuts bijectively and, for corresponding cuts c, c' , gives*

$$P^c(F) \sim_{E_{\mathcal{S}}} P^{c'}(F'), \quad R^c(F) \sim_{E_{\mathcal{S}}} R^{c'}(F'). \quad (90)$$

Consequently $I_{E_{\mathcal{S}}}$ is a homogeneous Hopf ideal, the quotient is connected graded, and filling, aligned context composition, and the $\mathcal{S}$ -orbit of the occurrence-resolved component semantics descend to the proof classes. If $E_{\mathcal{S}}$ itself is sectorwise in root boundary and global labelled premise word, the typed Faà di Bruno identity also descends in the form of Corollary 3.16.

Proof. For one move at vertex address v , replace the first premise-slot coordinate below v according to σ , leaving every other coordinate unchanged. This is a bijection of vertex and puncture occurrences preserving and reflecting the prefix order. It therefore bijects the prefix-free cut sets, without identifying two equal-looking edge occurrences.

If a cut contains v or one of its ancestors, the whole local move occurs in one detached factor and the remainders agree. Otherwise v remains in the retained factor; any selected descendants are reindexed as a forest, and the retained factors differ by the aligned corolla equation (89). This includes a cut containing several incomparable descendants of v . Hence (90); forest multiplication treats the connected factors independently.

Context composition, filling, and schematic substitution only place such a move at another vertex, while symmetry and transitivity compose the resulting address bijections. Hence every pair of $E_{\mathcal{S}}$ -equivalent forests has equal integral multiplicities of corresponding factor classes. Proposition 3.11 gives the Hopf quotient; vertex number is unchanged, so its filtration remains a grading. The component-orbit and context-composition claims follow from the local incidence hypothesis and the same transported witnesses. Under the stated global sector hypothesis, Corollary 3.16 gives the final assertion. $\square$

The result is stronger than quotienting by an equality already built into the carrier. Formula contexts and displayed hypersequents were multisets from the outset, but premise-slot indices were deliberately retained. Proposition 3.12 permits those indices to be forgotten only where the proof theory itself supplies a local symmetry, and it preserves the number of edge occurrences rather than merely the support of the coproduct.

Corollary 3.13 (Material-ground tag irrelevance). *For nullary material grounds, write $b \equiv_0 b'$ when b and b' are locally interface-isomorphic with the empty premise permutation. Thus they have the same boundary and the same local focus and incidence data up to the chosen output-occurrence reindexing; only the material tag or its local presentation may differ. Identify the members of each $\equiv_0$ -class, while fixing every positive-arity material occurrence and every logical or structural occurrence, and let E_{gr} be the resulting proof congruence. Form $\mathcal{B}_{\text{ext}}$ by replacing every nonempty class $C \subseteq \mathcal{B}_G^{(0)}$ by one ground $\star_C : () \rightarrow G$ with a fresh material tag and the interface of a chosen representative, and leave the rest of $\mathcal{B}$ unchanged. The label-replacement map*

$$\pi_{\text{gr}} : H_{\mathcal{B}, \mathcal{R}} \longrightarrow H_{\mathcal{B}_{\text{ext}}, \mathcal{R}}$$

is a surjective connected graded Hopf map with kernel $I_{E_{\text{gr}}}$. Hence it induces

$$H_{\mathcal{B}, \mathcal{R}} / I_{E_{\text{gr}}} \cong H_{\mathcal{B}_{\text{ext}}, \mathcal{R}} \quad (91)$$

compatibly with puncturing and filling. In the discrete nullary presentation (5), where the boundary is the only local data apart from the tag, this remembers only which primitive boundaries are supported. It does not identify arbitrary compound proofs of the same boundary, or grounds with observably different local interfaces.

Proof. The defining ground pairs are local interface symmetries, so Proposition 3.12 applies. Replacing every ground label in a class C by $\star_C$ defines a surjection on proof forests which commutes with context composition and with every address-indexed CK cut. Two basis forests have the same image exactly when they differ by finitely many such ground replacements. Its linear kernel is therefore $I_{E_{\text{gr}}}$, giving (91). $\square$

This positive quotient is stable under enlargement of the material base. For every persistent material inclusion $\iota : \mathcal{B} \hookrightarrow \mathcal{B}'$, persistence preserves local interfaces, so each old $\equiv_0$ -class lies in a unique new one and distinct old classes remain distinct. Choose the fresh tags coherently by interface; sending each $\star_C$ to the ground representing its enlarged class gives an injective map $\iota_{\text{ext}} : \mathcal{B}_{\text{ext}} \hookrightarrow \mathcal{B}'_{\text{ext}}$ and a commuting square of Hopf maps:

$$\pi'_{\text{gr}} \iota_* = (\iota_{\text{ext}})_* \pi_{\text{gr}}. \quad (92)$$

Because both ground quotients change only the corresponding nullary labels,

$$I_{E'_{\text{gr}}} \cap H_{\mathcal{B}, \mathcal{R}} = I_{E_{\text{gr}}}, \quad (93)$$

so the old-sector quotient remains injective after extension. Extensionalising material reasons therefore commutes with adding new ones; it does not retrospectively change the factorisation classes of old proofs.

For example, if $b_S, b'_S : () \rightarrow S$ are distinct but locally interface-isomorphic material reasons, the two running proofs $i(m(b_S, b_T))$ and $i(m(b'_S, b_T))$ remain different raw proof trees but become equal in the quotient. Their coproducts in (28) agree term-for-term after replacing b_S, b'_S by one class representative $\star_{[b_S]_0}$: the quotient forgets which reason token was used, not the position at which material support entered the proof.

As a second instance, suppose the calculus admits connected bar-assembly occurrences with the local symmetry

$$\text{Asm}_{G,H}(x : G, y : H) \sim \text{Asm}_{H,G}(y : H, x : G). \quad (94)$$

The conclusion $G \mid H$ is unchanged as a hypersequent multiset and the component relation is transported by the transposition, so the equation is CK-stable. This identifies the two premise presentations of one connected assembly proof; it does not identify that proof with the independent forest $x \sqcup y$. If the two premises are the same proof, or become equal in the quotient, the two singleton edge cuts still contribute multiplicity 2 after quotienting. The local symmetry forgets left-versus-right presentation without collapsing two proof uses into one. Any nontrivial premise permutation can exchange the global ordered premise-word sectors after punctured contexts are inserted, even when $G = H$: inherited words KL and LK need not agree. Its Hopf quotient remains valid, but the ordered Green-series statement of Corollary 3.16 requires the full congruence to be sectorwise, or else an orbit-indexed refinement of input words. The material-ground tag quotient fixes premise slots and is sectorwise.

Local interface symmetry therefore supplies a positive proof identity which preserves premise ancestry. The next example changes that ancestry, and the factorisation criterion detects the difference.

Example 3.14 (A proof-net permutation cannot preserve raw CK factorisation). In the one-sided unit-free multiplicative linear sequent calculus, take

$$x \vdash \Gamma, A, B, D, \quad y \vdash \Delta, C$$

and compare the standard $\mathfrak{A}/\otimes$ commuting conversion

$$\begin{aligned} s(x, y) &= \otimes_{D,C}(\mathfrak{A}_{A,B}(x), y), \\ r(x, y) &= \mathfrak{A}_{A,B}(\otimes_{D,C}(x, y)). \end{aligned} \quad (95)$$

Both contexts use the same two labelled inputs once and conclude $\vdash \Gamma, \Delta, A \mathfrak{A} B, D \otimes C$. They are the familiar two sequentialisations of the same MLL proof net [29]. Write $\bar{s} = s(x, y)$ and $\bar{r} = r(x, y)$ for the two unresolved contexts. Their proper CK factorisations are

$$\begin{aligned} \bar{\Delta}_{\text{CK}}[\bar{s}] &= [\mathfrak{A}_{A,B}(x)] \otimes [\otimes_{D,C}(\square, y)], \\ \bar{\Delta}_{\text{CK}}[\bar{r}] &= [\otimes_{D,C}(x, y)] \otimes [\mathfrak{A}_{A,B}(\square)]. \end{aligned} \quad (96)$$

This failure cannot be repaired by enlarging the generated proof congruence. Let E be any proof congruence containing the aligned equation $\bar{r} \sim_E \bar{s}$. The detached factors have flattened roots

$$G_s = \vdash \Gamma, A \wp B, D, \quad G_r = \vdash \Gamma, \Delta, A, B, D \otimes C. \quad (97)$$

They are unequal: after the common multiset Γ is removed, the left boundary contains two formula occurrences and the right contains $|\Delta| + 3$. By (83), the two detached factors cannot belong to one E -class. If

$$U = [\wp_{A,B}(x)]_E, \quad V = [\otimes_{D,C}(\square, y)]_E,$$

then the two one-cut calculations give

$$N_s^E(U, V) = 1, \quad N_{\bar{r}}^E(U, V) = 0. \quad (98)$$

Proposition 3.11 therefore rules out every CK-stable proof congruence containing this permutation, not merely the congruence minimally generated by it.

This distinguishes two notions which are easy to conflate. In *co-present puncture independence*, two holes already occur in one context and insertion modifies disjoint subtrees; Corollary 2.3 therefore gives the same raw tree and the same CK witnesses in either order. The $\wp$ - and $\otimes$ -links above are instead independent in the proof-net or *conversional* sense: proof theory permits their permutation, but each sequentialisation makes one rule occurrence an ancestor of the other and gives it a different intermediate boundary. They also share the input x , so component incidence alone neither states nor decides this conversional independence.

Remark 3.15 (Focusing selects presentations; it does not force descent). Focalization ensures that every cut-free provable linear-logic sequent has a focused proof [3]; it does not by itself choose a unique representative or establish an equation between derivations. More particularly, focusing cannot rescue descent for the iso-polar equations relevant to proof-net canonicity. For pairwise distinct atoms A, B, C, D and $x : \vdash \Gamma, A, B, C, D$, the erasures of two focused one-input contexts are

$$p = \wp_{A,B}(\wp_{C,D}(x)), \quad q = \wp_{C,D}(\wp_{A,B}(x))$$

whose focused lifts consist of one negative phase and differ by a negative/negative iso-polar permutation. Their sole detached factors have respective roots

$$\vdash \Gamma, A, B, C \wp D, \quad \vdash \Gamma, A \wp B, C, D,$$

so the argument of Example 3.14 applies unchanged: no CK-stable proof congruence can contain $p \sim q$. This is directly relevant to proof identity: in cut-free unit-free MLL, iso-polar classes of maximally multifocused proofs are in bijection with proof nets [16]. Selecting focused representatives therefore does not prove quotient descent. Nor does restricting the basis to complete end-to-end focused proofs give a CK subcoalgebra: an arbitrary cut can expose an open phase fragment. One may take a CK-hereditary closure of the erased fragments as a bare coalgebra, but an intrinsically typed focused filling and context calculus would require phase-indexed puncture data absent from the present requirement colour $\text{req}_R(h)$, which records only the boundary required at a puncture h .

Thus proof-net equality is an independently motivated proof identity which the raw tree coproduct cannot refine to these proof classes. The mismatch is structural: CK remembers sequential premise ancestry which the proof net deliberately forgets, and no additional boundary-preserving proof equations can identify the exposed factors in (97). This does not preclude a coproduct defined natively on proof nets; it precludes obtaining one by descending the present tree coproduct through this standard permutation.

Sectorwise preservation is substantive here. Without it, a premise permutation may identify a context in the global input-word sector KL with one in LK . The Hopf quotient may still exist, but the word-indexed Green-series identity is then no longer well typed unless one passes to orbits of input words.

Corollary 3.16 (Typed decomposition on proof classes). *Let E be sectorwise in root boundary and labelled premise word and satisfy Proposition 3.9. Every finite vertex-degree truncation of the typed CK-cut/context-composition identity (35) descends along $q_E \otimes q_E$. If the quotient Green series are coefficientwise locally finite, their completion therefore satisfies the same typed Faà di Bruno identity with every $\mathfrak{G}_K^G$ replaced by its E -class series.*

Proof. Apply the induced coproduct to the finite truncations of Theorem 2.8; local finiteness permits passage to the coefficientwise completion. $\square$

Remark 3.17 (Persistent base change of proof congruences). Let $\iota : (\mathcal{B}, \mathcal{R}) \hookrightarrow (\mathcal{B}', \mathcal{R}')$ be a persistent calculus inclusion, and let E, E' be proof congruences such that

$$F \sim_E F' \implies \iota_* F \sim_{E'} \iota_* F'.$$

Then $\iota_*(I_E) \subseteq I_{E'}$, so ι_* induces an algebra map between the two proof-class quotients. If both ideals pass the coideal and counit test, this induced map is a Hopf map by Theorem 2.16.

The induced algebra map is injective exactly under the conservativity condition

$$I_{E'} \cap \iota_*(H_{\mathcal{B}, \mathcal{R}}) = \iota_*(I_E). \quad (99)$$

When this holds, the proper-factorisation criterion for an equation between old forests is unchanged by extension. Indeed, Theorem 2.16 preserves all its cut factors, and (99) preserves exactly their equivalence classes. Hence the two multiplicity tests agree on the old sector.

Condition (99) is not automatic. An extended calculus may supply new substitution instances, fillers, and surrounding rule contexts for the generating equations. Their contextual closures may create new identifications even among old forests; generators involving genuinely new forests must in any case satisfy Proposition 3.9, or the relative criterion of Theorem 4.6 when common input regions are specified. Hence CK-stable proof congruences form a natural family under persistent extension only when this additional base-change condition is verified. Even when the old-sector test is preserved, it does not force the full extended ideal to be a coideal. Material-ground tag irrelevance is the concrete positive exception established above: (93) verifies the required intersection for every persistent material inclusion.

This is the proof-theoretic form of the standard coideal test: proposed synonyms must admit equally many decompositions into synonymous detached forests and synonymous punctured remainders. The statement concerns the collected proper-factorisation semantics, not the uncollected address-indexed witnesses.

4 Harmony, structural demand, and Gentzen Cut

Harmony is assessed relative to a fixed structural regime: the concrete structural rules present in the calculus and any independently justified structural proof equations. CK factorisation neither chooses that regime nor confers harmony. It tests what an already justified conversion or identity preserves, directly or relative to separately chosen CK-stable background proof equations.

A rule corolla is already a finite derivation context, punctured exactly at the ordered premises on which the concrete rule acts; a nullary corolla is complete. Final-rule extraction therefore exposes its premise telescope without a second construction. Fixing all but one position gives the one-puncture slice classified by (16).

Structural demand is likewise read from the concrete instance rather than from arity. For example, in the familiar shared-context rule $\Gamma \vdash A, \Gamma \vdash B / \Gamma \vdash f(A, B)$ the repeated metavariable Γ records sharing, whereas the split-context version uses distinct multiset metavariables. CK arity alone does not record this distinction; it belongs to the instantiated rule, not to a label copied onto its punctures.

The main question of this section is more general. Given two independently justified punctured rule contexts with the same input variables, which factorisations belong to the input proofs and which to the surrounding rule structure? The answer yields a relative test for normalisation, harmony equations, and macro rules without asking Hopf algebra to choose those equations. Here a context is *linear* in its labelled inputs when each input variable occurs exactly once.

For orientation, when Gentzen Cut is admitted, its simplest two-input context is the corolla

$$K_A = \text{GCut}_A(\Box_1, \Box_2), \quad K_A\langle\pi, \sigma\rangle = \text{GCut}_A(\pi, \sigma).$$

The machinery below becomes informative when a redex or reduct is a compound context around such inputs. Figure 3 is the concrete target to keep in view. Its boxed premise proofs are preserved by a principal tensor reduction, while the intervening rule context changes. The construction below first accounts for every cut lying wholly inside the boxed inputs, then records only the factorisations contributed by the surrounding rule vertices.

$$\begin{array}{c}
\boxed{\pi} : \Gamma \vdash A, \quad \boxed{\sigma} : \Delta \vdash B, \quad \boxed{\tau} : A, B, \Theta \vdash C \\
\hline
\frac{\frac{\boxed{\pi} \quad \boxed{\sigma}}{\Gamma, \Delta \vdash A \otimes B} (\otimes R) \quad \frac{\boxed{\tau}}{A \otimes B, \Theta \vdash C} (\otimes L)}{\Gamma, \Delta, \Theta \vdash C} (\text{GCut}_{A \otimes B}) \quad \Rightarrow \quad \frac{\boxed{\sigma} \quad \frac{\boxed{\pi} \quad \boxed{\tau}}{\Gamma, B, \Theta \vdash C} (\text{GCut}_A)}{\Gamma, \Delta, \Theta \vdash C} (\text{GCut}_B) \\
\hline
\text{principal redex} \qquad \qquad \qquad \text{standard reduct} \\
\hline
\{\otimes R\}, \{\otimes L\}, \{\otimes R, \otimes L\} \qquad \{\text{GCut}_A\} \\
\text{Cut}_{s_{\otimes}}(q) = 2q + q^2 \qquad \text{Cut}_{r_{\otimes}}(q) = q
\end{array}$$

Figure 3: The principal tensor Gentzen-Cut reduction. The common input proofs are boxed. The underlying proof theory justifies the input-preserving reduction; the cut polynomials record the change in its surrounding rule context, not a failure of harmony. The contexts $s_{\otimes}, r_{\otimes}$ are defined formally in Proposition 4.16.

4.1 The input-proof coaction and relative factorisation

The full CK coproduct sees cuts in both the input proofs and the fixed punctured derivation which uses them. To isolate those internal to the inputs, adjoin a formal puncture x to each root proof space. The subtraction below cancels from the linear observation the CK-cut terms already attributable to supplied inputs; it neither deletes nor reduces a proof. For $x : G$, set

$$\tilde{C}_{G,x} = kx \oplus C_G^{\text{con}}$$

and define

$$\lambda_{G,x}(x) = \mathbf{1} \otimes x, \quad \lambda_{G,x}([t]) = [t] \otimes x + \delta_G[t]. \quad (100)$$

The first term detaches the whole filler and restores the formal puncture; δ_G records the empty cut and every proper cut inside the filler. Expanded on a basis proof, this says simply

$$\lambda_{G,x}([t]) = [t] \otimes x + \mathbf{1} \otimes [t] + \sum_{c \in A^+(t)} [P^c(t)] \otimes [R^c(t)]. \quad (101)$$

For a labelled premise telescope $\Xi = (x_i : G_i)_{1 \leq i \leq n}$, put

$$V_{\Xi} = \bigotimes_{i=1}^n \tilde{C}_{G_i, x_i}.$$

Tensoring the pointed coactions defines a common input-proof coaction. We use Sweedler notation $\lambda(v) = \sum v_{[-1]} \otimes v_{[0]}$, meaning “sum over the finitely many terms of $\lambda(v)$ ”; the indices merely name the two tensor factors. Then

$$\lambda_{\Xi}(v_1 \otimes \cdots \otimes v_n) = \sum v_{1[-1]} \cdots v_{n[-1]} \otimes (v_{1[0]} \otimes \cdots \otimes v_{n[0]}). \quad (102)$$

In each term, $v_{[-1]}$ is the detached forest and $v_{[0]}$ is the retained pointed input from the finite list in (101); no further proof objects are added. Every positive-degree punctured rule context $s : \Xi \rightarrow G$ admitted by the current calculus has an evaluation

$$e_s : V_{\Xi} \longrightarrow C_G^{\text{con}} \quad (103)$$

by context composition; a complete input fills its position, while a formal input x_i leaves that puncture unresolved.

Lemma 4.1 (Fixed linear context replacement). *Equations (100) and (102) define left H_{CK} -coactions. If $s, s' : \Xi \rightarrow G$, changing the outer context while fixing $v \in V_{\Xi}$ preserves every tuple and iterated family of CK cuts chosen wholly inside the input proofs, including their detached factors and relative cut addresses inside each labelled input proof.*

Proof. The pointed coaction laws follow from $\Delta_{\text{CK}}[t] = [t] \otimes \mathbf{1} + \delta_G[t]$. Tensor products of comodules use multiplication on the detached factors, which gives (102). Changing s is literally the identity on each v_i , hence on every cut witness internal to it. $\square$

Definition 4.2 (Context-relative factorisation residue). For a fixed positive-degree linear punctured context s of root boundary G , define

$$\Omega_s = \delta_G e_s - (\text{id} \otimes e_s) \lambda_{\Xi} : V_{\Xi} \longrightarrow H_{\text{CK}} \otimes C_G^{\text{con}}. \quad (104)$$

The running tree through the residue. Return to $u = i(m(b_S, b_T))$ from (28), and regard

$$s(x) = i(m(x, b_T))$$

as a one-input context: b_S is the supplied proof, while the side proof b_T and the occurrences m, i belong to the fixed surrounding context. Since b_S is one-vertex,

$$(\text{id} \otimes e_s)\lambda_{S,x}([b_S]) = \mathbf{1} \otimes [u] + [b_S] \otimes [i(m(\square_{(1,1)}, b_T))].$$

Subtracting these two input-internal terms from $\delta_U[u]$ leaves

$$\begin{aligned} \Omega_s([b_S]) &= [m(b_S, b_T)] \otimes [i(\square_1)] + [b_T] \otimes [i(m(b_S, \square_{(1,2)}))] \\ &\quad + [b_S \sqcup b_T] \otimes [i(m(\square_{(1,1)}, \square_{(1,2)}))]. \end{aligned} \tag{105}$$

The three terms are exactly the cuts which enter the nonroot fixed context—at m , at the fixed side proof b_T , or at b_S and b_T together. The cut lying only in the supplied proof b_S has cancelled.

Proposition 4.3 (Support and defect of the relative residue). *If $v = v_1 \otimes \cdots \otimes v_n$ is a basis input and $T = e_s(v)$, tag the vertex set as*

$$V(T) = V(s) \sqcup \bigsqcup_i V(v_i).$$

Here $V(x_i) = \emptyset$ for an unresolved pointed input. For a cut c , let $\text{Sel}_T(c) = \{\text{the root vertex of } T|_p : p \in c\}$ be its selected vertex set. Then

$$\Omega_s(v) = \sum_{\substack{c \text{ a proper CK cut of } T \\ \text{Sel}_T(c) \cap V(s) \neq \emptyset}} [P^c(T)] \otimes [R^c(T)]. \tag{106}$$

Thus Ω_s records exactly the proper cuts whose selected vertex set contains a nonroot vertex of the fixed context s . Cuts selected entirely in an input region, including detachment of a whole input at its root, have been subtracted. Whether a later proof identity treats the remaining context-sensitive factorisations as significant or bureaucratic is expressed by the background proof-equation ideal I_{bg} below. It satisfies the comodule-defect identities

$$(\varepsilon_{\text{CK}} \otimes \text{id})\Omega_s = 0, \quad (\Delta_{\text{CK}} \otimes \text{id})\Omega_s = (\text{id} \otimes \delta_G)\Omega_s + (\text{id} \otimes \Omega_s)\lambda_{\Xi}. \tag{107}$$

Proof. The input coaction independently chooses the empty, proper, or whole-root cut in each input region. After evaluation, these are exactly the root-preserving CK cuts c of T with $\text{Sel}_T(c) \cap V(s) = \emptyset$: the term $[v_i] \otimes x_i$ handles whole-filler detachment. They cancel term by term in (104), leaving precisely (106). The counit and coaction equations for δ_G, λ_{Ξ} give (107) after the two middle terms cancel. $\square$

For every one-vertex corolla,

$$\Omega_{\text{Cor}(a)} = 0. \tag{108}$$

Indeed, its only fixed-context vertex is the root, which no proper CK cut may select. Thus Ω detects nonroot structure contributed by a surrounding context; it cannot distinguish two atomic rule choices, even when they share a typed interface. Atomic choices are instead exposed by final-corolla extraction and compared through their local κ_a -relations, whereas relative residues compare multi-vertex contexts with a common labelled premise telescope and root boundary.

Corollary 4.4 (Replacement residues and material naturality). *For two contexts $s, s' : \Xi \rightarrow G$, put*

$$\text{Res}_{s,s'} = \Omega_{s'} - \Omega_s. \tag{109}$$

Relative residues simply telescope:

$$\text{Res}_{s,s} = 0, \quad \text{Res}_{s,s''} = \text{Res}_{s,s'} + \text{Res}_{s',s''}. \tag{110}$$

For a persistent inclusion $j : \mathcal{B} \hookrightarrow \mathcal{B}'$, let j_{Ξ} be the tensor product of the pointed-input maps. Then

$$(j_* \otimes j_*)\Omega_s^{\mathcal{B}} = \Omega_{j_*s}^{\mathcal{B}'} j_{\Xi},$$

and the same equation holds for relative residues. Constructor deletion has the analogous naturality whenever the contexts contain no deleted occurrence.

Proof. All evaluations have the common domain V_{Ξ} , so the last equation is literal telescoping. Base naturality follows from Theorem 2.16 and Corollary 2.17; deletion naturality follows from Theorem 2.19, since no vertex of the fixed outer context is killed. $\square$

The subtraction and telescoping law are elementary coalgebraic defect identities; interacting tree Hopf algebras and substitution coactions are established constructions [12]. The proof-theoretic construction is the typed residue Ω_s : it separates the decomposition internal to the input proofs from cuts whose selected vertices enter the nonroot fixed context, uniformly over material bases and staged replacements.

We reserve *input-preserving* for a comparison in which both contexts insert every labelled input exactly once and identify the embedded input regions pointwise, including every CK cut at its relative address. More generally, a comparison is *Prot-protected* when only the inputs indexed by **Prot** are held fixed in this way; changed use of the remaining complete inputs stays in the residue. Input preservation is stronger than uniform syntactic substitution because it fixes the embedded input regions and their internal cuts. It does not make the whole proof equation CK-stable. Nor does it assert that an individual evaluation has zero defect Ω_s .

Definition 4.5 (Protected-input coaction and residue). Let $\text{Prot} \subseteq \{1, \dots, n\}$ be a chosen set of protected input positions. Algebraically, “protected” means that the pointed coaction of that factor is subtracted. It entails literal preservation only for the single-occurrence comparisons specified below. Put

$$W_i^{\text{Prot}} = \begin{cases} \tilde{C}_{G_i, x_i}, & i \in \text{Prot}, \\ C_{G_i}^{\text{cl}}, & i \notin \text{Prot}, \end{cases} \quad V_{\Xi}^{\text{Prot}} = \bigotimes_{i=1}^n W_i^{\text{Prot}}.$$

Give a protected factor its pointed coaction λ_{G_i, x_i} , and an unprotected complete-proof factor the trivial coaction $v \mapsto \mathbf{1} \otimes v$. Their tensor product is a coaction

$$\lambda_{\Xi}^{\text{Prot}} : V_{\Xi}^{\text{Prot}} \longrightarrow H_{\text{CK}} \otimes V_{\Xi}^{\text{Prot}}. \quad (111)$$

Thus $V_{\Xi}^{\{1, \dots, n\}} = V_{\Xi}$ and $\lambda_{\Xi}^{\{1, \dots, n\}} = \lambda_{\Xi}$, while $V_{\Xi}^{\emptyset}$ contains only complete inputs and observes none of them internally.

Let $e_0, e_1 : V_{\Xi}^{\text{Prot}} \rightarrow C_G^{\text{con}}$ be linear extensions of basis maps whose paired outputs are rule-rooted, positive-degree, and have the same root boundary and labelled premise telescope. Put

$$\Omega_{e_j}^{\text{Prot}} = \delta_G e_j - (\text{id} \otimes e_j) \lambda_{\Xi}^{\text{Prot}}, \quad \text{Res}_{e_0, e_1}^{\text{Prot}} = \Omega_{e_1}^{\text{Prot}} - \Omega_{e_0}^{\text{Prot}}, \quad \chi = e_1 - e_0. \quad (112)$$

The next criterion is exact for the algebra ideal generated by this displayed comparison. Its closure under arbitrary proof contexts remains the separate class-level test of Proposition 3.9.

Theorem 4.6 (Protected-input coideal criterion). *With the data of Definition 4.5, for every $v \in V_{\Xi}^{\text{Prot}}$,*

$$\Delta_{\text{CK}} \chi(v) = \chi(v) \otimes \mathbf{1} + (\text{id} \otimes \chi) \lambda_{\Xi}^{\text{Prot}}(v) + \text{Res}_{e_0, e_1}^{\text{Prot}}(v). \quad (113)$$

If J is the algebra ideal generated by $\chi(V_{\Xi}^{\text{Prot}})$, then

$$\begin{aligned} J \text{ is a CK coideal} &\iff \text{Res}_{e_0, e_1}^{\text{Prot}}(V_{\Xi}^{\text{Prot}}) \subseteq J \otimes H_{\text{CK}} + H_{\text{CK}} \otimes J \\ &\iff (q_J \otimes q_J) \text{Res}_{e_0, e_1}^{\text{Prot}} = 0, \end{aligned} \quad (114)$$

where $q_J : H_{\text{CK}} \rightarrow H_{\text{CK}}/J$ is the algebra quotient.

Proof. A tensor product of pointed and trivial coactions is a coaction. Subtracting the two definitions of Ω^{Prot} , and using $\Delta_{\text{CK}} z = z \otimes \mathbf{1} + \delta_G z$, gives (113). Its first two terms lie in the two ideal tensors, so coideal closure on the generators is equivalent to (114). Multiplicativity extends this to the generated algebra ideal, and the counit vanishes on differences of nonempty connected presentations. $\square$

The projected residue is therefore the complete obstruction to coideal closure for the displayed comparison ideal $J = \langle \chi(V_{\Xi}^{\text{Prot}}) \rangle$: after the protected-input coaction removes every factorisation assigned to the designated evidence regions, no further CK term remains for these generators. The full contextual proof congruence generated from the comparison must still pass Proposition 3.9.

Corollary 4.7 (Descent relative to background equations). *Let J be the comparison ideal of Theorem 4.6. The structural rules of the calculus are already represented in the carrier. Let $I_{\text{bg}} \subseteq H_{\text{CK}}$ instead be an independently fixed algebra ideal of accepted proof equations with*

$$\Delta_{\text{CK}} I_{\text{bg}} \subseteq I_{\text{bg}} \otimes H_{\text{CK}} + H_{\text{CK}} \otimes I_{\text{bg}}, \quad \varepsilon_{\text{CK}} I_{\text{bg}} = 0,$$

and suppose that I_{bg} does not relate basis contexts with different flattened roots or different inferred ordered, slot-addressed premise interfaces (factorwise for forests). Thus distinct root judgements and premise telescopes remain distinct. This sectorwise condition supplies the typed proof-theoretic reading; the coideal and counit conditions alone form the quotient bialgebra. Write $q_{\text{bg}} : H_{\text{CK}} \rightarrow \overline{H} = H_{\text{CK}}/I_{\text{bg}}$ and put $\overline{J} = q_{\text{bg}}(J)$. The quotient coproduct is characterised by the commuting square

$$\begin{array}{ccc} H_{\text{CK}} & \xrightarrow{\Delta_{\text{CK}}} & H_{\text{CK}} \otimes H_{\text{CK}} \\ q_{\text{bg}} \downarrow & & q_{\text{bg}} \otimes q_{\text{bg}} \downarrow \\ \overline{H} & \xrightarrow{\overline{\Delta}} & \overline{H} \otimes \overline{H}. \end{array} \quad (115)$$

The same comparison descends relative to these CK-stable background proof equations exactly when

$$\overline{J} \text{ is a coideal in } \overline{H} \iff (q_{\text{bg}} \otimes q_{\text{bg}}) \text{Res}_{e_0, e_1}^{\text{Prot}}(V_{\Xi}^{\text{Prot}}) \subseteq \overline{J} \otimes \overline{H} + \overline{H} \otimes \overline{J}. \quad (116)$$

Equivalently, with $q_{I_{\text{bg}}+J} : H_{\text{CK}} \rightarrow H_{\text{CK}}/(I_{\text{bg}} + J)$, the obstruction class

$$\text{Ob}_{\text{Prot}, I_{\text{bg}}}(e_0, e_1) = (q_{I_{\text{bg}}+J} \otimes q_{I_{\text{bg}}+J}) \text{Res}_{e_0, e_1}^{\text{Prot}} \quad (117)$$

*vanishes. The parameter **Prot** fixes the supplied reasons preserved literally, while I_{bg} fixes the surrounding factorisations already declared inessential.*

Proof. The hypotheses on I_{bg} make $\overline{H}$ a quotient bialgebra. With $\overline{\lambda}_{\Xi}^{\text{Prot}} = (q_{\text{bg}} \otimes \text{id})\lambda_{\Xi}^{\text{Prot}}$, the ideal $\overline{J}$ is generated by $q_{\text{bg}}\chi(V_{\Xi}^{\text{Prot}})$; applying $q_{\text{bg}} \otimes q_{\text{bg}}$ to the same identity and repeating the generator argument proves (116). Quotienting $\overline{H}$ once more by $\overline{J}$ is the map $q_{I_{\text{bg}}+J}$, so the tensor-ideal inclusion is equivalent to vanishing of (117). $\square$

For each fixed tuple of unprotected complete basis proofs, suppose e_j is obtained by ordinary single-occurrence context composition of the protected inputs into a fixed punctured rule context. Then $(\text{id} \otimes e_j)\lambda_{\Xi}^{\text{Prot}}$ is literally the family of cuts wholly inside those protected proof regions. Cuts in an unprotected input which e_j actually uses are not cancelled. Any difference between e_0 and e_1 in erasing or independently copying such an input is therefore retained in $\text{Res}_{e_0, e_1}^{\text{Prot}}$. Structural Gentzen-Cut cases can protect the premises they transport unchanged while treating a weakened or duplicated complete premise as unprotected. The set **Prot** is fixed by the reduction schema.

Taking **Prot** to be all input positions and $e_0 = e_s, e_1 = e_{s'}$ recovers $\text{Res}_{s, s'}$ and the earlier full-input criterion. Taking $I_{\text{bg}} = 0$ gives raw-tree descent. A nonzero raw residue may vanish after projection by q_{bg} , so proof identity can instead be tested relative to a chosen CK-stable background quotient. When that quotient is induced by an admitted proof congruence E_0 , the same theorem is applied inside $H_{\mathcal{B}, \mathcal{R}}$ with $I_{\text{bg}} = I_{E_0}$; Proposition 3.9 is exactly the eligibility test.

The protected-input parameter specifies the comparison domain and which pointed coactions are subtracted; under the single-occurrence hypothesis these are exactly the evidence regions held fixed. The background quotient by I_{bg} specifies which factorisation distinctions are identified.

For a comparison internal to the fixed calculus, restrict the construction to its basis inputs and outputs. The paired outputs have the same labelled premise telescope, so there is a least proof congruence $E[e_0 = e_1]$ containing those instances. Proposition 3.9 is the exact descent test for its contextual closure.

4.2 Gentzen Cut as a two-punctured rule context

When Gentzen Cut is present in the calculus, a formula-mediated concrete instance is the binary occurrence

$$\frac{\pi : \Gamma \vdash \Delta, A \quad \sigma : A, \Pi \vdash \Lambda}{\Gamma, \Pi \vdash \Delta, \Lambda} (\text{GCut}_A), \quad (118)$$

whose rule instance identifies the paired A -occurrences, the remaining multiset contexts, and the induced component relation. Its two-punctured corolla

$$K_A = \text{Cor}(\text{GCut}_A) = \text{GCut}_A(\square_1, \square_2)$$

is therefore a punctured rule context, and filling instantiates the rule:

$$K_A\langle\pi, \sigma\rangle = \text{GCut}_A(\pi, \sigma). \quad (119)$$

Cutting at the roots of the two immediate premise derivations gives

$$(\text{id} \otimes \text{pr}_1)\Delta_{\text{CK}}[\text{GCut}_A(\pi, \sigma)] = [\pi \sqcup \sigma] \otimes [K_A]. \quad (120)$$

Fixing either premise instead gives a one-punctured Gentzen-Cut context,

$$K_{A,\sigma}(x) = K_A\langle x, \sigma\rangle, \quad K_{\pi,A}(y) = K_A\langle\pi, y\rangle,$$

of the corresponding derivative sector. The reduced coproduct of the completed Gentzen-Cut proof contains the three immediate-input terms

$$[\pi] \otimes [K_{A,\sigma}(\Box)] + [\sigma] \otimes [K_{\pi,A}(\Box)] + [\pi \sqcup \sigma] \otimes [K_A]. \quad (121)$$

Thus Gentzen Cut closes these particular one-punctured derivations by formula-mediated composition, and CK cut exposes both an input proof and the context which uses it. The right factor records the concrete Gentzen-Cut occurrence. Equation (108) gives $\Omega_{K_A} = 0$; for a compound realisation of the same operation, Ω instead measures the cuts whose selected vertices enter that compound outer context.

Corollary 4.8 (CK exposure of a local Gentzen-Cut reduction). *Let $\ell, r : \Xi \rightarrow G$ be input-linear punctured contexts admitted over a common material base, with the same labelled premise telescope and root, meaning that each input variable occurs exactly once in each context. Suppose that ℓ contains a designated local Gentzen-Cut redex and r is an independently justified local reduct, possibly containing Gentzen Cuts of lower rank. For every admitted tuple of input proofs $\mathbf{t} = (t_i)_{i \in I}$:*

- (i) *every CK cut, and every iterated cut, lying wholly inside the named t_i is preserved at the same relative input address;*
- (ii) *with $\chi_{\ell,r} = e_r - e_\ell$, the remaining discrepancy is*

$$\Delta_{\text{CK}}\chi_{\ell,r}(v) = \chi_{\ell,r}(v) \otimes \mathbf{1} + (\text{id} \otimes \chi_{\ell,r})\lambda_\Xi(v) + \text{Res}_{\ell,r}(v), \quad (122)$$

and the algebra quotient by the aligned generator ideal preserves the full CK coproduct exactly when the residue satisfies (114);

- (iii) *reductions at address-incomparable subtrees with disjoint input-variable sets, whose applicability is determined by their displayed premise requirements, commute by associativity of context composition. A finite sequence whose every step preserves one common labelled premise telescope preserves all cuts internal to those input proofs and has the telescoping total residue.*

If the reduct context and fillers are Gentzen-Cut-free, the reassembled proof is Gentzen-Cut-free. A rank-lowering reduct inherits its decrease from the underlying proof calculus.

Proof. Lemma 4.1 gives (i), and Theorem 4.6 gives (ii). Context-composition associativity and Corollary 4.4 give (iii). The final claims inspect the constructor tags and Gentzen-Cut ranks of the fixed reduct. $\square$

This corollary represents the local Gentzen-Cut-reduction schemata supplied by a calculus. Any rank decrease, termination, or confluence property remains inherited from that underlying calculus. A global elimination procedure normally inspects the final rules above a Gentzen Cut; cutting down to those immediate subderivations presents each such local schema in the required form. For a structural case which erases or duplicates a premise, Theorem 4.6 protects the inputs transported unchanged and retains the remaining evidence-use change in the protected-input residue.

Remark 4.9 (The published Communication case and reused input). The standard Communication case in the Gentzen-Cut-elimination proof for GLC, the hypersequent calculus for Gödel–Dummett logic, is deliberately outside the input-linear corollary above. In the extended-multicut induction, the same proof is supplied to two induction-hypothesis calls; their outputs are then recombined by Communication and external Contraction [6, §3.1, proof of Theorem 11]. The extended multicut is a derived metarule rather than a concrete vertex of the present carrier. A CK residue calculation must therefore first choose an explicit expansion into ordinary Gentzen Cuts and structural constructors. With the reused proof left unprotected, the protected residue records the resulting copying. It should not be described as an automatically vanishing permutation residue; a presentation-independent treatment would require an explicit sharing convention or a sharing-aware carrier. An alias map on punctures can record that several independent tree copies came from one formal input; only a proof-DAG carrier can make those occurrences refer to one shared proof node. Filling and CK cutting would then have to distinguish a shared subproof from repeated tree occurrences, so this is not a local change to Definitions 1.3–1.4.

4.3 Substitution-uniform operations and CK resource use

The preceding coaction distinguishes linear transport from erasure and copying. We now make the corresponding positive invariant explicit. Fix $C = (\mathcal{B}, \mathcal{R})$, put $T_C(X)_G = \text{OpenPrf}_C(X)_G$, and let X be a finite family of sorted formal proof variables. Define the substitution category Sub_C by

$$\text{Sub}_C(X, Y) = \prod_{x \in X} T_C(Y)_{\text{sort}(x)}, \quad (123)$$

with simultaneous proof-term substitution as composition and $(x)_{x \in X}$ as the identity at X . The assignment $Y \mapsto T_C(Y)_G$ is the representable functor $\text{Sub}_C(\{z : G\}, Y)$.

Proposition 4.10 (Representation of substitution-uniform proof operations). *For every finite sorted X and boundary G , evaluation gives a bijection*

$$T_C(X)_G \cong \text{Nat}(\text{Sub}_C(X, -), T_C(-)_G). \quad (124)$$

The term R represents the family $\sigma \mapsto R[\sigma]$, and a natural family Θ is represented by the unique generic value

$$R_\Theta = \Theta_X((x)_{x \in X}). \quad (125)$$

Persistent reinterpretation sends the representing term R to the same term over the larger constructor signature.

Proof. This is the Yoneda lemma for Sub_C . Directly, naturality at a substitution $\sigma \in \text{Sub}_C(X, Y)$ gives $\Theta_Y(\sigma) = R_\Theta[\sigma]$, proving existence and uniqueness. $\square$

The proposition uses naturality under *proof substitution*, not merely under enlargement of the material base. Base-extension naturality alone would still allow an operation to inspect the root constructor of an input and branch on it. Proposition 4.10 is standard free-term infrastructure rather than a Hopf theorem. Its role here is to identify the proof-theoretically meaningful domain on which the CK resource observation acts.

Every $R \in T_C(X)_G$ may be displayed as an ordinary context $\widehat{R}$ with distinctly addressed puncture occurrences together with an *alias map*

$$\alpha_R : \text{Punc}(\widehat{R}) \longrightarrow X. \quad (126)$$

The map need not be injective or surjective. Filling all punctures labelled x_i by one proof inserts a fresh tree copy at each occurrence; it does not create a shared DAG node. Put

$$\nu_i(R) = |\alpha_R^{-1}(x_i)|. \quad (127)$$

Allowing arbitrary alias maps is an independent-copy structural envelope. In a substructural proof theory, its noninjective or nonsurjective members are diagnostics of contraction-like or weakening-like use, not structural rules automatically admitted by the object calculus.

Adjoin pairwise fresh nullary probes $\pi_i : () \rightarrow G_i$ in a formal persistent calculus extension C^π , and write

$$t_R = R[\pi_i/x_i]_i. \quad (128)$$

Over $k[\mathbf{z}]$, let $\varphi_{\mathbf{z}}$ be the algebra character on H_{C^π} which sends the connected one-vertex tree $[\pi_i]$ to z_i and every other connected tree generator to 0. Reuse the all-ones character $\zeta([F]) = 1$ from Theorem 2.12, after scalar extension, and define

$$\mathbf{Q}_R(\mathbf{z}) = (\varphi_{\mathbf{z}} \otimes \zeta) \Delta_{\text{CK}}[t_R] = (\varphi_{\mathbf{z}} * \zeta)([t_R]). \quad (129)$$

Theorem 4.11 (CK resource polynomial and compositional proof use). *For every finite open proof term R ,*

$$\boxed{\mathbf{Q}_R(\mathbf{z}) = \prod_i (1 + z_i)^{\nu_i(R)}}. \quad (130)$$

On the integral form, the coefficient of $\prod_i z_i^{r_i}$ counts the simultaneous CK cuts which detach exactly r_i occurrences of probe i and no nonprobe subtree; when $R = x_i$, selection of its sole root probe is represented by the whole-tree coproduct endpoint. In particular, the multidegree recovers the entire use vector $\nu(R)$.

If $S_i \in T_C(Y)_{G_i}$, then

$$\nu_j(R[S_i/x_i]_i) = \sum_i \nu_i(R) \nu_j(S_i), \quad Q_{R[S]}(\mathbf{z}) = \prod_i Q_{S_i}(\mathbf{z})^{\nu_i(R)}. \quad (131)$$

For a tuple of output terms $\mathbf{R} = (R_a)_a$, put $N_{\mathbf{R}}(a, i) = \nu_i(R_a)$, with rows indexed by outputs and columns by inputs. Then $N_{\mathbf{R} \circ \mathbf{S}} = N_{\mathbf{R}} N_{\mathbf{S}}$.

Proof. Freshness makes the probe vertices precisely the variable occurrences of R . If R is rule-rooted, every probe is a nonroot leaf, so every subset of probe occurrences is a CK-admissible cut. The character $\varphi_{\mathbf{z}}$ kills every cut whose detached forest has a connected component other than a single probe, while ζ sends every retained punctured remainder to 1. The surviving sum therefore chooses or does not choose each occurrence independently, giving (130). When $R = x_i$ is nodeless, the empty cut and whole-tree endpoint of $\Delta_{\text{CK}}[\pi_i]$ supply $1 + z_i$, so no exception is required.

Under simultaneous substitution, each of the $\nu_i(R)$ occurrences of x_i receives an independent copy of S_i . Counting the resulting variable occurrences proves the first equation in (131); applying (130) proves the second. $\square$

After base change to a general coefficient ring, the integer coefficients in (130) are observed through their images; in positive characteristic distinct cuts may therefore collect modulo that characteristic. Address-resolved witnesses retain the unreduced family. The top monomial remains monic, so the multidegree still recovers $\nu(R)$.

The relation with the context derivatives of Section 1.3 is exact after separating ordered pointings from unordered CK selections. Define the use monomial

$$M_R(\mathbf{u}) = \prod_i u_i^{\nu_i(R)}. \quad (132)$$

For a multi-index $\mathbf{r} = (r_i)_i$, let $\text{Pt}_{\mathbf{r}}(R)$ consist, for each i , of an ordered choice of r_i distinct puncture occurrences in $\alpha_R^{-1}(x_i)$. Write $D^{[\mathbf{r}]}$ for the multivariate Hasse derivative, characterised integrally by

$$D^{[\mathbf{r}]}f(\mathbf{u}) = [\mathbf{h}^{\mathbf{r}}]f(\mathbf{u} + \mathbf{h}).$$

Corollary 4.12 (Pointing–cut correspondence and binomial transform). *Work first over the integral form. The group $\prod_i \mathfrak{S}_{r_i}$ acts freely on $\text{Pt}_{\mathbf{r}}(R)$ by reordering the selected occurrences. Its orbit set is canonically the address-resolved family of coproduct choices which detach exactly r_i copies of probe π_i and no nonprobe subtree: these are ordinary probe cuts when R is rule-rooted, with the whole-tree endpoint providing the nonempty choice when $R = x_i$. Consequently*

$$Q_R(\mathbf{z}) = M_R(\mathbf{1} + \mathbf{z}), \quad [\mathbf{z}^{\mathbf{r}}]Q_R = (D^{[\mathbf{r}]}M_R)(\mathbf{1}) = \prod_i \binom{\nu_i(R)}{r_i}. \quad (133)$$

Over a $\mathbb{Q}$ -algebra, or whenever every displayed factorial is invertible, the middle expression is

$$\frac{1}{\prod_i r_i!} \left. \frac{\partial^{|\mathbf{r}|} M_R}{\prod_i \partial u_i^{r_i}} \right|_{\mathbf{u}=\mathbf{1}}.$$

Moreover,

$$M_{R[S]}(\mathbf{u}) = \prod_i M_{S_i}(\mathbf{u})^{\nu_i(R)},$$

so the use-monomial law is the exponent-vector shadow of the typed context composition in Theorem 2.8.

Proof. After the alias map, an element of $\text{Pt}_{\mathbf{r}}(R)$ is an ordered tuple of distinct occurrences of each formal input. Forgetting the orders gives precisely a subset of probe addresses. For a rule-rooted term every such subset is a CK cut because the probes are leaves; for $R = x_i$, the selected root probe is the whole-tree endpoint. The action is free and each orbit has cardinality $\prod_i r_i!$. Theorem 4.11 and the defining identity for Hasse derivatives then give the displayed equations. The last formula is the occurrence-counting part of (131). $\square$

The container derivative used earlier marks existing inputs of a sum-product grammar. Differential linear logic, the differential λ -calculus, and resource Taylor expansion instead make linear differential substitution and the approximation of reusable !-proofs part of the proof or term theory [22, 23, 24, 21]. The present construction introduces neither an exponential modality nor differential reduction: even if such constructors belong to a chosen equipped calculus, (133) is only the finite occurrence-counting shadow common to the two settings, enumerating subsets of copies already present in one represented proof operation.

For the running context

$$R_{m,i}(x_S, x_T) = i(m(x_S, x_T))$$

from (28), $Q_{R_{m,i}} = (1 + z_S)(1 + z_T)$. Its three nonconstant monomials are the probe cuts at the same three address selections as the two singleton ground cuts and the simultaneous two-ground cut displayed there. Fixing b_T gives the one-input slice $s(x) = i(m(x, b_T))$ with $Q_s(z) = 1 + z$, the same context used in (105).

For a binary rule occurrence $a : (H, H) \rightarrow G$, the aliased context $R(x) = a(x, x)$ gives

$$Q_R(z) = 1 + 2z + z^2. \quad (134)$$

Integrally, the coefficient 2 is not read from the label a : it is the collected multiplicity of the two singleton probe cuts at distinct premise addresses. The top term is the single simultaneous cut detaching both independent copies. More generally, $\nu_i = 1$ identifies linear use, $\nu_i \leq 1$ affine use, and $\nu_i \geq 1$ relevant use; unrestricted finite values describe the independent-copy envelope. These classify how one represented operation uses its declared proofs, not whole logics by rule name.

Corollary 4.13 (Resource-profile obstruction to CK-stable proof identity). *Let R, R' be substitution-uniform operations with the same sorted formal input family and root boundary. Let E be a CK-stable proof congruence on the fresh-probe extension, in the sense of Proposition 3.9, and suppose it is probe-conservative: the character $\varphi_{\mathbf{z}}$ factors through the Hopf quotient. If their probe completions satisfy $t_R \sim_E t_{R'}$, then*

$$Q_R = Q_{R'}, \quad \nu(R) = \nu(R').$$

Consequently, different use vectors obstruct this stronger, factorisation-stable proof identity.

Proof. If q_E is the quotient map and $\varphi_{\mathbf{z}} = \bar{\varphi}_{\mathbf{z}} q_E$, the all-ones character also factors as $\zeta = \bar{\zeta} q_E$. Hence

$$Q_R = (\bar{\varphi}_{\mathbf{z}} * \bar{\zeta})(q_E[t_R]) = (\bar{\varphi}_{\mathbf{z}} * \bar{\zeta})(q_E[t_{R'}]) = Q_{R'}.$$

The monic top multidegree in (130) recovers the use vector. Probe conservativity is essential: an equation identifying a fresh probe with an old same-boundary ground deliberately erases the observation. $\square$

Proposition 4.14 (Full-input detection of arbitrary independent copying). *Assume an input of type G admits a positive-degree filler t . For $r \geq 0$, define the basiswise map*

$$\text{Copy}_r : \tilde{C}_{G,x} \longrightarrow \bigotimes_{j=1}^r \tilde{C}_{G,x_j}, \quad x \longmapsto \bigotimes_{j=1}^r x_j, \quad [u] \longmapsto \bigotimes_{j=1}^r [u], \quad (135)$$

with the empty tensor at $r = 0$ interpreted as the erasure map to k carrying its trivial coaction. The map is not an input-comodule morphism for any $r \neq 1$. Thus substitution-uniformity does not entail preservation of proof use.

Proof. The source coaction has the whole-filler term $[t] \otimes x$ of first-factor degree $|V(t)|$. For $r = 0$ the target has no such positive degree term. For $r > 1$, its tensor coaction contains $[t]^r \otimes (x_1 \otimes \cdots \otimes x_r)$ in degree $r|V(t)|$, exceeding every first-factor degree in the source. Neither discrepancy can cancel against a proper-cut term. For $r = 1$, the map is the comodule isomorphism induced by relabelling x as x_1 . $\square$

The protected-input construction localises this discrepancy to the unprotected inputs. Theorem 4.6 compares such a transformation relative to the unaffected positions by placing the erased or independently copied premise outside Prot . The changed use then remains in the residue. A noninjective alias map records several copies generated from one formal input; coherent sharing still requires an explicit sharing or Contraction occurrence, or a proof-DAG carrier.

All results in this subsection use exact proof-boundary sorts and licensed constructor occurrences. They assume neither premise Weakening nor global Gentzen Cut and therefore survive a material support

relation which is nonmonotone in its formula premises. Proof-variable substitution fills an already licensed premise slot; it is not arbitrary formula substitution, which would require additional substitution invariance of the material base. Revisory changes which invalidate old grounds remain outside the persistent-extension naturality claim, exactly as explained after Definition 2.15.

Remark 4.15 (Harmony does not determine a use vector). Resource use is fixed by the surrounding structural regime, not by harmony alone. In a calculus with Weakening, a harmonious projection may erase one introduced proof; with Contraction, a harmonious β -reduction may substitute several copies of its argument; in a linear regime the corresponding input is required exactly once. Thus Corollary 4.13 gives a necessary condition for probe-conservative CK-stable identity, not a criterion for harmony and not a structural-regime-independent demand that $\nu_i = 1$. Harmony, use profile, literal input preservation, and CK descent therefore test four different aspects of a conversion.

4.4 Normalisation, harmony, and material scope

A principal β -reduction, an η -expansion, or a local Gentzen-Cut reduction is first justified in the underlying proof theory. The CK semantics then separates two further questions: whether the transformation preserves the named derivations on which it acts, and whether its aligned schematic equation preserves the collected CK factorisation modulo the equation itself and, when chosen, the background I_{bg} . Input preservation answers the first; the relative coideal criterion answers the second. If the equation is subsequently closed under arbitrary contexts, Proposition 3.9 tests that full congruence.

Each anchored rule context in an input-preserving comparison is uniform under persistent extensions of its licensing base:

$$j_*(s\langle \mathbf{t} \rangle) \cong (j_*s)\langle j_*\mathbf{t} \rangle. \quad (136)$$

New grounds may provide new fillers, but do not alter the punctured operation. An introduction or reduction schema thereby defines a stable way of using arbitrary material reasons, while its concrete rule instances state any sharing, erasure, separation, or component transport explicitly. An ordinary harmony datum—an independently justified introduction/use pair together with its principal β/η conversions or inversion principle—may therefore be tested for input preservation and CK descent without being redefined by either test.

Proposition 4.16 (Principal tensor Gentzen-Cut reduction). *In the unit-free multiplicative linear fragment, let*

$$\pi : \Gamma \vdash A, \quad \sigma : \Delta \vdash B, \quad \tau : A, B, \Theta \vdash C.$$

The principal tensor Gentzen-Cut redex and its standard reduct have rule contexts

$$\begin{aligned} s_{\otimes}(\pi, \sigma, \tau) &= \text{GCut}_{A \otimes B}(\otimes R(\pi, \sigma), \otimes L(\tau)), \\ r_{\otimes}(\pi, \sigma, \tau) &= \text{GCut}_B(\sigma, \text{GCut}_A(\pi, \tau)). \end{aligned} \quad (137)$$

They have the same three linear input variables and root boundary. Their non-cut boundary formula occurrences and derived component relations agree, while their intermediate rule vertices differ. The reduction $s_{\otimes} \rightarrow r_{\otimes}$ is input-preserving and replaces the Gentzen-Cut formula $A \otimes B$ by Gentzen Cuts on its proper subformulae.

On the unresolved input $v_0 = x_{\pi} \otimes x_{\sigma} \otimes x_{\tau}$, however, their raw context-relative factorisations differ:

$$\text{Cut}_{s_{\otimes}}(q) = 2q + q^2, \quad \text{Cut}_{r_{\otimes}}(q) = q, \quad \text{Res}_{s_{\otimes}, r_{\otimes}}(v_0) \neq 0. \quad (138)$$

Put

$$U_A = \text{GCut}_A(x_{\pi}, x_{\tau}), \quad V_B = \text{GCut}_B(x_{\sigma}, \square).$$

The unmatched local term is

$$\begin{aligned} (\text{pr}_1 \otimes \text{pr}_1) \text{Res}_{s_{\otimes}, r_{\otimes}}(v_0) &= [U_A] \\ &\quad \otimes [V_B] \neq 0, \end{aligned} \quad (139)$$

and it cannot be absorbed by enlarging the proof congruence. For every proof congruence E containing the aligned tensor equation,

$$N_{r_{\otimes}}^E([U_A]_E, [V_B]_E) = 1, \quad N_{s_{\otimes}}^E([U_A]_E, [V_B]_E) = 0. \quad (140)$$

Hence no CK-stable proof congruence can contain the principal tensor reduction equation.

Proof. Each input derivation occurs once, so Corollary 4.8 gives input preservation; the displayed reduct is the standard principal Gentzen-Cut reduction and lowers formula complexity. Direct inspection of the displayed rule instances shows that the same non-cut boundary formula occurrences are carried to the conclusion; composition of the derived component relations gives the same component transport. In the source context, the two nonroot occurrences $\otimes R$ and $\otimes L$ are incomparable, yielding two singleton cuts and their two-element cut. In the reduct, the only nonroot context occurrence is the inner GCut_A . The cut formula (106) gives (138). The reduct residue has the unique bidegree-(1,1) term displayed in (139): it cuts off U_A and retains the outer V_B corolla. The redex residue has only bidegrees (1,2) and (2,1).

It remains to show that no larger proof congruence can identify this term with a redex factorisation. The three detached redex factors are

	flattened root	number of punctures
$U_R = \otimes R(x_\pi, x_\sigma)$	$\Gamma, \Delta \vdash A \otimes B$	2
$U_L = \otimes L(x_\tau)$	$A \otimes B, \Theta \vdash C$	1
$U_R \sqcup U_L$	$(\Gamma, \Delta \vdash A \otimes B) \mid (A \otimes B, \Theta \vdash C)$	3.

(141)

The factor U_A has root $\Gamma, B, \Theta \vdash C$ and two puncture requirements

$$\{\{\Gamma \vdash A, A, B, \Theta \vdash C\}\}.$$

Its puncture number already separates it from U_L and $U_R \sqcup U_L$. If its root were equal to that of U_R , cancellation would give $C = A \otimes B$ and $\Delta = B, \Theta$. After cancelling the common requirement $\Gamma \vdash A$, equality of their puncture-requirement multisets would then require

$$A, B, \Theta \vdash A \otimes B = B, \Theta \vdash B,$$

which is impossible. Thus U_A has the open interface of none of the three redex factors. Equation (84) prevents any proof congruence from identifying them, proving (140); Proposition 3.11 gives the conclusion. $\square$

At the coarser resource level both contexts have

$$Q_{s_\otimes} = Q_{r_\otimes} = (1 + z_\pi)(1 + z_\sigma)(1 + z_\tau).$$

They therefore pass the resource-profile obstruction. Thus the reduction is harmonious, linear in all three named inputs, and input-preserving, but no boundary-and-interface-preserving proof identity which contains it can retain raw CK decomposition. It preserves the supplied derivations, non-cut boundary incidences, and component transport while changing intermediate ancestry—the feature recorded by raw CK decomposition. A more general algebraic Hopf ideal could annihilate the unmatched factor or cross its proof-interface sector, but such an ideal is not induced by a proof congruence in the sense of Definition 3.8.

Reassigning degrees cannot create the missing factorisation. For example, let $\text{rk}(A)$ be the connective complexity of A , put $w(\text{GCut}_A) = e_{\text{rk}(A)}$ in $\mathbb{N}^{(\mathbb{N})}$, put $w(a) = 0$ on every other rule occurrence, and set $w(F) = \sum_{v \in V(F)} w(a_v)$. Retain vertex degree alongside w . Then

$$w(F) = w(P^c(F)) + w(R^c(F))$$

for every CK cut, because the two factors partition the vertices. The resulting pair gives an auxiliary connected Hopf multigrading but leaves the coproduct and its unmatched support unchanged. The usual cut-rank measure also makes the displayed reduction decreasing; in the associated graded of the inhomogeneous relation ideal, however, the initial relation annihilates the higher redex term rather than identifying its decompositions with those of the reduct. At the still cheaper level of the cut-size diagnostic, weighting a cut c by $q^{|c|}$ multiplies the unmatched singleton term by q ; $q = 0$ hides it only by deleting every proper decomposition of a connected proof. Even after multiplicities are forgotten, the two sets of factor-class pairs differ. A rank-sensitive restriction changing which cuts are admitted would instead define a new coproduct and would require its coassociativity and descent to be proved afresh.

Example 4.17 (Disharmony precedes CK descent). In the singleton-hypersequent sector, Prior's connective tonk combines the rules

$$\frac{\pi : \Gamma \vdash A}{\Gamma \vdash A \text{ tonk } B} (\text{tonk}R) \quad \frac{\tau : B, \Delta \vdash C}{A \text{ tonk } B, \Delta \vdash C} (\text{tonk}L)$$

[38]. Their principal Gentzen Cut has inputs $\pi : \Gamma \vdash A$ and $\tau : B, \Delta \vdash C$, but no uniformly typed lower-rank reduct composes them: one supplies A , where the other requires B . There is therefore no independently justified input-preserving comparison for the CK residue test. Exchange or resource-management equations cannot supply the missing A -to- B bridge; a background which already does so merely masks, rather than repairs, the disharmony.

The examples give a four-stage diagnostic rather than a classification of reductions by rule name.

- (i) Independent refinements at distinct punctures satisfy the strict interchange of Corollary 2.3; both schedules are the same proof context, so no quotient test arises.
- (ii) A locally interface-isomorphic rule presentation gives genuinely distinct raw trees but transports their ancestry and cuts bijectively; Proposition 3.12 supplies a nontrivial CK-stable quotient.
- (iii) A genuine commuting conversion or principal reduction compares distinct but uniformly typed contexts. Input preservation and CK descent are then separate questions. The MLL permutation of Example 3.14 and the tensor reduction above both preserve their named inputs, but no proof congruence containing the displayed equations can retain raw CK descent.
- (iv) For tonk, no uniformly typed principal comparison exists, so the factorisation test never begins.

The two middle stages locate the boundary of the construction: local presentation symmetry is harmless to recursive factorisation, whereas a well-typed ancestry-changing conversion requires the protected residue and class-level criterion and can meet an absolute obstruction.

The preceding examples concern proof transformations supplied by the calculus. Readers concerned only with proof transformation may skip the algebraic calibration in Proposition 4.18; its proof-theoretic conclusion is that the canonical projector rejects Gentzen-Cut-containing presentations instead of choosing reducts for them. Birkhoff factorisation instead combines a Hopf decomposition with a target-algebra projection which marks unwanted terms. Taking “contains a Gentzen-Cut occurrence” as the unwanted property is the most direct constructor-level attempt to make this standard subtraction machinery return normal proofs. The following calibration shows exactly why it does not.

When an algebra is a direct sum of two subalgebras, projection onto either summand is an idempotent Rota–Baxter operator in the convention used below. This algebraic splitting is the input required by the standard Birkhoff recursion; it says nothing yet about reducts of proof trees.

Proposition 4.18 (Birkhoff collapse to constructor deletion). *Let $H = H_{\text{CK}}$, $D = \text{GCut}$, and*

$$H_{\text{cf}} = \text{span}_k\{[F] : F \text{ contains no Gentzen-Cut occurrence}\}, \quad I = I_{\text{GCut}},$$

let $\Pi_{\text{cf}} : H \rightarrow H_{\text{cf}} \hookrightarrow H$ be the idempotent Hopf endomorphism induced by constructor deletion, and put $\Pi_{\text{bad}} = \text{id} - \Pi_{\text{cf}}$. The two summands in $H = I \oplus H_{\text{cf}}$ are subalgebras, so Π_{bad} is the idempotent Rota–Baxter projection satisfying

$$\Pi_{\text{bad}}(x)\Pi_{\text{bad}}(y) + \Pi_{\text{bad}}(xy) = \Pi_{\text{bad}}(\Pi_{\text{bad}}(x)y + x\Pi_{\text{bad}}(y)). \quad (142)$$

Convolution combines two maps by applying them to the two factors of every CK decomposition and multiplying the results. Regarding H as both source Hopf algebra and target algebra, this is

$$f \star g = m_H(f \otimes g)\Delta_{\text{CK}},$$

and use the Birkhoff convention $\phi = \phi_-^{\star-1} \star \phi_+$. For $\phi = \text{id}_H$, the factorisation relative to Π_{bad} is

$$\phi_+ = \Pi_{\text{cf}}, \quad \phi_- = \Pi_{\text{cf}} \star S_{\text{CK}}. \quad (143)$$

Here S_{CK} is the antipode, the convolution inverse of id_H . Thus the positive part fixes a Gentzen-Cut-free forest and annihilates every forest containing a Gentzen-Cut occurrence.

Proof. The Rota–Baxter identity and existence and uniqueness of the displayed factorisation for a splitting into two subalgebras are standard [19]. Since I is a Hopf ideal,

$$(\Pi_{\text{cf}} \star S_{\text{CK}})(\ker \varepsilon_{\text{CK}}) \subseteq I.$$

Associativity of convolution and $S_{\text{CK}} \star \text{id}_H = u\varepsilon_{\text{CK}}$ give

$$(\Pi_{\text{cf}} \star S_{\text{CK}}) \star \text{id}_H = \Pi_{\text{cf}}.$$

Thus $\Pi_{\text{cf}} \star S_{\text{CK}}$ and Π_{cf} have the required negative and positive target parts and satisfy the stated convention; uniqueness gives the result. $\square$

No finite-type hypothesis is used here. The homogeneous k -modules may have infinite rank—for example, the degree-one piece already does so if the calculus contains a distinct nullary initial id_A for every formula A of an infinite language—but every basis forest has only finitely many admissible CK cuts. Hence the antipode and the convolution values above are pointwise finite and require neither a graded dual nor a completion.

The proposition marks the exact limit of constructor-based subtraction: the antipode recursively cancels CK factorisations, but the projector records only constructor membership. The resulting Birkhoff factorisation therefore recovers the deletion map of Theorem 2.19; it contains no typed choice of a Gentzen-Cut reduct. For this canonical constructor-membership splitting, Birkhoff subtraction is a rejection filter, not a Cut-reduction procedure.

Remark 4.19 (What a normalisation map additionally requires). If the calculus supplies boundary-preserving typed local reducts, a well-founded complete strategy, and determinism or confluence, recursive reduction on complete proof forests gives an idempotent algebra retraction

$$N : H^{\text{cl}} \longrightarrow H_{\text{cf}}^{\text{cl}}.$$

A Gentzen-Cut-containing proof normally has a nonzero normal form, so $\ker N \neq I_{\text{GCut}} \cap H^{\text{cl}}$. Nor does N extend freely to punctured contexts: a bare Gentzen-Cut corolla generally cannot reduce before the final rules of its fillers are known. The calculus-supplied measure need not be CK vertex number. The protected residue of Theorem 4.6 is therefore the local interaction of N 's reducts with CK factorisation, while global descent still requires the proof-congruence criterion.

Open problem: rank-sensitive Birkhoff data. For a fixed normalising calculus, one may ask whether there are a target Rota–Baxter algebra (A, P) , a character $\Phi : H \rightarrow A$ encoding typed local redex data, and a decoding of the positive Birkhoff factor Φ_+ which agrees with a chosen normalisation strategy on complete proofs. The Cut-rank multigrading above could organise such a recursion, but it supplies neither the target projection P nor the typed choice of reduct. Projecting only by constructor membership or positive rank again gives deletion, while Proposition 4.16 prevents normalisation equality from being imposed as a boundary-preserving quotient of the raw tree coproduct. A positive solution therefore needs genuinely new target data, not merely a regrading of H .

The tensor calculation and the tonk contrast therefore concern logical harmony relative to a fixed structural regime. The next section instead examines an external structural rule whose component action is proof-theoretically significant: Communication.

5 Communication and analytic hypersequent proof

Gentzen Cut supplies the principal proof-transformation example. We now turn to a rule for which the displayed hypersequent structure is itself essential. Here *analytic* means the availability of a direct Gentzen-Cut-free proof operation, not merely derivability of the same boundary by some proof. Communication remains an admitted connected vertex, not a quotient equation: its immediate-input CK cut detaches the two premise proofs while retaining the crossed Communication corolla. Avron's separated-tail Communication rule for single-conclusion hypersequents is

$$\frac{p : \mathcal{G}_1 \mid \Gamma_1 \vdash A_1 \mid \mathcal{H}_1 \quad r : \mathcal{G}_2 \mid \Gamma_2 \vdash A_2 \mid \mathcal{H}_2}{\mathcal{G}_1 \mid \mathcal{G}_2 \mid \Gamma_1 \vdash A_2 \mid \Gamma_2 \vdash A_1 \mid \mathcal{H}_1 \mid \mathcal{H}_2} (\text{Com}). \quad (144)$$

The passive tails may be empty [4, 5]. Unlike a presentation with a shared passive side, this form has two literal whole-hypersequent premise positions and requires no implicit sharing. Its concrete vertex also selects the active component occurrence c_i in each premise.

5.1 Crossed component transport

Let c_i be the active component of premise i , and let $d_1 = (\Gamma_1 \vdash A_2)$, $d_2 = (\Gamma_2 \vdash A_1)$. When both antecedent blocks are nonempty, the active component relation is

$$\kappa_{\text{Com}}^{\text{act}} = \{(c_i, d_j) : i, j \in \{1, 2\}\} = \mathbf{K}_{2,2}. \quad (145)$$

At formula-occurrence level there is nevertheless neither copying nor erasure:

$$\Gamma_1 : c_1 \rightarrow d_1, \quad A_1 : c_1 \rightarrow d_2, \quad \Gamma_2 : c_2 \rightarrow d_2, \quad A_2 : c_2 \rightarrow d_1. \quad (146)$$

Each arrow abbreviates the appropriate address-preserving bijection. If an active block is empty, its corresponding occurrence-induced edge is absent, so $K_{2,2}$ is a property of the displayed nondegenerate instance rather than of every substitution instance.

For comparison, the connected bar-assembly Asm_1 of (69) places two hyperproofs beneath one rule root and transports their component addresses bijectively; it is not commutative forest multiplication.

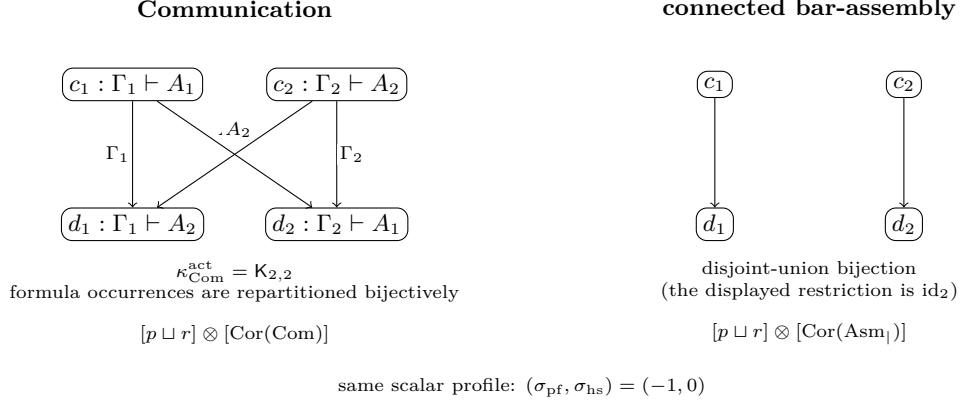


Figure 4: Component transport at the immediate-input CK cut. In the displayed nondegenerate Communication instance, each input component contributes disjoint formula occurrences to both outputs, giving $K_{2,2}$ at component level without copying or erasing any formula occurrence. Connected bar-assembly instead transports the two component occurrences bijectively.

Proposition 5.1 (Scalar collision, relational separation). *Communication and connected bar-assembly both have*

$$(\sigma_{\text{pf}}, \sigma_{\text{hs}}) = (-1, 0).$$

On the nondegenerate active instance, however, $\kappa_{\text{Com}}^{\text{act}} = K_{2,2}$, which is neither functional nor inverse-functional, whereas bar-assembly has a disjoint-union bijection. Their full formula-occurrence transports are bijective.

For compatible complete proofs p, r , cutting at their roots gives

$$\begin{aligned} (\text{id} \otimes \text{pr}_1) \Delta_{\text{CK}}[\text{Com}(p, r)] &= [p \sqcup r] \otimes [\text{Cor}(\text{Com})], \\ (\text{id} \otimes \text{pr}_1) \Delta_{\text{CK}}[\text{Asm}_1(p, r)] &= [p \sqcup r] \otimes [\text{Cor}(\text{Asm}_1)]. \end{aligned} \tag{147}$$

The corresponding immediate-input entries of CT therefore have the same cut addresses and scalar size. Their restrictions to the two active input and output components carry, respectively, $K_{2,2}$ and id_2 ; the full relations additionally carry every passive component occurrence.

Proof. Both occurrences have two proof premises and preserve the total number of displayed components, giving the scalar pair. Equations (145) and (146) give the two incidence claims. Bar-assembly sends each premise component to its own output component. Final-rule extraction and Proposition 3.6 give the displayed CK and component-trace statements. $\square$

Corollary 5.2 (Component-transport obstruction to macro-realisation). *Let $\mathcal{S}$ be a fragment whose every concrete rule occurrence has a functional component relation. Then every proof context generated by $\mathcal{S}$ has functional composite component transport. Hence no context in $\mathcal{S}$ with the typed interface of a nondegenerate Communication instance can realise its active $K_{2,2}$ -transport. Dually, the same conclusion follows if all generating relations are inverse-functional.*

Proof. Functionality and inverse-functionality are preserved by disjoint union and relational composition (Lemma 3.3), whereas $K_{2,2}$ has neither property. $\square$

Thus the component trace is not merely descriptive: it obstructs macro-realisation of Communication in a fragment whose local rules are uniformly splitting-free (all local rule traces are functional) or uniformly merger-free (all local rule traces are inverse-functional). The uniformity is essential: Example 5.3 shows that mixed splitting and merging traces can compose to $K_{2,2}$ at the level of bare component transport. This defeats that bare relational obstruction only; it does not construct a correctly typed Communication macro.

Example 5.3 (Splitting calibrates the component criterion). Avron’s intuitionistic Splitting rule in GLC is

$$\frac{\mathcal{G} \mid \Gamma, \Delta \vdash A \mid \mathcal{H}}{\mathcal{G} \mid \Gamma \vdash A \mid \Delta \vdash A \mid \mathcal{H}} \text{ (SI)}$$

[5, §III.1.2]. If c is the active input component and d_Γ, d_Δ the two active outputs, then

$$\kappa_{\text{SI}}^{\text{act}} = \{(c, d_\Gamma), (c, d_\Delta)\}, \quad (\sigma_{\text{pf}}, \sigma_{\text{hs}}) = (0, +1). \quad (148)$$

Thus Splitting is nonfunctional but inverse-functional on active components; at formula-occurrence level it also copies the succedent occurrence A . It has the same scalar pair as external Weakening which adds one component, but that rule is functional and has an unsupported output, hence is nonsurjective. The relation separates the two structural actions which the scalar pair identifies.

This also calibrates the limit of Corollary 5.2. Communication-free GLC still contains Splitting and external Contraction. At the level of bare component relations, a contraction trace $\{(c_1, e), (c_2, e)\}$ followed by the splitting trace $\{(e, d_1), (e, d_2)\}$ composes to $\mathsf{K}_{2,2}$. Component transport alone therefore does not prove the Gentzen-Cut-free nondefinability of Communication in that fragment; the analytic-support separation below is strictly stronger.

Why the residue is not the separator here. The relative residue from Section 4 compares contexts only when they share one typed premise telescope and root boundary. Generic Communication and bar-assembly do not share such an interface: bar-assembly retains the two premise hypersequents as disjoint output blocks, while Communication repartitions their active formula occurrences. Even when interfaces do agree, every one-vertex corolla has zero residue by (108). Final-corolla extraction and the CK-resolved component relation are therefore essential here: they distinguish root-level component transport which scalar CK cut data and the input-relative residue leave unresolved.

The scalar defects check boundary arithmetic; the component relation classifies the relevant transport. The immediate-input cut exposes two independent hyperproofs together with the two-punctured Communication corolla whose concrete rule instance crosses their active components.

5.2 Prelinearity and a resource-sensitive check

Communication yields the familiar prelinearity derivation

$$\begin{aligned} A \vdash A, B \vdash B &\xrightarrow{\text{Com}} A \vdash B \mid B \vdash A \\ &\xrightarrow{2(\rightarrow R)} \vdash A \rightarrow B \mid B \rightarrow A \\ &\xrightarrow{2(\vee R), \text{EC}} \vdash (A \rightarrow B) \vee (B \rightarrow A). \end{aligned} \quad (149)$$

Here EC is external Contraction. The soundness, completeness, and Gentzen-Cut elimination of Avron’s hypersequent calculus GLC for Dummett’s logic LC, and the connection of Communication with prelinearity, are inherited results [4, 5, 6]. The present calculation is proof-intensional. Let $C_{\text{lin}}(-)$ be the outer context formed by the two $(\rightarrow R)$ -steps, the two $(\vee R)$ -steps, and external Contraction, and put

$$t_{\text{lin}} = C_{\text{lin}}(\text{Com}(\text{id}_A, \text{id}_B)).$$

The CK cut at the two identity roots contributes the explicit tensor

$$[\text{id}_A \sqcup \text{id}_B] \otimes \boxed{[C_{\text{lin}}(\text{Com}(\square_A, \square_B))]} \text{ to } \Delta_{\text{CK}}[t_{\text{lin}}]. \quad (150)$$

The boxed retained factor carries two premise obligations, the $\mathsf{K}_{2,2}$ Communication relation at its embedded corolla, and the later component-local logical and contraction steps. Retrospectively it records how the two identity proofs were recombined; prospectively it can accept any proofs of the same two hypersequent premises.

The same active-component relation is not an artefact of internal Weakening or Contraction. HUL, the hypersequent calculus for uninorm logic, is based on multiplicative–additive intuitionistic linear logic and has no general internal Weakening or Contraction rules. Its empty-shared-side Communication instance is

$$\frac{p : \Gamma_1, \Delta_1 \vdash \Psi_1 \quad r : \Gamma_2, \Delta_2 \vdash \Psi_2}{\Gamma_1, \Gamma_2 \vdash \Psi_1 \mid \Delta_1, \Delta_2 \vdash \Psi_2} (\text{Com}_{\text{UL}}). \quad (151)$$

Whenever both premises contribute occurrences to both output components, its active relation is $K_{2,2}$ and its occurrence relation is bijective. In particular, the identity instance

$$B \vdash B \quad A \vdash A \quad \Longrightarrow \quad A \vdash B \mid B \vdash A \quad (152)$$

has this profile. No internal Weakening or Contraction occurrence appears in the calculation. We rely on the established metatheory of HUL [35, 7].

5.3 Consequence equivalence versus analytic proof structure

Communication is consequence-conservative in full GLC but analytically intensional in its Gentzen-Cut-free fragment. Let C^+ be GLC and let C^- be the system obtained by deleting Communication while retaining all other GLC rules, in particular conjunction, Splitting, and Gentzen Cut. For any equipped calculus C , write H_C for its proof-tree Hopf algebra and put

$$\begin{aligned} \text{Conseq}(C) &= \{G : \text{Prf}_C(G) \neq \emptyset\}, \\ \text{An}_D(C) &= \{G : \text{Prf}_C^{-D}(G) \neq \emptyset\} \\ &= \{G : (H_C/I_D)_{\text{cl}}^{\text{con}}[G] \neq 0\}. \end{aligned} \quad (153)$$

The second set is the support of direct connected complete proofs avoiding the constructor family D ; its quotient presentation uses the sector notation fixed at the start of Section 3.

Proposition 5.4 (Communication as analytic intensional enrichment). *On GLC hypersequent boundaries,*

$$\text{Conseq}(C^-) = \text{Conseq}(C^+), \quad \text{An}_{\text{GCut}}(C^-) \subsetneq \text{An}_{\text{GCut}}(C^+). \quad (154)$$

The first equality holds because Avron derives Communication in C^- using conjunction, Splitting, and Gentzen Cut; the second says that this macro-realisation is unavailable analytically. For distinct atoms $p \neq q$, take

$$G_{p,q} = p \vdash q \mid q \vdash p \quad (155)$$

as a witness to strictness. It has the direct Gentzen-Cut-free proof $t_{p,q} = \text{Com}(\text{id}_p, \text{id}_q)$ in C^+ , but no Gentzen-Cut-free proof in C^- .

The direct proof has reduced CK coproduct

$$\begin{aligned} \overline{\Delta}_{\text{CK}}[t_{p,q}] &= [\text{id}_p] \otimes [\text{Com}(\square_1, \text{id}_q)] \\ &\quad + [\text{id}_q] \otimes [\text{Com}(\text{id}_p, \square_2)] \\ &\quad + [\text{id}_p \sqcup \text{id}_q] \otimes [\text{Cor}(\text{Com})]. \end{aligned} \quad (156)$$

The constructor-deletion quotient of Theorem 2.19 expresses the inherited separator as

$$\begin{aligned} (H_{C^+}/I_{\text{GCut}})_{\text{cl}}^{\text{con}}[G_{p,q}] &\neq 0, \\ (H_{C^-}/I_{\text{GCut}})_{\text{cl}}^{\text{con}}[G_{p,q}] &= 0 \end{aligned} \quad (157)$$

Proof. The consequence and Gentzen-Cut-free claims are Avron's derivability and separation results [5, §III.2.1, Theorem 1]. The three coproduct terms are the three nonempty CK cuts of the two identity leaves. The quotient reformulation follows because deleting the Gentzen-Cut constructors kills exactly the proof presentations which use that rule. $\square$

Thus equality of ordinary consequence support does not imply equality of the direct analytic proof operations represented in the constructor-free quotient. Constructor deletion sends a Gentzen-Cut-containing proof to zero; Gentzen-Cut elimination constructs a new proof in the surviving carrier. Avron's Communication elimination requires Gentzen Cut and sufficient logical structure, while the atomic separator rules out a Gentzen-Cut-free replacement in the smaller fragment. Avron supplies that separation; CK exhibits the direct proof's three punctured factorisations and crossed final-corolla relation.

6 Conclusion

The CK structure has two positive proof-theoretic readings. First, the ancestry-forgetting map is a Hopf morphism from typed proof forests to the poset Hopf algebra of their dependency orders. Its cut terms are presentation-causal frontiers; coproduct iterates encode staged premise-first assemblies—equivalently, reversals of backward-refinement histories—while convolution powers of ζ and ω_1 count weak layerings and fully sequential schedules, respectively. Via the standard order-polynomial reciprocity theorem, the antipode evaluates signed strict layerings; it is not a normalisation procedure. Histories differing only by the scheduling of independent refinements present the same context. Thus the proof-context carrier records no chosen temporal schedule, while the CK coproduct retains and enumerates the presentation causality on which the later proof-net obstruction turns.

Second, substitution-uniform hypothetical proof operations are represented by open proof terms, and CK gives those operations a compositional resource observation. The polynomial $Q_R(\mathbf{z}) = \prod_i (1 + z_i)^{\nu_i(R)}$ is obtained by filling formal inputs with probes and evaluating the actual coproduct. Integrally, its coefficients count probe selections at distinct occurrences, so the coefficient 2 for a duplicated input is not inferred from a contraction label. Use vectors compose under substitution by matrix multiplication, and probe-conservative CK quotients must preserve them. More precisely, if $M_R(\mathbf{u}) = \prod_i u_i^{\nu_i(R)}$, then $Q_R(\mathbf{z}) = M_R(\mathbf{1} + \mathbf{z})$; Hasse derivatives of M_R at $\mathbf{1}$ count unordered CK probe selections, whereas ordinary iterated derivatives mark ordered pointings. This is a finite occurrence-counting bridge to the combinatorics familiar from differential linear logic, not a differential or Taylor semantics for the present carrier. The representation statement is standard free-term infrastructure; the incidence morphism, convolution counts, resource polynomial, and comodule obstructions are the specifically Hopf-algebraic content.

Proof identity is chosen independently of CK factorisation. Proof classes always support filling; they support recursive CK decomposition if and only if equivalent proofs have the same multiplicities of equivalent detached evidence and equivalent punctured remainders. In that case the algebra of proof classes is the universal connected filtered Hopf quotient realising precisely those equations. Local interface symmetries give a nontrivial positive family: they may forget a local evidence tag or permute genuinely symmetric premise slots while transporting every CK cut and its multiplicity. Material-ground tag irrelevance is the canonical instance. For the discrete nullary material presentation its quotient passes from tagged reasons to extensional primitive support, commutes with persistent base extension, and retains every compound logical and structural vertex. The resource polynomial supplies a further necessary test: any probe-conservative CK-stable identity must preserve the input-use vector. Harmony, input preservation, linear use, and CK stability therefore remain four distinct requirements; the harmonious tensor reduction passes the first three in the displayed linear instance and still fails the fourth.

The negative boundary is equally exact. No proof congruence containing the standard MLL proof-net permutation or the harmonious principal tensor reduction can retain the raw CK coproduct: their exposed factors lie in different proof-interface sectors. For local conversions, the protected-input coaction first removes factorisations internal to designated supplied proofs, and its residue localises the remaining change; the class-level criterion then proves the global no-go. Thus the tensor reduction preserves its input reasons, non-cut boundary incidence, and component action but not recursive tree factorisation. tonk fails at the earlier proof-theoretic stage because no uniformly typed principal reduct exists.

The construction begins by decomposing a typed and decorated hypersequent proof tree into detached subproofs and a punctured root context. Premise-slot types and component incidence make the retained factor an operation on a typed filling fibre. Generic-filler separation shows conversely that one tuple of fresh, distinctly tagged reasons determines the raw context which uses them. The typed CK-cut/context-composition bijection linearises to the Faà di Bruno identity, and its singleton sector pairs prospective one-slot insertion with retrospective one-address cutting.

Component transport has its own compositional semantics. The occurrence-resolved κ -map respects typed context composition, and the congruence given by equality of κ -values gives exactly the quotient which forgets everything except net component movement. CK-resolved traces refine that extensional quotient by retaining where the movement occurs. In particular, the crossed $K_{2,2}$ trace of Communication both separates it from bijective bar-assembly and obstructs its macro-realisation in a uniformly splitting-free or uniformly merger-free fragment. Splitting calibrates the limit: Communication-free GLC already has enough relation-level splitting and merger to compose a $K_{2,2}$ trace, so Avron’s analytic separator is strictly stronger than that local obstruction. At class level, C^- and C^+ have equal ordinary consequence support but strictly different Gentzen-Cut-free analytic support.

The standard constructor quotient and its truncated grammar respectively reject and enumerate

constructor-free presentations. The associated Birkhoff factorisation, for the canonical constructor-membership splitting, collapses to the same deletion projector; it does not normalise a proof. Typed reduts, a well-founded strategy, and confluence or determinism remain proof-theoretic data. A nontrivial rank-sensitive Birkhoff semantics would additionally require a target Rota–Baxter algebra, a character which encodes typed redex data, and a theorem decoding its positive factor as a chosen normal form; cut rank alone supplies none of these.

None of the fixed-calculus causal or resource results assumes monotonicity of the material support relation in its formula premises. They use exact typed boundaries and already licensed proof occurrences, not premise Weakening. Persistent base extension may add grounds while retaining the old ones and therefore acts by injective Hopf maps. A genuinely defeasible change which withdraws an old ground is revisory instead: the fixed-fibre theorems remain valid, but no forward naturality map is claimed. Constructor deletion can remove affected presentations, and punctured contexts expose replacement obligations; no general dynamics of defeat is claimed.

Two limitations of the tree carrier point toward one graph-level problem. The proof-net obstruction arises because a tree chooses an ancestry between rules which the net treats as independent; the extended-multicut Communication case arises because a tree represents reuse by copied occurrences rather than one shared input. A proof-DAG or net carrier with explicit sharing and native subnet decompositions could address both, but it would require new notions of admissible cut and filling and a new coassociativity proof. It is not obtained by a further quotient of the present tree coproduct. A tracelet-style route [9] is one concrete candidate: present proof construction as typed rewrites, quotient histories only by a proved shift equivalence of independent rewrites, and define decomposition on the resulting causal traces rather than attempting to descend raw CK through proof-net equations. Formula interfaces and hypersequent-component data would constrain that independence, but cannot decide it by themselves. The decisive open test is whether such classes admit typed puncturing and filling and give the two MLL sequentialisations a common native decomposition.

A second natural step is a first-order refinement with explicit eigenvariable and witness data and freshness support, modulo α -renaming. Fully instantiated quantifier occurrences still have nondependent premise telescopes; only a schematic presentation in which an exposed name or witness parameter determines later premise requirements would require a genuinely dependent refinement of the present judgement-indexed CK-cut/context-composition identity. The present framework therefore supplies positive factorisation-stable proof identities while distinguishing equality of consequences, identity of proof uses, harmony, and stability of recursive proof decomposition without reducing one to another.

Notation index

Symbols are grouped by their main scope. Roman capitals such as G, H, K denote hypersequent boundaries unless a more specific meaning is displayed; lower-case letters are local and are redeclared when their role changes.

scope	symbol	meaning
global	$\mathcal{L}, \mathcal{B}, \mathcal{R}$	formula language, material base, and admitted rule stock
global	$G, H, K \in \mathbf{HSeq}$	proof-boundary hypersequents; a sequent is a singleton boundary
global	$\text{Comp}(G), \text{br}(G)$	component occurrences of G , and their number
global	$a : (G_1, \dots, G_n) \rightarrow G, \text{Cor}(a)$	concrete rule occurrence and its one-vertex corolla; $n(a)$ is its proof-input arity
global	κ_a	local relation from premise-component occurrences to conclusion-component occurrences
global	$t, u; R$	derivation trees, complete unless stated; a punctured derivation or proof context, respectively
global	$F, \sqcup, \mathbf{1}$	proof forest, forest juxtaposition, and empty forest
§1	$\square_p, \text{Punc}(R), \text{req}_R(p)$	puncture at address p , puncture set, and required boundary
§1	$\text{Tel}(R), \text{Fill}(R)$	ordered premise telescope and complete filling fibre
§1	$\text{Prf}(G), \text{Ctx}(\mathbf{H}; G)$	complete G -proofs and contexts with labelled premise family $\mathbf{H}$
§1	$\text{OpenPrf}(X), \mathcal{O}(\mathbf{K}; G)$	open proof-term grammar and contexts indexed by requirement word $\mathbf{K}$
§1	$\text{bd}(t), \text{BRoot}(F), \text{Root}(F)$	root boundary of a derivation, block-root profile, and flattened forest boundary
§2	$p; c \in \mathcal{A}(t)$	slot address; CK-admissible cut (neither denotes Gentzen Cut)
§2	$\mathbf{P}^c(t), P^c(t), R^c(t)$	indexed detached subderivations, their forest product, and punctured remainder
§2	$\mathbb{W}_c(t)$	address-indexed CK-cut witness retaining the matching of factors and holes
§2	$[F], H_{\text{CK}} = H_{\mathcal{B}, \mathcal{R}}, \Delta_{\text{CK}}, \varepsilon_{\text{CK}}$	basis vector, the proof-tree Hopf algebra with structural or calculus-relative emphasis, CK coproduct, and empty-forest counit
§2	$\overline{\Delta}_{\text{CK}}, D_F, \mathfrak{G}_{\mathbf{K}}^G$	proper-cut coproduct, premise-ancestry poset, and context Green series
§2	$\text{Cut}_i(q), \Delta_{\text{CK}}^{(1)}, \triangleright$	cut-size polynomial, singleton-cut coproduct, and one-slot insertion product
§2	$\text{Hist}_C, \text{Anc}, H_{\text{ForPos}}^{\mathbb{Z}}$	backward-refinement histories, ancestry-forgetting Hopf map, and poset Hopf algebra of rooted forests
§2	ζ, ω_1	all-ones character and degree-one infinitesimal character; their convolution powers count causal layers and schedules
§2–5	I_D, q_D	constructor-deletion ideal and quotient; deletion rejects rather than reduces
§3	$C_G^{\text{con}}, \delta_G$	connected G -rooted derivation space and its root-preserving coaction
§3	$\sigma_{\text{pf}}, \sigma_{\text{hs}}, \kappa_R, \text{CT}(t)$	scalar defects, composite component transport, and CK-resolved transport profile
§3	$E, \sim_E, I_E, q_E, A_E$	proof congruence, its ideal, quotient map, and algebra of proof classes
§3	$\text{Int}(F), \mathcal{S}, E_{\mathcal{S}}$	open proof-interface signature, local interface symmetries, and their generated proof congruence
§3	$E_{\text{gr}}, \mathcal{B}_{\text{ext}}$	material-ground tag irrelevance and the extensionalised nullary material base
§3	$N_F^E(U, V)$	multiplicity of proper CK factorisations into E -classes U, V
§4	$\Xi = (x_i : G_i), V_{\Xi}, \lambda_{\Xi}$	labelled input telescope, pointed input space, and input coaction
§4	$\text{Sub}_C, T_C(X)_G, \alpha_R, \nu_i(R)$	proof-substitution category, open proof terms, independent-copy alias map, and input-use number
§4	$\varphi_{\mathbf{z}}, \mathbf{Q}_R(\mathbf{z})$	probe character and CK resource polynomial
§4	$M_R(\mathbf{u}), \text{Ptr}(R), D^{[\mathbf{r}]}$	use monomial, ordered pointings, and multivariate Hasse derivative
§4	$e_s, \Omega_s, \text{Res}_{s, s'}, \text{Res}_{e_0, e_1}^{\text{Prot}}$	context evaluation, context-relative residue, full-input replacement residue, and protected replacement residue
§4	Prot, χ, J	protected input positions, comparison difference, and generated comparison ideal
§4	$I_{\text{bg}}, q_{\text{bg}}, K_A$	accepted background ideal, its quotient map, and Gentzen-Cut corolla
§4	$\Pi_{\text{cf}}, \Pi_{\text{bad}}, \star, S_{\text{CK}}$	constructor-free/bad projectors, convolution, and antipode
§5	$\text{Com}, \text{Asm}, \text{SI}$	Communication, connected bar-assembly, and Splitting vertices
§5	$\text{Conseq}(C), \text{An}_D(C)$	ordinary consequence support and direct D -free analytic support

Two conventions are especially important. A proof forest is a multiset of proof-labelled derivation trees, not an ordinary displayed hypersequent. A CK cut decomposes an existing tree; Gentzen Cut is a rule occurrence; filling or context composition inserts a derivation at a typed puncture. Finally, $[F]$ is a linear basis vector, whereas $[F]_E$ is an E -equivalence class.

Several familiar glyphs are deliberately local. The Δ in a sequent such as $\Gamma \vdash \Delta$ is a formula multiset,

unlike Δ_{CK} . The proof polynomial $P_{\mathcal{B},\mathcal{R}}$ is unrelated to the detached factor $P^c(t)$. The letter C may denote a displayed proof context or, when declared as $C = (\mathcal{B}, \mathcal{R})$, an equipped calculus. Finally, D_F is a premise-ancestry poset, whereas D in I_D and An_D is a family of constructor tags.

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